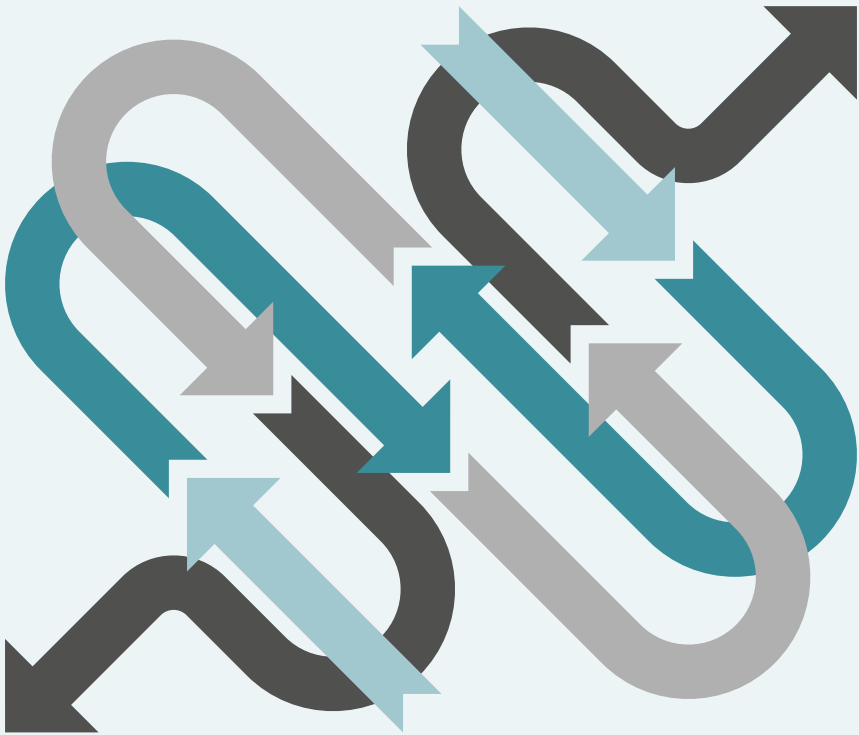


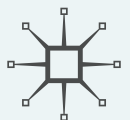
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USEFUL GAME THEORY

Fundamentals of
Decision Making



Jay Prag
Amanda Ishak Prag



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Jay Prag · Amanda Ishak Prag

Useful Game Theory

Fundamentals of Decision Making

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macmillan

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ISSN 2731-5614

ISSN 2731-5622 (electronic)

Palgrave Executive Essentials

ISBN 978-3-031-75153-0

ISBN 978-3-031-75154-7 (eBook)

<https://doi.org/10.1007/978-3-031-75154-7>

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Synopsis

Using parables, famous songs, and classic expressions, *Useful Game Theory: Fundamentals of Decision Making* takes readers on an exploration of human nature and the long road of choices known as “life.” Prag and Prag turn a mathematical discipline into a digestible (dare we say “enjoyable”) read, with a little wit, a lot of sarcasm, a humanist approach—and almost no math. From nuclear disarmament to Friday movie nights, each chapter guides you through a Game Theoretic analysis of the decisions humans make every day.

A grandmother's wit and wisdom are never lost, especially if her grandkids are writers.

*Granny Jean
Granny Sue
Sitou Augenie
Sitou Assene*

We love and miss you.

Robert, Thomas, and Julianna: it's all for you.

—J&A, 2024

Preface

Using parables, famous songs, and classic expressions, *Useful Game Theory* takes readers on an exploration of human nature and the long road of decision-making known as “life.” Prag and Prag turn a mathematical discipline into a digestible (dare we say “enjoyable”) read, with a little wit, a lot of sarcasm, a humanist approach—and almost no math. From nuclear disarmament to Friday movie nights, each chapter guides you through a Game Theoretic analysis of the decisions we make each day.

While this book will occasionally wander into the realm of controversy, our hope is that it makes some of the world a little less confusing. Game theory as an intellectual exercise is math-heavy, hypothetical, and full of controls that can render it almost useless. This book takes a more practical approach, applying the tools and lessons of game theory to the real world—a place that is controversial, complicated, uncontrolled, and very confusing.

In these pages, you’ll find an approach that’s relatable and not heavily analytical. Our goal is to enhance your understanding of game theory by applying it to the decisions and problems humans face every day. Some applications in this book are uncomfortable and challenging to navigate, topics like politics, religion, violence, and societal dysfunction. There are also unconventional uses of economics, sociology, public relations, management, and other academic disciplines. Game theory exists at the nexus of these practices, and it is at the heart of the humanity-based decisions we make every day. We didn’t want to show only the good stuff. After all, the human experience is both good and bad.

For those in academia who are uncomfortable with our anecdotal adoption of your areas of study, we offer this advice: focus on the goals of game theory. The discipline grew from a desire to formalize how human beings (from individuals to groups to companies to countries) make decisions. If you're so into math you forget it's about the human experience, you've missed the point. All academic disciplines stem from casual observations of the real world. So, focus on game theory, chill out, and remember: game theory is just another way of thinking about situations. It exists side-by-side with other ways of thinking. It is not a substitute; it is a complement. If applied science is a test of empirical knowledge, let's take the test.

For students of game theory, a bit of advice as well: we're going for breadth, not depth. In the pantheon of game theory games, this book focuses on a relatively small number. There are many complex games that are, more or less, for AI only. There are others that are less complex, but also less relevant to the choices you make every day. We are going to analyze a dozen or so games that resonate with your day-to-day life, and we'll apply those results to quite a few real-world situations.

Upland, USA

Jay Prag
Amanda Ishak Prag

Acknowledgments This book was made possible through the partnership and commitment of Bronwyn Geyer and the editorial team at Palgrave Macmillan.

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1

What Is Game Theory?

Your Granny knew game theory. She may not have called it “game theory,” and she probably didn’t review a whiteboard of advanced calculus to make every decision... But, Granny used it every day. We know this because she understood people. From the nosy neighbor to the annoying aunt to the helpful clerk at the grocery store, Granny knew how to navigate the world and get what she needed. She did it with a lot of wit and the goal of being helpful. We’re willing to bet you’d take your Granny’s advice over the predictions of a mathematical model any day.

Granny really knew game theory. So did Shakespeare, Aesop, Confucius, Machiavelli, most country singers, and the writers of every major children’s book in history. From *The Art of War* to “In the Jailhouse Now,” many of the songs, expressions, and parables we heard growing up are actually solutions or approaches to the games we play in everyday life.

Game theory, also known as strategic decision-making, is a simple concept: you and another person make decisions, and together your choices determine a specific outcome. That’s really it.

So, why do game theory books have so much darn math? Well, it turns out that all the little things matter. For example: *How many people are making decisions? Are they all choosing at the same time? Do I get to choose first and, maybe, more than once? How often am I choosing, and with whom?* These factors (and so many more) impact how a decision turns out for you, and math can help us quantify and model these factors to get to a predictable outcome. Math is useful, just not in this book. Here, we’ll focus on understanding the basics. So, game theory helps us understand the structure and

probable outcomes for things like which freeway to take, how to get a raise, and how to choose a movie on a Friday night.

Game theory doesn't teach you how to *win* games. It teaches you how to set them up to maximize the likelihood of your best option. What's the difference? Well, as much as mathematics and logic can provide humanity, they're not foolproof. A game theory model can do its best to include all possible variables, but when human beings are making decisions in the real world, things get missed. When people read a book or take a class in game theory, they are often disappointed to discover they aren't learning how to win every game. There are no books or classes to teach you such strategies. All game theory can provide is a way to understand the structure of a game and ways to increase the *likelihood* of your best outcome.

Long before Brouwer, Morgenstern, von Neumann, and John Nash, governments and societies set rules to increase the likelihood of best societal outcomes for situations that happen every day. Laws determine how people in modern society are allowed to play a game. When you drive up to an intersection with stop signs on every corner (a four-way stop), why not just always go through first? That's an approach to solving the game, right? Well, yes, but it's a unilateral decision. It doesn't consider what other players might do (and if everyone followed that approach, there would be a lot of crashes at that intersection). So, the government says you have to stop at a stop sign and look around, and society has given you a preferred way of making a choice (usually, the first person to arrive or the person to the right—in the US—goes first). Rules help set up a game by providing a framework for the players, and any decision strategy is bound by those rules.

Organizations do it too. Disneyland welcomes thousands of visitors daily—visitors from different cultures and countries with different expectations of waiting in line. The thing is, it's Disney. You're going to wait in line. So, how does the mouse make sure things run smoothly? Each park sets up clear boundaries with ropes, walls, fences, and hedges, to guide visitors where they need to go. A trip around the Magic Kingdom shows you there is no ambiguity about what visitors are expected to do. Compare this level of control to your local grocery store the Wednesday before Thanksgiving and you'll start to see organizational boundaries in action.

Unilateral decisions, rules, organizational boundaries—now we're getting somewhere. Game theory is very much a byproduct of human evolution. History, anthropology, and biology are full of examples where unilateral decisions (ex, I should kill you before you kill me) gave way to more cooperative solutions that helped our species thrive. When other people make choices that ultimately determine an outcome, self-interest can turn out poorly for

everyone. It's why Granny told us "Do unto others as you would have others do unto you." Granny knew game theory.

Strategic decision-making is not only limited to games here and now. On the contrary, many of the choices we make are playing games with our "future selves" (as Doc Brown might have said to Marty McFly: you have to think fourth-dimensionally). Your choices today combine with your future choices to determine an outcome later in life. The marketing slogan "What happens in Vegas stays in Vegas" is an attempt to convince you that your current choices do *not* affect your future outcomes. One trip to Sin City makes it clear that the slogan isn't really true. Most of the time the only thing that stays in Vegas is the money that you gambled away.

Game theory in a classroom is rooted in mathematical theory. Strategic decision-making, on the other hand, is the system we humans have grown up within. You learn a lot about life through experience and daily trial-and-error. That's the point of many of Granny's expressions. She's trying to save you from some potentially painful learning curves. On a good day, Granny understood context and how to set up a game for her benefit. When you told her you were going to Vegas, she might have said, "waste not, want not." She was warning you about the value of saving and the risks of gambling. On the other hand, if she was really cool, she might have just asked you to put \$10 on red.

Granny and all of the purveyors of worldly wisdom give us approaches to problem-solving, but they didn't have time to teach all of the nuances: when does that approach work? Why is it better? Why does setup matter? This book will fill in a few of those details.

Many games that Granny played during her lifetime look different today. We'll teach the game theory behind the expressions to see when, or if, they still apply in the modern world. When Granny's Granny taught her "be careful what you ask for, you might get it," there was no Internet, Facebook, AI, or screenshotting. Granny might not know that in the current world, the expression is cautionary in a lot of new ways.

"You can't tell a book by its cover" is an expression that teaches us not to overstate the importance of external factors. Arguably, this was Granny's way of saying "don't be prejudiced," but it's going to have special meaning in this particular book. This book's title makes game theory seem folksy and perhaps even easy. It is not. Game theory is complicated, subtle, and easy to get wrong. Granny was smarter than any PhD. So, we might as well start with her best hits.

What Is Game Theory?

Spoiler Alert: it's not really a game, and it isn't really a theory.

Game Theory is an attempt to formalize the structure and outcomes of a situation when two or more people are making choices that together lead to an outcome. No one player can unilaterally determine the outcome; it depends on the decisions of both people. *Strategic Decision-Making* is the process of making choices and decisions knowing that there are other “players” making their own choices and decisions, and that the combination determines how you all end up. Game theory is the discipline that looks at how these decisions are made, with an eye toward how to achieve mutually beneficial outcomes.

That's easy! In everyday life, we always think strategically, analyze a situation accurately, run the numbers, and make the best choices, right? Nope.

In the real world, we are emotional, spontaneous, unpredictable, and sometimes wrong. You use game theory every day. It's innate, it's automatic. You do it when you're in a relationship. You do it when you are a parent. You do it when you are driving. You do it when you decide to have another helping of ice cream *with* sprinkles. Game theory began by observing human decision-making and applying math to explain patterns and predictability. At its foundation, though, is the thought process (or lack thereof) we engage in multiple times a day.

Let's do the four-way stop game, now with more detail.

A Four-Way Conundrum

You're driving your new car, and you arrive at an intersection at the same time another car arrives perpendicular to you (across from you) signaling its intention to turn across your path (in the US, it has its left blinker on). Clap those hands, it's game theory time.

If you both go, you're going to crash. Let's assume that's not the outcome either of you wants.

If neither of you go, you'll just wait there forever. That too is not, you know... a great outcome.

You might be thinking “there are driving rules and regulations that tell us what to do.” Are you sure both of you know the same rules and always use them correctly?

You're not in a hurry and you're driving your brand-new car, so you wave at the other driver to go. They turn out of your way first, you drive through

the intersection second, there are no collisions, and everyone makes it to their destination.

You just did a lot of game theory! You and the other driver made choices—you waited and the other driver proceeded—and that combination of choices led to a good outcome. You and your car made it to your destination and you're happy with how things turned out. In the game's payoff matrix, a list of potential outcomes, you are both better off. The other car used a literal signal to tell you where it needed to go. You used a signal, a wave, indicating you would let them go first. All is well.

The structure of the four-way stop above is what we call a two-player, two-choice, simultaneous game: you each made choices at the same time. Game theory will help us break down other variations of this everyday choice. For example, would you have done the same if you were in a beat-up, hand-me-down car instead? If you were in a rush? As we will see, structure combines with preferences to determine an outcome.

Four-Way Stop—Rush Hour Version

While the four-way stop is a nice, generally painless type real-world game, we can complicate things by adding more information. When you and another driver get to a four-way stop at 1:30pm on a lazy Thursday afternoon, there's usually a lot of cooperation and the game solves quickly and easily. If you play the same game with the same people and at the same intersection at 7:45am when people are trying to get to work or get their kids to school (or both!), things are very different.

As the clock ticks toward "you're late," you find yourself blowing your horn at drivers. That's a very different type of signal. You might also coast through the sign, rather than stopping. Your nice, generous wave changes to a not-so-nice one-finger gesture. Time is just not on any player's side.

There's a deeper issue here. As we discuss our games and work through the paths to solutions, think about how choices and outcomes change with factors like duress. How you perceive and play games changes with the circumstances under which the game is being played.

Game Structure

When we talk about structure, we mean all of these nuances and factors that can affect players, choices, and outcomes. Let's not wander too far from our first game, the four-way stop. What was the structure of the game? There were two drivers, so two decision-makers; they had two choices (wait or go); and they both had some internal desired result. They also had constraints: they were in a hurry, or they worried about their new car. Implicit and important to the outcome is an assumption of what's often called rationality. There's more to rationality (a lot more), but for now, let's just say neither had an incentive to crash.

Zero-Sum Games

The payoff of a game should be familiar for most of us: it is the set of possible outcomes for each player based on their decisions. If you're a sports fan, a payoff is the potential score from a play or series of games. More often, though, it's simply a matter of somebody winning and somebody losing.

This is called a zero-sum game, where there's one winner and one (or many) losers. Zero-sum games got their name by applying numbers to a win/loss combo: a win is $+1$, a loss is -1 , so, technically, the sum is zero. There is no middle ground. There's no possibility for everyone to come out a winner. These definitions are perfectly fine when we're thinking about payoffs (ex, 0/1), but the vast majority of interpersonal games we play every day, the sort of games we'll be discussing, are not zero-sum games.

Arguably, zero-sum games are easier to figure out. More points = win; not very much to unpack here. Having no middle ground between winning and losing removes having to think in terms of "better off" or "worse off." Real-world problems are not so easy to solve. When *you* "get your way" in a decision with your significant other, are you both always better off? You won! But it doesn't always feel like a win, right? Does getting your way ever conflict with your relationship dynamic?

Sometimes, we wish we could resolve every conflict with a game of pickup basketball, but that's not real life. The thought process you just went through is precisely why zero-sum games are easier than most interpersonal games, and why most games in life are not zero-sum.

What's in a Game?

The traditional approach to game theory starts by defining three elements present in every game: the **players**, their **choices** (often called a “choice set”), and the **outcomes**. We call these the “who,” “how,” and “what” of a game, something like this (Fig. 1.1):

Most books start with the outcomes. These are called payoffs, or simply the results of the game. Remember, the goal of game theory is to find the “best possible outcome” and see how, if at all, players can get there.

Starting with outcomes is, well, backwards.

Humans evaluate their environment every minute of the day. *Then*, they make choices. In this book, we show that the setup of the game—understanding the players and the choices—is, in fact, the most important aspect for players. If we want to test theories and apply them to the real world, we should take the same approach humans do in everyday life. Think of it like this: you can negotiate a four-way stop better now than you did when you first started driving. Why? Even though the outcome is the same (don't crash), your older, wiser, more-experienced self knows how to read other players and the potential choices better. Understanding players and choices *first* helps you win more games.

Our goal is to examine the “best possible setup,” to achieve an outcome that may change over time. So, let's start with the Players, then Choices. In theoretical games, the outcome only needs be proven (usually mathematically) once. In applied game theory, results need to be repeated to have any validity. At the end of this book, we hope the reader has a broader understanding of how to evaluate players and choices toward better outcomes across many types of games.

WHAT'S IN A GAME?

PLAYERS

- Preferences
- Biases
- Needs
- Capabilities

CHOICES

- Frequency
- Timing
- Location
- Availability

OUTCOMES

- Payoff / Results

Fig. 1.1 What's in a game? Elements of games in game theory

Understand the Players

One of the things that we learn and teach in the realm of game theory is that you have to know things about the other players before you can determine your approach to the game. This issue gets us into a lot of tricky, complicated problems. After all, we're dealing with people, and human beings are not so easy to fit into neat little predictable groups (no matter how hard Marketing and Advertising firms try).

Consider the difference between parenting an infant and parenting a fully verbal (perhaps precocious) toddler. The infant has no ability to process or even react to your choices in a strategic way. She will cry until her problem is fixed (she is fed, her diaper is changed, she gets all the snuggles). Sometimes, she'll just cry for no obvious reason. Waxing poetic about the trials of everyday life probably isn't going to get you any peace and quiet. The difference in communication methods is a clear barrier, and you have to evaluate the game based on the infant's capabilities.

A toddler, by comparison, is learning about games and game theory. He won't be very good at it yet, but he is starting to make active choices that he *thinks* will get his best possible result. He'll test boundaries, play repeated games to see what affects the outcome, and learn what works. He also starts to understand the consequences of his actions. Maybe waxing poetic doesn't work here either, but a stern talking to just might impact the outcome. Playing the "just give me a minute!" game with a toddler could work. Parenting strategy is different for parents of infants (who don't understand consequences) and toddlers and teenagers (who, hopefully, do). Without a sense of time and outcomes, an infant is rarely ever "playing" a game.

We say "rarely," because, at some point, instinct does seem to kick in. By a baby's rationale: "if I cry, I get held. If I cry, I get food. If I cry, I get warm." At a certain point, however, this becomes learned behavior, and it begins to influence how a baby navigates his world. Indeed, some child development experts say to refrain from rewarding crying after a certain age. If it becomes standard practice to always "win" by crying, a toddler grows up characterizing the game very differently than you might want him to. Babies begin pretty self-interested (and they're so cute!), but raising a self-interested child is not usually a goal of most parents.

Selfishness or self-interest poses different levels of risk depending on your relationship with the other player. Strangers and friends are very different players in Strategic Decision-Making. In this book, we'll talk a lot about the Golden Rule: "do unto others as you would have others do unto you." Sure, we wish we could always make the cooperative, best-for-both-of-us choice in

all games, and we hope the other player will be doing the same. That's a lot more likely to work if the other player is a friend, a family member, or just someone you interact with on a regular basis (neighbors, co-workers, etc.). It may not work so well with a passing stranger, an enemy, or a faceless gnat in a comments section. It's harder to see direct consequences if another individual just isn't visible or known to you.

Total strangers aren't necessarily mean or universally self-interested, they just don't know you and you don't know them. So, if the downside of acting cooperatively when the other player is being self-interested is relatively big, neither player will act cooperatively. As we'll see, this outcome varies if the players are strangers or friends, or if they're playing the game once or repeatedly over time.

Wisdom and Truth

Familiarity affects another issue: truthfulness. As Granny used to say, "out of mouth of babes..." Young children are often credited with having clarity and intelligence that belies their age. Not wishing to deflate the proud parents of wise children, what you're often hearing is really straightforwardness, not wisdom. While they are learning about strategic decision-making, children usually don't have the capacity to be political. They are simply more inclined to state the truth than they are to finesse reality.

Suppose Mommy wants Daddy to take her to her office Christmas party, but he really needs a haircut first. Daddy isn't prone to getting haircuts. Mommy says, "darling, I'm swinging by the barbershop later—can I set up an appointment for you?" Junior says, "Dad, you look like a yeti!" Mom is acting strategically; Junior is telling the truth. So, when you hear the expression "honesty is the best policy," remember that in a game-theoretic context, honesty—best policy or not—and wisdom are two different things.

Rules Versus Rules of Thumb

In formal, what might be called "classroom" game theory, we follow approaches to games that lead to solutions in a mathematical way. In that context, we set up mathematical barriers, or rules. In the real world with real people and impossible-to-know things like precise outcomes of a game, the

rules that we've learned in the formal setting often become joined with rules of thumb. Academics call these heuristics. Rules of thumb are just inclinations (fancy people call this instinct), rather than absolute ways to make choices. They're the guidelines we hone over time for how to make our decisions.

Instinct and self-interest go hand-in-hand. A baby knows herself based on satisfaction. You grow into an adult knowing yourself first. You know what makes you happy and sad. Over time, you get to know about those close to you, what makes *them* happy and sad. This is how your rules of thumb develop, and they help you achieve an optimal outcome pretty well when the other player is a relative. The less familiar the other players are, however, the less you know about them. In that case, you'll have to rely more on set rules for making your decisions.

Rules of thumb change games via familiarity. We should note, however, that something *unfamiliar* is not the same as something *different*. You can play games with rules of thumb even when the other player is different if you have a pretty good idea of "what makes them tick." If you are a kind, caring, giving person, but you know the other person in the game is selfish and uncaring, you won't make cooperative choices even though you would like to do so. The selfish person may be familiar, maybe a relative or somebody known well to you; but you are diametrically opposed in how you operate. Different and unfamiliar are not the same.

If used incorrectly, rules and rules of thumb can give you inferior outcomes for no good reason. Such is the case with bad assessment criteria like prejudices. Prejudice makes you *think* you are familiar with a person's choice set, but you are often wrong. Information is key in playing a game correctly, and an incorrect perception of another can lead to bad information and bad outcomes.

You Can't Teach Height

All rules are made up—some in alignment with game theory and others not so much. In this book, you'll see examples of rules that exist to take away or create strategic (game-theoretic) advantages, such as laws that determine who goes first at a stop sign, who gets to move first in chess, blocks you can or can't rezone, or what happens when you take things that don't belong to you.

One big example of a strategic advantage is inherent skill. Many rules are in place to limit (or at least control) the impact of skill on a game. Rules

in competitive sports, for example, focus on leveling the playing field (get it?) regarding athleticism, age, weight, and training time. A player or team with the most skill is supposed to win most of the time, right? Without rules, probably not.

Take basketball.

Basketball's origins date back to the 1890s when a Massachusetts gym teacher wanted to give people an indoor sport to play during winter months. He put a fruit basket on the wall and gave players a point for getting a ball into the basket.

From there, things were formalized (and improved, considering the original basket didn't have a hole in the bottom), and eventually we got to modern professional basketball with a hoop with a net set ten feet above the ground. Jump ahead (seriously, get it?) a hundred years or so, and you find many of the successful basketball players are, well, tall. *If you ignore the other rules*, the easiest way to get a point is to drop the ball directly in the hoop. All the training in the world doesn't matter compared to that ability. Granny says: "you can't teach height."

Without rules, basketball teams would compete to recruit *only* the tallest individuals; not the fastest, not the strongest, not the most agile. But there are other rules. For example, you get more points if you make a basket from far away. Interestingly, the popularity of the game increased when they changed to "three points beyond the free-throw line." Shorter, skilled, agile players can succeed if they are good at making "three pointers." The rule made skill, training, and agility relevant again.

game theory is concerned with how your choices can change outcomes, and that will always depend on the rules of the game. If rules are written (more or less) entirely to benefit a set of inherent skills, you won't have an opportunity to use much game theory.

Food for thought: Given the importance of training, who's probably better at putting together a successful basketball team—the coach or the general manager?

Choices, Choices, Choices!

In your everyday interactions, choices can be simple (regular or decaf?) or extremely complex (move the family to Houston?). When we find ourselves in a situation that calls for the tools of game theory, we depend on our

ability to define a clear set of choices. As humans, we parse decisions out into a million different ways of actually doing them (e.g., you can roll forward slowly instead of coming to a full stop), but for ease of analysis, we'll assume choices are binary. This or that (e.g., everything that allows the other driver to go first counts as "waiting" on your part, whether you're creeping forward or not).

We acknowledge that there is always an alternative to all of the choices in a game: you could choose not to play. This and other options will be discussed later, but it's important to recognize this is almost always an option in daily life. Sometimes, Granny advised us to "just walk away."

Perhaps the most important lesson of growing up is how to effectively navigate multiple choices. Not playing is always an option; but there are several others (procrastination: choose later; abdication: make someone else choose; impatience: choose quickly and get it over with) we employ. Life exists somewhere in between a two-choice, neat, straightforward game on a whiteboard and the processing of *millions* of pieces of information we do every day.

Our games will be as applicable as we can make them, but there's always room to interpret choices for yourself.

Payoffs and Outcomes

We've defined players and examined how choices are organized. When it comes to payoffs, however, the discussion is a bit more complex. Your average game theory book shows you matrices and tree diagrams with several possible outcomes. We'll use a few of those, sure. Sometimes, payoffs are presented in currency or revenue or cost. Other times, they are communicated in something less tangible called a "utils," short for a measure of utilization. We can thank Economics for that one. (Side note: we have yet to meet anyone who can accurately describe what a "util" really is. We think it's a type of guinea pig or maybe a small fruit.)

Payoffs and outcomes, however you specify them, are just numbers that illustrate how players *feel* about a particular decision.

Outcomes are tied to the setup of each game. They come from the preferences, biases, hopes, dreams, and prayers of each player. If you buy a donut for \$2 and you love donuts, your payoff (your happiness) might be 100. If your friend buys the same donut for the same cost and she doesn't like donuts at all, her payoff might be -50 (because she spent money on something she doesn't even like). If your brother kind of likes donuts, but not as much as you, his payoff is a solid 60. Each of you paid \$2 for each donut, but your

happiness—your utility—is vastly different. The payoff is not your financial gain or loss; it is how much happiness you gain/lose based on how the game plays out.

It might be easier to explain it the other way: if each game's payoff was structured *only* in dollars, you wouldn't have enough information to make a real decision. How much do those dollars *matter* to each player? Is the person money-driven or motivated by something else? Until we understand how each player feels about possible outcomes, we can't figure out how a game works. Different players—different human beings—have individual choice sets, so the outcomes depend on your ability to understand and process these preferences in real time.

The Payoff Matrix

When we organize our thoughts around Players, Choices, and Outcomes, we often simplify things in a Payoff Matrix—an easy way to show the structure of a game. Throughout this book, we'll employ many matrices to show different games. Figure 1.2 shows a sample Game Structure matrix (Fig. 1.2).

First, on the outer edge, we identify the players in each game. For this foundational text, we'll usually use two players, but you can always create a matrix with more.

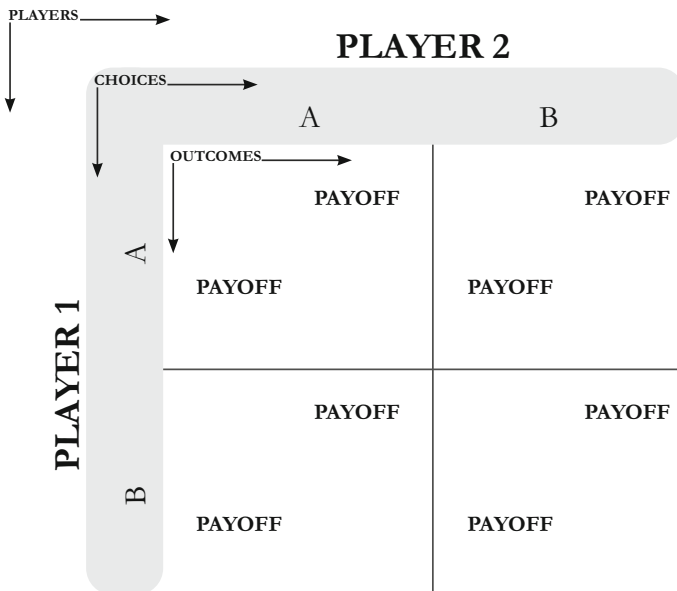


Fig. 1.2 Game Structure

Second, choices are presented in the gray bar. As we've said, these are the only things you can actively decide on in a game.

Finally, payoffs are the outcomes for each player based on potential choices, presented in the matrix itself.

The matrix form of games (spoiler alert: there are other ways to illustrate games!) is one of the easiest and most straightforward ways to visualize choices and outcomes. Don't be fooled by the presence of payoff information. Remember: even though you have full information, the only thing you control in the game is choice (the gray bar). Payoffs *inform* your choice, but you can't directly choose what you get.

Learning Versus Learning Game Theory

"Learning by doing" is an expression that has as many limits and exceptions as it does applications. "Learning the hard way" means you're learning by way of painful failure or mistakes. You might learn the hard way that some people are more selfish than you, that you can't always be selfish, and that you can't make choices with all people the same way. That's learning about the other players, choices, and outcomes in your games.

We're going to focus on learning how to play games, and how to specify the factors of each situation you're in. With that knowledge, you should be able to skip a lot of mistakes that come from misunderstanding the choices before you. You might find yourself in a predicament one day with someone at work and a friend will say "they're playing you." That's your friend telling you that your work colleague better understands the game-theoretical situation than you do. It's a good time to apply some game theory.

You might also learn when *not* to apply game theory. You'll sometimes hear sports commentators say that a coach or a player is "being too cute." This odd expression actually means they are doing something tricky and outside of the playbook. It's unsanctioned, off-strategy—it goes against everything the payoff matrix tells you to do—but it just might work. Game theory and Strategy can be thrown out the window sometimes.

For movie buffs, there's a famous scene in "The Princess Bride" when the bad guy, Facini, is trying to figure out which of two wine goblets is poisoned. Facini uses a lot of game-theoretic reasoning, ultimately making his choice after believing he tricked the masked man into revealing the answer. When Facini keels over dead, the masked man reveals they both had poison, but he was immune. In fact, the game was won when the masked man convinced Facini it was a game and not a foregone conclusion.

Practice Makes Perfect

Here's some circular logic for you: strategy can beat skill, but you have to learn strategy like a skill. It is important to learn, evaluate, and revise your strategy over sequential games (and, given long life, it's always a sequential game).

Skill is based on repetition, and the goal is to have it become innate, unrehearsed, and automatic. Athletes, coaches, and commentators use metaphors like “muscle memory” to describe the desired outcome of practice, which isn't exactly to “make perfect” but rather to automate performances so they're not affected by the stress of the game. Similarly, you'll hear people talk about “clutch performance,” good performance under stress. Is being clutch a strategic choice, or simply the result of repetitive practice and innate skill?

A good strategy, on the other hand, is adaptive. The goal of this book is to show good examples of “think about it, then decide.” Mastering it means entering a mental state where you are ready to make a decision. In studying, practicing, and understanding game theory, you'll make better choices even in stressful situations. The choices are not automatic, but the process of thinking through your choices is.

Think everyday decisions aren't stressful? Think again! Game-theoretic situations, as we'll see, can be literal life-or-death. In basketball, musical performance, treaty negotiations, and bedtime stalling tactics, routine and practice make perfect. Game theory can help change your performance and better your outcomes even against more “skilled” players because you've learned how to evaluate a game.

What About the Numbers?

A question that game theory students frequently ask is “does it have a lot of math?” In its commonly taught form, yes, tons. In its most useful form, no. You usually don't take a graphing calculator along to get engaged.

When we start describing and solving games, we use a “payoff matrix” that has a few numbers. In showing a game's payoffs, we'll use numbers of something (like dollars) that each player will get depending on who does what. We'll then try to determine what each player should do by assuming they will want their best number (usually the biggest number). Saying that

ten is bigger than five is not math. It's a simple comparison. So, assuming you know that ten is bigger than five, this book won't use a lot of math.

You might say "hey, you told me this didn't have a lot of math!" The matrix is just an illustration. This game theory book does not have calculations, it doesn't take derivatives, and it won't require anything more than what most people can do in their head. If you know 3 is less than 4, you're all set.

There's a whole other part of game theory that's full of math. It works out complicated versions of some of the problems we'll be looking at and other problems that are beyond the scope of this book. The math version of game theory has its place (proving Granny right in a mathematical sense). Other times it's proving things that theorists like to prove for the sake of proving them. Either way, math has its place, but not in these pages.

We're taking a humanist approach to game theory: learn how to identify the situation you're in—the players, the payoffs, and the rules—and make a strategic, beneficial choice. You already do this daily, and you learned to learn over time. To get to a more strategic level, we'll use a few numbers, but not math.

Even with our approach, though, game theory is a tricky discipline. In a classroom, we show students a bunch of games with straightforward rules and known outcomes and illustrate how different approaches work for solving those games. In doing this, we are practicing. We are abstracting from reality and doing drills. We are in the batting cage, running scrimmages, practicing our free throws, on a driving range, and hitting tennis balls shot to us by a machine. We are doing "wax on, wax off"—practicing the mechanics of decision-making so we're ready when a real-world opportunity presents itself.

If that reference made you smile, you already know what a lightbulb moment is. For the uninitiated, this is a famous quote from the 1984 film *The Karate Kid*. An old karate expert tells the teenager that he will teach him karate, but first, he has to do a lot of chores. Clean the floor, paint the fence, and wax the cars. In each case, there was a specific way the teenager was told to do the task: back and forth, up and down, circles to the right then circles to the left with each hand. Exhausted from days of chores and having seen no karate lessons, the young man tells the old teacher he's had enough and he's leaving.

The teacher says, "I have taught you karate!" He proceeds to show the boy how each chore was training his body to do certain things. "Wax on, wax off," circles to the right and circles to the left with each hand, is how you deflect your opponent's attacks. Aha!

Once the lightbulb was on, the young man did what he was asked to do knowing it had a point and it was teaching something important that had

nothing to do with waxing or painting. Why can't we teach every skill and lesson directly, without drills? To quote a proverb, "the journey of a thousand miles begin with the first step." If you told a 6-year-old, relatively new piano student they were going to play Rachmaninov's Concerto Number 3 and you showed her the sheet music, she might quit on the spot. So, you start with chopsticks and eventually, she plays a concerto.

Such is the case for most game theory examples. We aren't doing the Prisoner's Dilemma game to teach you what to do when you're being interrogated by the police. (hopefully, that particular situation won't ever apply to many of you). The setup, approach, and ultimate solution to that particular game shows up many times and in many places, so we'll refer to it a lot.

Often, learning game theory feels like trying to learn the unlearnable and knowing the unknowable: what people want, believe, know, and care about. Is it easier to play games with family members than with strangers? Sometimes. Is it easier to do games that are one time only as opposed to repeated play games? Sometimes. Is it easier to get a straight answer to a question from anyone other than these authors? Okay, well this—sometimes—is what the book is all about. Game theory is rooted in context. "When" does "what" work—and "why."

A Real-World, Nontraditional Application: Leadership

We characterize a lot of this introduction as games between equals or people of equal power (and, yes, we are arguing that a parent and a crying baby have equal power). However, game theory is often used in hierarchical relationships, with leader(s) and follower(s) and mutual decisions and outcomes. A lot of what's involved in successful leadership is best explained using a little game theory. Recall the alternative name for game theory is strategic decision-making. That process (really, all decision-making processes) involves information, delegation, trust, relationships (do people like me?), and many other factors.

Consider information, the topic of Chapter 4. Followers are often closer to the front lines and thus have critical information. Suppose that reliable information is crucial for a leader's ability to make correct decisions and for a situation—a game—to reach its best outcome. Now, suppose the leader hasn't developed trust or has proven to be too self-interested. Will the followers tell the leader that crucial information? Suppose earlier in the relationship, a follower did communicate relevant information, and it wasn't used correctly or ignored. Will the follower try again?

Along similar lines, economics uses a concept called Agency Theory (aka, the Principal – Agent Problem) to discuss challenges in delegating responsibility to those under you. Leaders should ask themselves: what do your team members (followers) want to see happen to *you*? If they don't like, trust, or believe in you, will they tell you what you need to know to allow you to succeed? This is game-theoretic stuff. As we'll see, there are important connections between leadership, trust, and game theory.

What *Isn't* Game Theory

There are many times when you and another person interact to determine an outcome, and it has nothing to do with game theory. When you run to the end of the block faster than your 3-year-old, you didn't use game theory; you used the physical skill that you possess. You're bigger and stronger. That's all. One might argue that you made a bad game-theoretic choice, depending on how it made your toddler feel or what she did when you left her behind, but that's not what we were talking about. We're saying that some outcomes between people are determined purely by physical differences or differences in skills. Those aren't the subject of game theory.

Game theory also doesn't justify statistical choices, like buying a lottery ticket every week from the same store at the same time. You never win the big jackpot. There's nothing wrong with your ticket buying "strategy," it's just not the sort of strategy that we're talking about in this book. You're basically being superstitious about something that is actually determined entirely by statistics and probabilities. There are a lot of books that can help you with that kind of game, but this isn't one of them.

Game theory is used when the choices and decisions made by you and another person determine the outcome together. Knowing that you're in such a situation and being aware of how your choices interact is the key to using game theory.

Of course, we sometimes attribute success to luck when in fact you stumbled into a successful game. If you "get lucky" repeatedly, you're probably playing a game that's part skill, part strategy. When you go out to a social gathering with your friends and you say you hope you "get lucky," you really mean you hope you can play the game. Your payoff at the end of the night has very little to do with luck.

Conclusion

Game theory is everywhere! It is family dynamics, work dynamics, political negotiations, grocery shopping, bathtime with Junior, and landing on the moon.

So, we'll relate many popular expressions to game theory. If your interpretation of timeless wisdom is different from ours, don't let that prevent you from seeing the usefulness of our alternative view. We're certainly not saying that the *only* thing Granny knew was game theory.

Game theory and Granny's wisdom—both—teach you to be considerate. It's why people like living where other people are nice to them, and it's why we are taught to trust, respect, and care about others. This book also explores the downside of human interaction, how we treat those who are naïve and gullible. Strategic decision-making, like all things, can be misused.

Ultimately, game theory is a process. It is an approach to situations that makes you ask things like “what do you know?”, “what makes people tick?”, and “what are my choices?” It will explain how, why, and when commitments, promises, and threats work. It will make you think a little harder about those words of wisdom that you heard your entire life and get a better sense of their value. Hopefully, by the end of these chapters, you'll understand when these expressions apply and when they do not.

Questions

- 1.1. Parenting is all about game theory. Most of us have our first experience with strategic decision-making when we were parented, but we didn't know it. Let's try to look back and figure out our parents' strategy:
 “Do your homework! Practice your piano or your gymnastics or your free throws! Do something comparatively boring instead of playing that game on your smartphone.” The 12-year-old thinks that her parents don't understand her or love her. The opposite is actually true. How are these commands part of a game-theoretic situation? What “advantage” do parents have for helping their kid make better choices.
- 1.2. Granny gave you two bits of advice: “A fool and his money are soon parted” and “Do unto others as you would have others do unto you.” One day, you were swindled out of \$100 by a woman who told you (falsely) that she could tell your future by reading your palm. Which of Granny's tidbits of advice was not heeded by the palm reader?

- 1.3. Congrats—you're in a long-term relationship with someone you've fallen in love with. Discuss what's wrong with the popular expression "love at first sight." (Hint: what information do you have?).
- 1.4. You took a class in game theory (or maybe finished reading this book) and decided to apply your new approach in Las Vegas. Not being experienced in many of the games there, you decide to play the lucky number game; you put a pile of chips on a number and the woman behind the table spins a giant wheel. Now, reread this setup carefully: which word lets you know that game theory—strategic decision-making—is not going to be useful?
- 1.5. "The Game of Life" is an old box game, now a computer game, and a book title. "The Games People Play" and "The Circle Game" are both song titles. If we were writing a book of lists, you could make a long one of titles that include the word "game." Most of these references are actually to game theory sort of games, not competitive sports games. As you understand it at this point, what's the fundamental difference between the two?



2

No Honor Among Thieves: The Prisoner's Dilemma

We're going to learn game theory by way of the familiar: typical expressions, known life lessons, and easy stories. As we invoke these familiar things, we add in some unknowns and ambiguities, helping us build an approach to problem-solving that does two things. First, it will help you recognize game-theoretic situations—circumstances where you alone cannot determine the outcome. Second, it will highlight the deeply personal factors that we must consider if we're going to succeed at real-world, non-mathematical game theory.

Let's hit the ground running with the most famous game of all: The Prisoner's Dilemma.

The Prisoner's Dilemma is a good starter because it gives us a set of parameters, structure, that is relatable and applicable to real-world situations. Structure, as we will see, is the most important part of any game. This setup is the foundation for understanding returns, expectations, and goals of each player. Prisoner's Dilemma will also allow us to explain and contextualize many oft-heard expressions in future chapters.

Before we get started...

... as suggested in the introduction, we have to get used to unpacking expressions. The borrowed imagery of the word “unpacking” makes us think about a travel bag or suitcase tightly stuffed with various items, each used only after the bag is *unpacked*. Consider, for example, the expression “no honor among thieves.” A good expression, like a good travel bag, this one holds a lot of stuff that needs to be unpacked before we can put it to good use. We need to think about each word, what goes into it, its context, and how it helps explain a result.

Honor

Historically, societies have created different approaches to get people to follow through on things like promises, commitments, and day-to-day life (which we discuss in detail in Chapter 7). One approach that has endured over time is a thing we call honor. It's a small word but with large connotations. Let's unpack it.

How people feel about outcomes, what economists usually call the utility that people get from their choices, is actually a programmed response. We begin as babies with sensory pleasures—how things taste or feel or smell or sound or appear—and our reactions are natural and inherent. In subsequent years, we “develop a taste” for certain things. This is the learned behavior part of it. Combining our inherent reactions with developed preferences, we figure out what a certain thing really gives us. Roller coasters might not be fun as a toddler, but your teen self loves the experience. Your utility—the benefit you receive from doing something—changes over time. This change is a well-accepted part of life, and it's helped along by reinforcing good choices and good behaviors.

The same learned-taste approach can be used to convince people that they *don't* get pleasure from some things; and, therefore, they shouldn't do them. This applies to things like stealing, murder, and speeding through a crosswalk. Stealing *feels* wrong, intrinsically. Also, society put rules in place to reduce people's incentive to do it. Even if you can't possibly be caught, you (hopefully) don't steal. That stealing is “bad” is an inherent reaction, and society's rules reinforce it.

People are trained that the happiness one gets from consuming something that was stolen doesn't give you the same amount of pleasure that you get when you acquire the same good in a more appropriate manner (buying it, making it yourself, bartering for it). One word that we use to explain the lack of happiness we get from consuming ill-gotten goods is honor. In other words: acquiring things by stealing or deception is not honorable, so this approach should not give you pleasure.

Thieves

By definition, a thief is a person who steals things. So, the expression “no honor among thieves” tracks. Let's unpack it even more. This expression might mean different things to different people, based on your pre-conceived notion and individual biases.

Here's a question: A 6-year-old boy takes another kid's lunch money 1 day in first grade. Does that make him a thief?

Before you answer, consider your own feelings on the following question. When and how do we learn about the *inappropriateness* of things? Should all first graders know right from wrong when it comes to stealing?

Let's add more information: the boy's teacher saw him take the lunch money and told his father, who promptly sat the boy down on the stairs of their house and had a calm conversation about how and why stealing is bad. The father taught him the basic lessons of right and wrong, good and bad, and other things we might ultimately call honor. The boy didn't steal again (and went on to coauthor a book on game theory). Most people would agree the boy is not what they think of as a "thief." He's not the person the expression "no honor among thieves" is referring to. When we unpack Granny's expression and try to use them to understand game theory, we can't be cavalier about the meaning of the words. We cannot assume an all-encompassing definition of each word. We have to be deliberate about the application of certain terms and clear about what is and isn't relevant.

When Granny said, "No honor among thieves," she was talking about hardened criminals who also happen to be the basis of one of our best known and beloved game theory games. On to the Dilemma.

The Prisoner's Dilemma

If you're a fan of police dramas, you'll be familiar with the setup of this famous game. A bank is robbed, and two people, Butch and Sundance, have been arrested. There were no eyewitnesses because the perpetrators were wearing masks. Police need to solve this case.

Hoping to get one of them to confess, detectives put Butch and Sundance in separate rooms and offer each of them the same deal:

- (a) If you confess and your partner does not, you'll get 1 year in prison and your partner will get 10 years.
- (b) If you both confess, you'll both get 5 years in jail.
- (c) If neither of you confesses, you'll both get 3 years because you resisted arrest anyway. (this is a very important part of the game and its outcome!)

As a reader or a student, you're probably saying "well, obviously, neither will confess. It's best for both!" Here's where game theory is useful. Butch and Sundance are evaluating the same choice *in separate rooms*. They don't know what each other will do. The thought process goes like this:

“If I don’t confess and my partner does, I’m going to jail for ten years. If he does confess, I’m a lot better off confessing too. Five is less than ten!

“If he doesn’t confess, I should probably still confess anyway, though. Either way, I’ll be going to jail for three years or one year. One is a lot less than three, and they’re both less than ten years. I’ll confess.”

Player one, let’s say Butch, has reasoned that he’s better off confessing regardless of what Sundance does. Sundance, in another room, thinks the same. So, they both confess because it’s best for them, given what the other *might* do. The Prisoner’s Dilemma’s famous result is that both players confess and get 5 years, even though they could have both gotten 3 years. There was an outcome that made *both* of them better off, but they didn’t get there.

Game Setup Matters

Butch and Sundance couldn’t talk to each other; they decided in a vacuum. In this case, the setup of the game makes reaching a superior result difficult. That’s why game theory is so great. We examine the setup (or structure), walk through the players and their choices, and see how we arrive at an outcome. Often times, this outcome doesn’t seem logical to the reader—but you’re not a player yet. Sometimes, the way a game ends up isn’t particularly ideal for anyone involved. It’s just the solution.

Outcomes can be portrayed in words, or a little more formal in a payoff matrix. Remember: game theory is all about players, choices, and outcomes, known *before* we start playing. You can think of game theory as the art of navigating a payoff matrix. The two choices detectives gave the gentlemen are illustrated in Fig. 2.1.

Using this visual representation of the game, we can see how each player approaches his decision. If one confesses, the other would be better off also confessing. Both choose to confess, and both get 5 years.

Why? Because neither player can be *sure* what the other will choose. The setup prevents collusion. Butch and Sundance both (logically) prefer to reduce their jail time (notice, the Detectives never say they can walk free; that’s not an option). So, each man will choose to confess.

Look at it another way: Butch cannot look at the outcomes in the payoff matrix and say, “I’m going to get 3 years.” He *and Sundance* are making choices (confess or don’t confess) that can lead to many outcomes, and the final outcome depends on *both* player’s choices.

Let’s practice using a payoff matrix. Look “up and down” in Fig. 2.1 to see Sundance’s choices and outcomes. If Butch confesses—the left column in the

		BUTCH	
		CONFESS	DO NOT CONFESS
SUNDANCE	CONFESS	BUTCH 5 YEARS SUNDANCE 5 YEARS	BUTCH 10 YEARS SUNDANCE 1 YEAR
	DO NOT CONFESS	BUTCH 1 YEAR SUNDANCE 10 YEARS	BUTCH 3 YEARS SUNDANCE 3 YEARS

Fig. 2.1 The Prisoner's Dilemma, payoff matrix

matrix—Sundance will either get 5 years or 10 years. He'll choose to confess, since $5 < 10$. If Butch does not confess—the right column—Sundance will get either 3 years or 1 year. $1 < 3$, so he'll confess again. Regardless of what Butch does, Sundance is better off confessing. In game theory parlance, we say Sundance has a dominant strategy to confess.

Run it the other way for Butch, looking right and left. If Sundance confesses—the top row—Butch will either get 5 years or 10 years. He'll choose to confess, because $5 < 10$. If Sundance does not confess—the bottom row—Butch is choosing between 1 and 3 years. He'll confess again. Butch also has a dominant strategy to confess. The game solves at Confess/Confess, with each man serving 5 years in prison.

Confound It!

The Prisoner's Dilemma is a great way to start learning game theory. It is well-structured, has a small number of easy-to-understand choices, a small number of players, and the famous result that both players end up worse off than they could have been if they both had chosen differently. Based on their dominant strategies they will both be going to jail for 5 years, but we see from the payoff matrix that they both could have gone to jail for 3 years. Annoying result, but true.

Well... why didn't they just talk about whether to confess before they committed the crime? Aha, that's a cooperative solution and we'll talk about that in Chapter 5. The implication is, if they could have cooperated, they would have both chosen not to confess and they both would have gotten the shorter, 3-year prison sentence... maybe. The problem: the decision to confess is actually driven by self-interest.

Let's prove that by loosening the rules a little bit. Suppose Butch knows that Sundance will not confess. Maybe the police let him see the interrogation and it's clear he isn't "going to break." Butch will still confess; 1 year in jail is still better for Butch than 3 years. You'd get the same result the other way around, if Sundance knew with certainty that Butch was not going to confess. As it turns out, the driving force for each player to confess is their desire to minimize *their own* time in jail. The outcome is dependent on the other person, but the choice is not.

Empathy and the Prisoner's Dilemma

You can show the importance of self-interest in determining the outcome of the Prisoner's Dilemma in another way. Suppose Butch and Sundance made their choice based entirely on getting the best possible outcome for *the other person*. They want to minimize *their partners'* time in prison and don't care about their own outcome at all. This is tricky; using the payoff matrix again. Now, they make the choice that's best for the other player.

Here we go: if Sundance confesses, Butch can either get him—get Sundance—a sentence of 5 years or 1 year. Butch is still looking side-to-side at his choices but he's looking at Sundance's outcomes. Butch would not confess in order to get Sundance the 1-year sentence. If Sundance does not confess, Butch can get Sundance either 10 years or 3 years. Butch would not confess again and get Sundance 3 years. In this case, Butch—with his singular focus on Sundance's well-being—has a dominant strategy to *not* confess.

Now, let Sundance also be only interested in his partner's outcome (and remember, you're now looking up and down). He too has a dominant strategy to not confess and—voila!—they both get 3 years! If Butch and Sundance play the game with an eye toward the *other's* outcome, they are both better off.

Before we pick apart the game or find limitations to its applicability, let's consider how useful the Prisoner's Dilemma and thus game theory can be in dealing with day-to-day situations. You make a lot of choices in situations that involve one other player and a small number of choices. Most romantic or familial relationships, transactions, and negotiations fit this description.

Here's an example: on any given day, after dinner, you can choose to sit and watch TV—your preferred choice if your partner does the dishes—or you can help clean up. Your partner can make the same choices and has the same preference. Even if you don't want to watch the same thing on TV (which might complicate the decision process), if no one *ever* cleans the dishes, you're going to be watching TV in a pretty nasty smelling house.

We make choices like these multiple times a day, every day—usually with similar parameters. It's a big part of adulting. Being completely self-interested will probably get you into couple's therapy or in search of a new partner. While the therapist might help you communicate better, the only real solution is to find a partner who isn't always self-interested. Empathy promotes a solution that is better for all parties—but everybody has to get on board with that approach for it to be effective.

What about Butch and Sundance? Well, they're thieves first. The point of the expression “no honor among thieves” tells us a lot. If you're in a game where your choice and the ultimate outcome depends on your belief in the *honor* of the other player, their tendency toward empathy matters. Granny was saying, if you're playing with a thief, you can safely assume self-interest will always drive the process.

The Payoff Matrix and Jollies

We've already done one important tweak on the Prisoner's Dilemma game. When Butch and Sundance chose without regard to their individual outcomes and focused on minimizing the sentence of their partner, they achieved the optimal outcome. In essence, their own time in prison didn't affect their happiness, but the amount of time their partner spends in prison did. Time each other spends being in prison became a rule of thumb.

The Prisoner's Dilemma can be a little confusing since higher numbers (years in prison) are bad. Let's switch to something where higher numbers are really good: pizza.

You consume pizza and that makes you happy. Economists try to quantify that happiness in a measure they call utils. Since nobody knows what a util is, we created our own measure: jollies. Your consumption of pizza results in a certain number of jollies, and more is better. We swear this relates to the Prisoner's Dilemma. Bear with us.

Imagine you and your partner are going out to dinner at an Italian restaurant. You agree to order together and share. You really *really* like pizza, with a

		JAY	
		PIZZA	SALAD
AMANDA	PIZZA	JAY 25 JOLLIES AMANDA 25 JOLLIES	JAY 30 JOLLIES AMANDA 60 JOLLIES
	SALAD	JAY 50 JOLLIES AMANDA 40 JOLLIES	JAY 20 JOLLIES AMANDA 20 JOLLIES

Fig. 2.2 Dinner out, part I

side salad. He prefers a big salad with a half slice of pizza. Your payoff matrix might look something like Fig. 2.2:

In the top left and bottom right quadrants, you’re both ordering pizza or salad. So, nobody’s really getting the whole meal they prefer. In the top right corner, Amanda gets to order her favorite pizza and take a few pieces of lettuce from Jay. She’s happy. In the bottom left, Jay gets to order the super big salad and take a small slice of pizza from Amanda. He’s happy.

It turns out, we need some help to solve this game, and it’s right there in the setup: you can’t order at **exactly** the same time. This is a sequential game (which we’ll discuss in Chapter 6). Amanda takes a look and says “If I want to maximize my jollies, I should choose Pizza (since 60 is the highest number on the board). If I choose Pizza, Jay is choosing between 25 and 30 jollies. He’ll choose salad (30), and we both get what we want in the top right corner).

It works the other way too: if Jay goes first, he sees his largest number is in Pizza (50). He knows, if he chooses Pizza, Amanda will choose salad (40) over another Pizza (25). The game solves at the bottom left corner.

(Pro note: if you’ve taken a game theory class before, you may be saying “hey, this game has two Nash Equilibria,” and you’d be correct. We’ll talk about that in Chapter 5).

Jollies are just a placeholder for how much you prefer one outcome over another. The numbers can be called anything. They are not prices or revenues or costs, per se. They are a quantitative measure of perceived benefit in the outcome.

Having numbers in a payoff matrix actually allows us to represent some cool things, like preference. In this case, we see that Jay and Amanda would be okay only having salad or only having pizza, but they clearly prefer a mix of food. The numbers also indicate empathy: all jollies are positive, which means there's no resentment between Amanda and Jay if only one person gets their way.

The number of jollies aren't meant to be literal units of happiness. This is just our approach to show a person's preferences, or relative happiness, when deciding.

"Can't Buy Me Love"

Granny was a HUGE fan of The Beatles. Paul McCartney and John Lennon probably didn't realize when they composed it, but "Can't Buy Me Love" is applicable to game theory in a big way. Money doesn't motivate people to make choices very well. Beyond a certain point, people will have enough money. Money isn't usually represented on payoff matrices because it is only part of the set of factors that impact happiness. Money isn't love.

If you want to grasp the depth of this problem, ask yourself this: what is love? This is going to be, for most people, what we call a Potter Stewart definition. This former US Supreme Court Justice famously said, when asked to define pornography, "I'll know it when I see it." Such is the case with love. You can't really define it, but you know it when you feel it.

Money and many other tangible possessions are at best a proxy for the things that truly motivate people—the results and outcomes that we deal with in game theory. Emotional motivators are ultimately more important. Trying to motivate a player with money will not get you cooperative behavior if you treat them with disrespect. Work-life balance has emerged as a more influential factor for recruiting—more than base salary and additional compensation. John and Paul were right, and we'll come back to this issue frequently throughout the rest of the book.

Back to Prisoner's Dilemma: Let's restate the payoff matrix in terms of empathy; each player now *loses* one jolly for every year *the other* has to spend in jail.

The advantage of reimagining the payoff matrix with empathy (Fig. 2.3) is that we no longer have to look at the other player's outcome to make a

		BUTCH	
		CONFESS	DO NOT CONFESS
SUNDANCE	CONFESS	BUTCH -5 JOLLIES SUNDANCE -5 JOLLIES	BUTCH -1 JOLLY SUNDANCE -10 JOLLIES
	DO NOT CONFESS	BUTCH -10 JOLLIES SUNDANCE -1 JOLLY	BUTCH -3 JOLLIES SUNDANCE -3 JOLLIES

Fig. 2.3 The Prisoner’s Dilemma, restated

better choice. Butch sees: if Sundance confesses, I lose 5 jollies or 3 jollies. I’ll choose the lesser of the two evils, not to confess, and lose only 3 jollies. If Sundance does not confess, I will either lose 10 or 3 jollies. I will not confess and lose only 3. My dominant strategy is to not confess.

An empathetic Sundance will do the same and the game ends up with no one confessing. Both players get the optimal, 3-year sentence.

Unfortunately, restating the payoff matrix in terms of jollies moves the game into a much fuzzier place. It’s easy to say just make the choice that minimizes your years in prison. It’s much harder to know how that relates to *happiness units*. We assumed that every year in prison led to a decrease in jollies of one unit. That’s just our estimate, but we can’t know that for sure.

Still, most games’ payoff matrices have to be done with jollies, and not the years in prison, dollars, or whatever else the original game is structured around. Jollies are what we ultimately get from our choices. Happiness is why we, as humans, make decisions. If that makes games harder to play, so be it. Even the most penny-pinching Grannies prioritized happiness once in awhile.

Honor, Reputation, and Lunch

We’ve talked a lot about honor in this chapter, and we’ll talk about reputation over and over throughout the rest of the book. Let’s add something to the pizza story:

		JAY	
		PIZZA	SALAD
AMANDA	PIZZA	JAY 25 JOLLIES AMANDA 25 JOLLIES	JAY 30 JOLLIES AMANDA 60 JOLLIES
	SALAD	JAY 50 JOLLIES AMANDA 40 JOLLIES	JAY 20 JOLLIES AMANDA 20 JOLLIES

Fig. 2.4 Dinner out, part II

Suppose Amanda and Jay are on their way to dinner. During the drive, they have a chance to discuss what they're going to get. Look at the payoff matrix in Fig. 2.4.

Again, neither player has a dominant strategy and games like this are harder to solve. One way that this game could be solved is by direct communication—someone simply says what they're going to get. That should work, right? Well, kind of. What if Amanda changes her mind when they get into the restaurant?

Let's add another factor. Amanda has a powerful sense of honor and she's always worried about her reputation. Specifically, she loses jollies if she lies or, in this case, changes her mind. Sadly, this is not true of Jay (at least, not when food is involved).

Amanda loses 20 jollies any time she lies about anything. She announces, as they approach the restaurant, that she's going to get pizza. Let's incorporate her honor—her potential lost jollies—into a new payoff matrix in Fig. 2.5:

Jay still doesn't have a dominant strategy, but now Amanda does. She's going to order pizza. Knowing this (presumably knowing about Amanda's personal honor code), Jay is getting salad. This would have worked equally well if Amanda had said she was getting salad. Salad would be her dominant strategy and Jay would order accordingly.

Honor and reputation have real effects on our decision-making in the real world. They are forms of commitments because they change the payoff matrix. We'll talk more about this in Chapter 6.

		JAY	
		PIZZA	SALAD
AMANDA	PIZZA	JAY 25 JOLLIES AMANDA 25 JOLLIES	JAY 30 JOLLIES AMANDA 60 JOLLIES
	SALAD	JAY 50 JOLLIES AMANDA 20 JOLLIES	JAY 20 JOLLIES AMANDA 0 JOLLIES

Fig. 2.5 Dinner out, with honor

Honor and reputation are not exactly the same thing. Reputation is usually applied to situations in which you expect to interact with the same player again at some point in the future. For Jay and Amanda, that fits nicely: they’re married. Reputation is a factor in repeated games. Honor is more internal, more endogenous, and immovable. Reputation asks, “what do people think of me” and honor asks “what do I think of myself.”

Turns out, reputation and honor also affect how the game is played if a player changes. Amanda may be more likely to impose the jolly penalty if she lies to her best friend (reputation is the driving force) rather than a perfect stranger she’ll never see again (honor is the driving force).

The Golden Rule and the Prisoner’s Dilemma

Granny must’ve said this a thousand times: “Do unto others as you would have others do unto you.”

That’s a simple life strategy to be sure. The question is, when does it work and (if it does work), why? This is the basic question we come back to time and time again in this book. When you’re taught a solution to game-theoretic situations: why and when do they work?

The Golden Rule actually helps Prisoner’s Dilemma type games. It can help take a noncooperative, often repeated game with a dominant strategy solution that is inferior for both players and bring it to a cooperative solution.

What does all that mean?

Suppose you are 12 years old, and your sibling is thirteen years old. You share a bathroom in your family home. After you take a bath, you leave your towel and clothes on the floor and the bathroom is a complete mess. Your grandmother, upon seeing this, says “do unto others as you would have others do unto you.” Being an inquisitive, polite person, you say “huh?”.

Granny explains: ... you know your sibling is going to use the bathroom next and will have to clean up your mess before they can take a bath. Your messiness has made the bathroom experience a little better for you, but worse for your siblings. That's all well and good today but what about tomorrow? Your sibling might use the bathroom before you and they will follow your lead, leaving the bathroom in equal disarray. Tomorrow, your bathroom experience will be the one that's diminished (apparently your grandmother talked like an economist...).

“But Granny,” you point out, “when we were all staying in that hotel in Orlando a while ago everyone - including you - left the towels on the floor. Why is that different?”

This would seem to be a “gotcha” moment, but Granny can explain the difference. In that case, there's someone who is paid to clean the bathroom. Those people would have to clean the bathroom thoroughly whether you left it a mess or folded every towel. That's not the same game and you don't play it the same way.

Let's think about what else mattered in this messy bathroom game, specifically the age of the players. Age matters for two reasons: you and your sibling, both young, are going to be around awhile. Logic tells you if you act in your own best interest in the short run, and the other player does as well, you will *both* be worse off in the long run. You're going to be sharing a bathroom with each other or somebody else—for a long time.

Age also matters on the other end of the spectrum. Suppose the players are both in their nineties. While you might get the same cooperative result, you might also get a couple of people who correctly say, “what long run?” The point: cooperative behavior is most effective (results in more good payoffs) as a strategy when it's adopted early in a long-term, repeated, sequential game.

Put Me In, Coach

In basketball, there are five players on each team, all trying to score more baskets than the five players on the other team. Team A has five players of varying skill levels who each shoot the ball to score every time they get possession. Team B has “role players.” Some are good at passing the ball, others are good shooters, etc.

It’s not hard to imagine that Team B will score more points and win more often than Team A. If everyone on a team gets jollies *only* from winning the game, then we can see how giving (passing the ball instead of shooting the ball) on the part of some players can lead to winning and thus to more happiness.

This particular example might feel like a bit of a reach. Game theory isn’t really about sports, and there are a lot of strategies for success in team sports. But, the deep logic of the Prisoner’s Dilemma is valuable. Players of different skill learn that success comes from cooperation rather than self-interest. When the five play as a team, rather than individual shooters, they reach their optimal outcome.

This is the goal of coaches and mentors. Is winning everything, or is there a greater purpose to team sports? Do you want one star player or a five-person team? These are the questions of coaches the world over.

Dog Walking: The Game

When are dog owners more inclined to pick up their pet’s poop: if they own a home in a small, gated community? Or if they rent an apartment in a massive complex?

As we’ve seen, you don’t need to specify the structure of the game explicitly. This is a Prisoner’s Dilemma-style game where the players have learned that they are only better off cooperating—cleaning up their dog’s poop—if everyone *else* cooperates. Your surroundings reinforce the correct decision when you specify a payoff matrix with jollies. We don’t even need to provide numbers: dog owners lose jollies when neighbors look scornfully at them for letting their dog do its business in their front yard and not cleaning it up.

Homeowners frequently establish their own rules about such things just in case your jolly calculator isn’t the same as the community norms. Sometimes, they enforce these rules with fines. Do fines work as well as glares? As with

The Beatles, people are better motivated by their own standards, their own jollies. In this case, jollies come through a sense of belonging to the community, and belonging, like love, can't be bought. People in a gated community don't smile at you or wave to you because you're the richest person there. They have to like you. So, you pick up the poop.

Timing

We've been playing games with different expectations of the timing of players' choices. There are two obvious ways that the timing of players' choices could be made. When players choose at the same time, we call this a simultaneous game. When they choose in a pre-established order, we call it a sequential game. As described, the Prisoner's Dilemma is a simultaneous game. Both players make their choices before they know what the other player has chosen.

Sequential games are ones like checkers and chess where someone goes first then someone goes second and that repeats until the game is done. We'll talk about sequential games in detail in Chapter 6. If Amanda and Jay are going to get food, that's a sequential game. One person orders before the other.

Timing is usually part of the structure or rules established when the game is first created. Interestingly, changing the timing from simultaneous to sequential (or vice versa) can alter outcomes from the same payoffs. In some cases, it can help a game turn out better for all players.

Prisoner's Dilemma as a Sequential Game

What if the Prisoner's Dilemma was played sequentially? Recall that we were really specific that Butch and Sundance choose at the same time in separate rooms. It's the reason we get to the "both-will-confess" outcome.

Or is it? Let's play the game sequentially, and with self-interested players. We haven't done any sequential games yet, but this one is easy.

Suppose Butch goes first; he will choose before Sundance and his choice will be known to Sundance. Butch is thinking about choosing "Don't Confess," figuring that Sundance will then do the same and they'll both get 3 years. But ultimately, a self-interested Sundance will only look at his own outcomes. So, if he *knows* that Butch did not confess, Sundance will choose to confess and get a sentence of 1 year instead of 3 years. Knowing that Sundance is likely to confess, Butch would never choose "Don't Confess" when he makes the first choice because he'd end up with 10 years, not three. No honor among thieves.

		BUTCH	
		CONFESS	DO NOT CONFESS
SUNDANCE	CONFESS	BUTCH 5 YEARS SUNDANCE 5 YEARS	BUTCH 10 YEARS SUNDANCE 1 YEAR
	DO NOT CONFESS	BUTCH 1 YEAR SUNDANCE 10 YEARS	BUTCH 3 YEARS SUNDANCE 3 YEARS

Fig. 2.6 The Prisoner’s Dilemma, sequentially

Same thing happens if Sundance chooses first. (Pro Tip: run the last paragraph through Fig. 2.6 again to see the numbers).

If the players are self-interested, the result will be the same whether the game is simultaneous or sequential. This result is the embodiment of Granny’s expression “no good deed goes unpunished.” In games where other players are entirely self-interested, any attempt on your part to reach the cooperative outcome unilaterally will fail and you will actually end up worse off.

Repetition

Another timing factor that has to be discussed in the setup of a game is played is frequency. Is this game going to be played just one time or will it be repeated many times (perhaps even forever)? As we’ll see in Chapter 8, this issue matters a lot in many games, and it can even push games like the Prisoner’s Dilemma toward the cooperative outcome.

Why a whole chapter for repetition? Well, repetition is tricky. Will we play the game repeatedly with the exact same players or will it be the same game with new players? If Butch and Sundance literally play this exact same game repeatedly and they know they will be put in the same jail together, wouldn’t that incline them to make different choices? To communicate? To collude? (or, just to steal better and stop getting caught?) We have to amend each game to show new factors each time it’s played.

Prisoner's Dilemma is a great framework for understanding the who, how, and what of a game, and we'll come back to it a lot in future chapters.

Conclusion

As Granny said, a journey of a thousand miles begins with the first step. Thus, we begin our journey into game theory. We worked through our first game, the famous Prisoner's Dilemma. We've talked about game structure—the players, their choices, the payoff matrix, and timing. We learned how to approach and solve a game and discussed dominant strategies, cooperation, and honor.

We waded into deeper, harder issues like what's really in the payoff matrix? Is it the explicitly stated payoff, jail time in the case of the Prisoner's Dilemma, or the real payoff, jollies? Turns out, this game is applicable to basketball, pizza, and walking your dog. We looked at games with many choices and, in doing so, decided what's for dinner.

With this foundation, we can discuss other types of games that feature heavily in our lives, games that don't solve out as easily. Most games have a few possible solutions and thus need additional information to find a solution. This type of game will allow us to define and employ the Nash Equilibrium—a way of knowing if a particular outcome is “final.” Sequential games have a player's choices being made in a prespecified order and repeated games. All of these factors increase the fun in future chapters.

Game theory, for most people most of the time, is informal. Every day, you are put in the position of making choices knowing that others are making their own choices, and together you determine the outcome. The point of game theory is to expand the information we seek, see the genius in game setups, and better evaluate the choices that determine the outcomes in your life.

Questions

- 2.1. You had an opportunity to get someone fired at work, which would have gotten you a promotion and a nice raise. This did not involve lying or doing anything inappropriate. Still, you chose not to do it. What aspect of your payoff matrix told you to, correctly, not make the choice that would have given you more money for firing someone?
- 2.2. In searching for a dominant strategy in a game, you look at your best outcomes under all possible choices the other player could make. You

say to yourself, “if Bill does this, my best choice is that; if Bill does the other, my best choice is whatever....” Which word in your thought process is, arguably, the cornerstone of game theory?

- 2.3. You’ve been neighbors with Janet for 12 years and you can now successfully navigate most game-theoretic situations with her. Janet has a lot of dominant strategies and you’ve figured out her payoff matrix. You’ve been butting heads a bit with the new guy who just moved into the house on the other side of yours. Why can’t you work it out with him as easily as you work with Janet? What’s the difference?
- 2.4. Coordination games exist everywhere—even on a preschool playground. Three- and four-year-old kids need a lot of guidance and intervention to enjoy many of the things on a playground. When they are older, however, kids become able to play what are called “pickup games”—spontaneous games often with total strangers. Why can you do this as a teenager but not a toddler? (hint: think about what you need for a coordination game).
- 2.5. Too Many Choices. It can be a problem in many situations (not all game-theoretic). Think about how you approach your decision about what to have for dinner at a restaurant that has thirty choices. Do you do anything to narrow your options? Having done that, do you ever proceed to make it a game-theoretic decision? How did the first step help the second?

Appendix 1: More Choices

An easy complaint about classroom game theory is that most of the games only involve two choices and two players. Life’s games, one could argue, are rarely as simple as the ones we analyze on the marker board (or, frankly, in most of this book). The fact is games with many choices are much harder to solve in the classroom and in real life. If the number of choices is too big, people are tempted to throw up their hands and make decisions haphazardly. When the choice set is large, people tend to stop evaluating payoffs and look for *other* ways to choose (like what’s the fastest, first, clearest, or otherwise easiest evaluation). This is certainly a strategy, but not a very good one.

Good news: the approaches we’ve developed can be used to solve bigger games. In fact, we can turn a game with many choices into a game with only a few legitimate choices that our human minds can understand. Let’s see how that might work. And yes, we’re talking about food again.

Suppose a couple has agreed that each person will shop for part of dinner on the way home from work, and they also agreed not to communicate; they want to surprise each other. So romantic.

Sam is going to buy the protein and Dan is going to buy the vegetables. Let's suppose these people know each other well and they know their own preferences as well as their partner's preferences for all possible combinations of protein and vegetables. Their potential payoff matrix is Fig. 2.7.

Ew, numbers, gross.

What's your instinct when you face a complicated problem in life with what appears to be too many choices? Well, you might try to eliminate some of the choices you hate and make your menu smaller. Game theory takes the same approach.

		DAN			
		CHICKEN	BEEF	FISH	LIVER
SAM	BEANS	DAN 2 SAM -5	DAN 3 SAM -5	DAN 0 SAM -5	DAN -1 SAM -2
	ASPARAGUS	DAN 3 SAM 5	DAN 2 SAM 15	DAN 10 SAM 10	DAN 0 SAM -15
	PEAS	DAN 4 SAM 0	DAN 20 SAM 0	DAN 2 SAM 10	DAN 1 SAM -20
	BROCCOLI	DAN 15 SAM 5	DAN 4 SAM 10	DAN 3 SAM 0	DAN 2 SAM -10
	BEEETS	DAN 6 SAM -10	DAN 5 SAM -10	DAN 4 SAM -10	DAN 30 SAM -30

Fig. 2.7 Winner, winner; what's for dinner—in jollies

First, let's take a look at a dominant strategy: is there a choice a player would always make? Spoiler alert: nobody has a dominant strategy in this game (you can run the matrix to double check, if you want). The jollies from each meal combination are so intertwined, no single strategy emerges for either player.

So, we'll take another approach to solving this game: is there a choice a player would *never* make?

Look closely at the bottom row. Sam is never going to choose beets. Look at Sam's jollies for any protein choice that Dan makes and there's never a case in which Sam would choose beets. In game theory parlance, beets are dominated by all of the other choices. Functionally, that means we can cross that row off. Act as if that row isn't there. Beets are NOT on Sam's menu.

That leaves us with a somewhat smaller version of the game, shown in Fig. 2.8.

In this smaller game, we can see that Dan will (now) never choose liver. The only time that he would have chosen liver was if he could pair it with beets, but he knows (because he knows Sam's jollies) that Sam will never buy

DAN

		CHICKEN	BEEF	FISH	LIVER
SAM	BEANS	DAN 2 SAM -5	DAN 3 SAM -5	DAN 0 SAM -5	DAN -1 SAM -2
	ASPARAGUS	DAN 3 SAM 5	DAN 2 SAM 15	DAN 10 SAM 10	DAN 0 SAM -15
	PEAS	DAN 4 SAM 0	DAN 20 SAM 0	DAN 2 SAM 10	DAN 1 SAM -20
	BROCCOLI	DAN 15 SAM 5	DAN 4 SAM 10	DAN 3 SAM 0	DAN 2 SAM -10

Fig. 2.8 No beets for you!

beets. Liver is also a dominated choice. We can remove that column. Sam can also say that Beans give no pleasure at all (-5 jollies across the board). So, we can remove that row as well. The result of these dominated choices is shown in Fig. 2.9.

So, we’ve been able to remove two rows (beets and beans) and one column (liver) and make the game look and feel easier to work with. Note, this isn’t a sequential game; they are choosing—shopping—simultaneously, but they can certainly do these eliminations in a step-by-step way. On their own.

Dominated choices only get us so far, though. If we look at the resulting subgame shown in Fig. 2.9, we see that Dan still doesn’t have a dominant choice. If Sam chooses asparagus, Dan prefers fish; if Sam chooses peas, Dan prefers beef; and if Sam chooses broccoli, Dan prefers chicken. So, let’s look at the vegetables again and see if we can make headway there.

As it turns out, Sam now has a dominant strategy to choose asparagus (and not a moment too soon). If Dan picks chicken, Sam is indifferent between broccoli and asparagus (5 jollies in each case). If Dan picks beef, Sam prefers asparagus outright. If Dan picks fish, Sam is indifferent between peas and asparagus. Asparagus is either a win or in a tie through all three potential combinations. Knowing Sam the way he does, Dan will figure out that Sam is buying asparagus, Dan will buy fish, and dinner is as healthy as can be.

		DAN		
		CHICKEN	BEEF	FISH
SAM	ASPARAGUS	DAN 3 SAM 5	DAN 2 SAM 15	DAN 10 SAM 10
	PEAS	DAN 4 SAM 0	DAN 20 SAM 0	DAN 2 SAM 10
	BROCCOLI	DAN 15 SAM 5	DAN 4 SAM 10	DAN 3 SAM 0

Fig. 2.9 No liver, no beans, no beets, oh my!

Notice, we were able to solve this game with only one player having a dominant strategy (Sam) and what we called full information. Even though Dan doesn't have a dominant strategy, he knows what Sam's dominant strategy is and he simply picks what's best for him given that knowledge.

This process feels artificial to most readers; no one really does this, right? Actually, you do it quickly and automatically every day. You reduce the choices in your daily decisions to a small, solvable game by process of elimination. Familiarity is the only requirement for doing this. This example works on an assumption of full information. Sam knows Dan, and Dan knows Sam. It's easier to narrow down multiple options when you're working, playing, living, and partnering with people you *know*.

Indifference is the key issue here. Indifference, or a "tie," is a very real emotion in daily decisions. We usually deal with this by engaging in comparisons to get to a conclusion that "feels" right. We've formalized the process here with numbers in this example, but you do it ten times a day by evaluating your emotions toward particular choices.

Lest you think this is too analytical to be realistic (game theory teachers have probably all heard students say "no one really does that"), think about how you shop differently when you are shopping for your family than you did when you were shopping only for yourself. You almost certainly incorporate other people's jollies into your decision process now. Maybe you put the jar of Nutella back more than you used to. Maybe you buy more apple slices instead of Nutella, or Diet Coke instead of Mountain Dew. Indeed, what we call "compromise" is nothing more than a streamlined, multi-choice game.

Our Modern World: The Dark Side of "Too Many Choices"

Multiple choices can seem daunting, frustrating, and confusing. Why do there have to be so many options? Economists will tell you that variety is good, economically. It gives everyone in a society a chance to find something that makes them happy (maximizes their jollies), which lowers per-unit prices and raises the jollies of society overall.

Some retailers have turned "too many choices" into a marketing strategy. The retailer (or restaurant) has made an active decision to give you so many choices that occasionally you buy the item they want you to buy. Store brands that look almost identical to name brands are placed alongside their counterparts on crowded shelves in the hopes that you grab the "wrong" bottle. We don't normally want to consider games in which

one player counts on the other player to regularly make mistakes, but the too-many-choices game might be an exception.

Other retailers take the opposite approach: they remove choices for you. Large chains like Costco and Trader Joes famously have only 1 or 2 options for each product. From an economics perspective, this means there is limited variety, so quality must be higher to ensure consumers continue to buy. For the consumer, though, removing too many choices is beneficial; the choice set returns to “buy” or “don’t buy,” rather than a more complicated and stressful matrix.

Go Up A Hill...

Having more than two choices doesn’t always mean the problem is difficult. Consider the game in Fig. 2.10:

Jack and Jill are choosing between three items each: A, B, or C for Jill, and X, Y, or Z for Jack. When we use the process that we learned to search for a dominant strategy, we see that Jack does not have a dominant strategy—but Jill does. Jill will always choose C. She is indifferent between A, B, and C if Jack picks X; she is indifferent between B and C if Jack picks Y; and she

		JILL		
		A	B	C
JACK	X	JILL 5 JACK 10	JILL 5 JACK 0	JILL 5 JACK 4
	Y	JILL 0 JACK 5	JILL 10 JACK 1	JILL 10 JACK 4
	Z	JILL -4 JACK -2	JILL 0 JACK 20	JILL 4 JACK 1

Fig. 2.10 Jack & Jill

prefers C if Jack picks Z. The logic is tricky, but she's always getting the best possible outcome when she picks C.

Surely the fact that Jill has a dominant strategy makes this game easy to solve. In the Shopping Game, we saw that all we need is one player to have a dominant strategy and full information and we can find a solution. The player that does not have full information knows what the other player is going to choose. But this game is a little trickier.

Take a look at Fig. 2.10 again. In this case, knowing that Jill will choose C leaves Jack indifferent between X and Y. That means **the best we can do** is say this game will end up at either 4\5 or 4\10. (Jack\Jill).

We've actually made this game trickier still. While Jack does not have a dominant strategy, he does have a *dominated* strategy: Y. Regardless of Jane's choice, Jack would never pick Y. If Jack's first approach to playing this game is to see if he has dominant then dominated strategies, he will find that he has the latter. Then, when he discovers Jill's dominant strategy, he will choose X. But, suppose his first approach is to see if Jill has a dominant strategy. Then the game could have either of two solutions 4\5 or 4\10.

On seeing this, we're tempted to say that Jack will steer the result to 4\10 because he's no worse off and Jane is better off. Two things work against that. First, if we've specified the game correctly, everything Jack cares about is in his number and that means he is completely indifferent. Second, if he did look for a dominated strategy before looking for Jane's dominant strategy, he eliminated B and thus made 4\10 impossible. There is no rule about what you do first when you try to solve a simultaneous game. Everyone is different.

The point of this example is to show that these formal approaches to game-theoretic situations—dominant strategies, dominated strategies, etc.—still have potential potholes. You can learn to do games “right” and still find games that are hard to solve.



3

Know Thyself: The Players

Traditionally, we teach game theory by formalizing the structure of the game—the players, the choices, the payoffs first. Then, we talk about approaches for and possible solutions to various games. We assume that all of that structure is identifiable and in general, independent. That means that each player is choosing on their own, without any type of cooperation.

But players are people: human, living, breathing people with biases and perceptions and attitudes and bad days. Sometimes, you're in the mood to play a long, drawn-out game (like a salary negotiation). Other times, you're hangry and you want a snack now. These preferences and tolerances directly affect how a game is set up by changing the available choices and expected payoffs. Let's discuss the humanist issues that affect games and outcomes in a variety of ways.

We started down this path in Chapter 2 when we said that the payoff matrix really needed to be specified in jollies—happiness units—because anything else (dollars, prison time, etc.) effectively assumes that all players like, dislike, or are motivated in exactly the same way. Fundamentally, we know people's preferences are different. Economists have long worried that measuring and modeling happiness is difficult, if not impossible. It's why they use utils and jollies to specify happiness. Doing game theory with anything else in the payoff matrix will certainly have you playing games incorrectly; you will not make the optimal decision, and you will not get the results that you expect.

In all human interactions, we make assumptions about other players. We project a value proposition on them in every activity (and, sometimes, we're

wrong). Game-theoretic models are often inapplicable to real-world decision-making because they've assumed away everything that makes us human. Why is it easier to interact in a series of games with your friends, family, co-workers, even members of your church, club, or organization? Well, you share a source of jollies with these groups, and that gives you more insight into their humanity.

Players and the payoff matrix are inexorably connected. Traditionally we separate the two when we teach game theory, but that's not the way things work in the real world. Knowing the players in a game means knowing what they value. Knowing what they value means you can more accurately evaluate choices and outcomes *before* you play the game.

What Makes People Tick

In traditional game theory, we say all players are rational.

Well, there's problem #1 right there.

The problem with the word *rational* is it overreaches. When we call somebody "rational," we think they are perfect, with an ability to make unemotional, intentional decisions. This is the adopted meaning of the word, what it has come to describe in the modern lexicon. In game theory, *rational* means something very different.

The easiest way to remember what rationality means in game theory is to recall it is a subdiscipline of mathematics. In math, rational means *consistent*. So, let's just restate the word rationality with something less academic: we assume players have consistent preferences in their games.

What's the difference? Well, somebody can make consistently bad choices. They can always choose the most expensive item, the sugary drink, or the trip with the most layovers. Somebody can be consistently risk averse or risk tolerant. He can always act on emotion, rather than cold, hard facts. She can go with the flow, never showing care or concern. When we say people (players) are rational, we do not mean they always do the right thing. We just mean they are consistent and predictable, which helps us better specify their preferences and payoffs.

What does an *irrational* person look like? In a word: chaotic. An irrational player does not have easily identified motivators, a set of heuristics, biases, or preferences that allow us to construct an accurate payoff matrix. When dealing with an irrational player, there isn't enough information for you to evaluate their jollies. To quote Granny: we just don't know "what makes them tick."

Leaders, therapists, and significant others use this phrase more than “rationality,” because we don’t expect others to be perfect. It’s a little less analytical, but it still implies a person has an “order,” or a level of predictability. When all is said and done, that’s all rationality means: a consistent choice. It means that a person presented with the same choices and the same circumstances will consistently make the same decisions. It does *not* mean that everyone uses the same approaches or “gets jollies” from the same thing. It does *not* mean that these choices are consistently bad or good. Only that there’s a little predictability to the player that allows you to assess the game.

Are your employees motivated by money, benefits, or mobility, or something else (like respect)? Is your significant other happy with your choice of wall art, or is she just choosing not to fight that particular battle? Does your boss actually like your idea, or is he placating you so you’ll stay? *Communication* is the key to many game-theoretic situations. It may not be verbal (actually, it almost never is), but communication happens between players before, during, and after a game. This transmission of information is the lifeblood of game theory. It can reveal, conceal, reinforce, invalidate, and directly affect the setup of a game.

What makes people tick?

Sometimes, you meet people who are motivated by things you can’t comprehend. Remember, *rational* just means predictable. It doesn’t mean *you* would choose the same things as another person. Their choices do not need to seem “good” to you. Even if a person’s preferences are incomprehensible to you, you can still play the game because you can specify it.

But, what if the person’s choices are all over the place? It’s not just that your preferences don’t align, you don’t even know what the person *wants*. Unpredictability, irrationality, a person who has no discernable sense of order, is hard to play with. You can’t win if you can’t even figure out what it looks like when *they* win.

Along these lines, a popular relationship-based cliché is “opposites attract.” For magnets, sure. In relationships? Eh, not so much. The cliché is probably Granny’s mistaken observation. It’s not opposites who are attracted to each other, but complements. Players who are better described as complements are arguably trying to reach a cooperative outcome that’s better for both parties. Each player in the relationship is looking for a partner to fill in their gaps, but they’re still trying to achieve similar goals.

People who are true opposites might find that a relationship is pretty challenging. Every game-theoretic decision—and there are many in a relationship—would find the couple at odds with each other because payoffs are,

well, inverse. Interestingly, they might be able to predict each other's choices; exact opposites would assume the other player made the opposite choice from theirs every time. But, it's hard to imagine either of them would be happy with the outcomes, even if they could play the game.

Better Safe than Sorry: Maximin

Until now, we've been navigating games based on what a person wants. The direction: find the best possible outcome. Is this always how we approach a choice?

Is your priority always to achieve the best outcome?

Sometimes, rather than making choices based on the best outcomes and your knowledge of the game, you consider things like worst-case scenarios. Even if one particular outcome is unlikely (or statistically shouldn't happen), you might avoid the choice that *could* lead to it just to make sure it can't happen. Instead of finding the best possible outcome, you're trying to avoid the worst possible outcome.

We call this strategy maximin; a fancy way of saying that you are making *sure* the worst-case scenario cannot happen. (We'll see this again when we do sequential games in Chapter 6.)

Consider the game in Fig. 3.1:

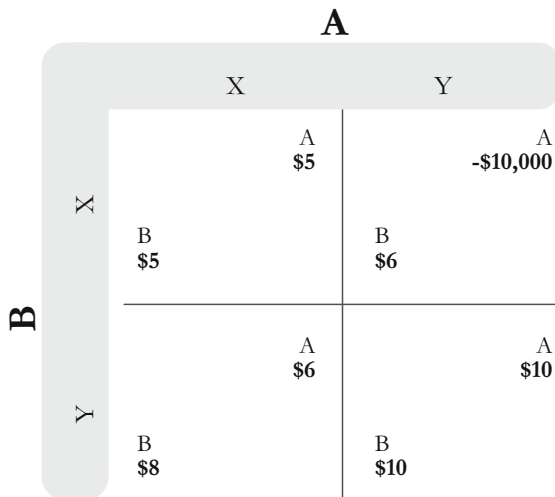


Fig. 3.1 A big loss

Two players, A&B, are choosing simultaneously and with full information, meaning both players know the entire payoff matrix.

Player B has a dominant strategy to choose Y. Regardless of Player A's choice, Player B is better off choosing Y.

Player A doesn't have a dominant strategy. If Player B chooses X, Player A would choose X; if Player B chooses Y, Player A chooses Y.

With full information, Player A knows that Player B will (should) choose Y, so Player A will also choose Y and the game should solve out with both players getting \$10, right?

Maybe.

This nice, neat, game-theoretic process puts player A in a precarious position. They are making the choice (X) that *allows the possibility* of their worst-case scenario. If Player B, for whatever reason, chooses X, then Player A will end up — \$10,000.

Game theory academics say that won't happen; players always will and should pick their dominant strategy. Oh, but Player B is human. It *could* happen. Player B might make a mistake or not understand the game or have a different preference for money and think \$10 k isn't that big of a loss. Any way you slice it, Player A *could* lose a lot of money, even though she shouldn't.

Whatever the situation is, you can't say Player A should always trust Player B just because he has a dominant strategy. People go against game-theoretic strategy all the time. Real-world probabilities tell us there's always a chance the worst thing can happen.

But, Granny already knew that—and she prepared you for it: “better safe than sorry.”

She was specifically telling you to make choices that avoid the worst-case scenario, even if it didn't bring you or the other person the best return.

Note: it doesn't have to be about the other player making mistakes playing the game or wanting to see you suffer. We just can't know with 100% certainty what the other player's payoffs really are. Any outcome—however statistically improbable—is possible.

Maybe we can get a little closer to understanding, though. Money doesn't incorporate all of the potential emotions that could lead to an unfavorable outcome (like spite). But, jollies do. Jollies allow us to incorporate things like risk aversion, caution, and sensibility. It also allows us to evaluate the scale of preferences; how drastically different a particular outcome will make somebody *feel*.

Let's respecify the payoff matrix incorporating Player A's fear of loss. This will allow us to get the maximin outcome without messing up our game theory approach.

		A	
		X	Y
B	X	^A 5 JOLLIES ^B 5 JOLLIES	^A -10,000 JOLLIES ^B 6 JOLLIES
	Y	^A 6 JOLLIES ^B 8 JOLLIES	^A 5 JOLLIES ^B 10 JOLLIES

Fig. 3.2 Maximin Jollies

We've made two changes in the payoff matrix in Fig. 3.2. First, we've converted dollars into jollies, and we've taken half of the jollies away from player A in the bottom right corner. Why? If A chooses Y, they'll face anxiety (because there's a possibility of getting - 10,000 jollies). This feeling factors into a payoff matrix with jollies, since it accounts for the relative happiness in each outcome.

Player A now has a dominant strategy to choose X, and Player B still has a dominant strategy to choose Y. The game solves out at the bottom left corner, the previous maximin solution. Player A gets 6 and Player B gets 8, and this is an equilibrium (nobody wants to change). Correctly specifying and using jollies, the things that make people tick, is key to using game theory correctly.

A Method to the Madness: Irrationality

Irrational behavior is present when a player doesn't seem to have an ability to make choices that align with any interests in a game. Arguably, this is the very nature of mental illness. To an extent, any mental illness that renders one player completely unpredictable makes a game-theoretic approach almost impossible. For most of what we're doing in this book, that's just a limitation for the application of game theory we take as given.

Societal rules come from well-accepted individual beliefs, and individuals learn that most aspects of game theory work differently (or don't work at all) if a player doesn't have an ability to make rational choices. Punishment

is intended to dissuade a player from making noncooperative choices. Basically, if a player makes a choice that *intentionally* takes jollies away from another player, society punishes him. A person who has no ability to comprehend rational choice, however, won't likely respond correctly to the threat of punishment. So, societies state that this difference in rationality should be treated as a special case with different provisions. In many societies, there are legal exceptions for those who make self-interested choices as a result of mental illness.

There's a lot of alternative social policy here and we'll leave it to the reader to think about this area using game theory, but let's go in a different direction. The fascinating expression "There's a method to his madness" invokes the possibility that the irrational player is, in effect, gaming these provisions. "Method" implies a predictable pattern. A person who has a method to their madness might be what's sometimes called an "evil genius;" someone who is very good at being badly unpredictable. Arguably, that player is completely responsible, just very different from most other players, and much harder to play against in any game. You never know what he might do.

The Control Freak

Control is an ability to determine, or at least affect, outcomes. It sounds like something that's entirely outside the realm of game theory. After all, the fundament of game theory is to explore situations where you alone cannot determine the outcome.

There are choices that are unilateral, though. We make them all the time. In a game, a person makes a choice, another makes a choice, and they determine the outcome together. Unilateral choices and games usually have different outcomes, different benefits. That's the nature of having others involved. Watching a movie by yourself isn't the same thing as watching a (possibly different) movie with someone else.

If the payoff is specified in jollies, in happiness, we have to allow for the possibility that some people are happier making decisions alone. Some people get jollies from the feeling of control. That is, interestingly, absent from formal realm of game theory most of the time, but it still affects games. How?

Remember: you can simply decide not to play a game. Alternatively, you can decide you will only play games in which you have a dominant strategy, but recall, that does not assure that you control the outcome. The control freak's payoff matrix is structured such that, whenever possible, she will only

play games with people who allow her to determine the outcome. Here's the thing: if you're the control freak, you don't usually recognize you're acting with this motivation. Control tendencies are usually recognized from the perspective of other players who cannot seem to affect their outcomes.

If you're playing against a control freak, you should recognize that your choices don't affect the outcome. Knowing this, you might not play. A few examples: if Mary likes driving the car once in a while, she probably won't end up in a relationship with Harry, a notorious control freak. If your friend never lets you choose where to eat, you might stop going (or stop caring where you eat). You always have the option to not play the game.

Patriarchy, Hegemony, etc.

It's easy to relegate the discussion of a control freak to the category of rare exceptions or perhaps cautionary tales. Unfortunately, history has taught us otherwise.

Many countries and cultures actively remove some of their citizens from the game as a measure of control. Concepts like patriarchy, hegemony, and even colonialization are all situations where a small group of players usurp the decision-making ability of a large swath of the population in order to maintain control.

These approaches spring from the well-known concept of parental approval. In many situations and real-world games, we don't let children make their own decisions independently. Repressive control mechanisms are as insulting as their history. Treating other adults like children in a decision-making context is wrong in more ways than we can describe. But, in a game-theoretic context, it is quite literally imposing the decision-maker's source of jollies on everyone else.

Having laws and rules that don't allow people to vote, work, drive, etc. are systemic versions of the control freak. While the players who are allowed to make all of the choices can tell themselves that they are taking care of everyone else, the fact is, they are not ever allowing people to choose. As we have seen and will see in repeated games, this policy often leads to inferior outcomes. We will revisit this issue in Chapter 4 and connect it to information and access to education.

You'll Understand When You're Older: Rationality and Age

An easy example of how rationality is relative can be seen when you think of... well... your relatives. Your *older* relatives, to be precise.

We hope to all be there; we all get older and change. When a teenager looks at mom trying to figure out Tiktok, they might say “I’ll never be so incompetent.” Over time, though, there’s no way to know. Saying you “won’t be like that” is assuming you even know what “that” is. You are who you are: your age, your tastes, your concerns, your feelings. You probably won’t be like “that” in the sense that you won’t be like your elders are right now, but they—and Granny—probably said the same thing once upon a time.

Some would classify the actions of an older loved one as “irrational,” but now we know that doesn’t fit the definition in game theory. Priorities of any age, based on age, can be understood and preferences of any predictable nature can be worked into a payoff. You might try to understand the perplexing approaches or thought processes of an older relative, but remember: they aren’t irrational. They just aren’t...you.

Here’s a fun one: try to figure out a payoff matrix when you go out to eat with an older relative. The answers might surprise you, but the process will enlighten you.

If game theory is applicable to seniors, why not to babies? Well, as we saw in the introduction, babies have not yet figured out how to make decisions. They are purely reactive beings, responding to positive and negative stimuli without an agenda. Babies won’t respond to game theory the way a learned adult might.

Know Thyself

When we discuss sequential games in Chapter 6, we will include games that you play with yourself. These are not the same as unilateral, controlled games. Rather, we mean games where your decision today can change your outcome of a future game. In essence, you’re playing against the needs of your future self.

When we do this in game theory, we are not simply talking about growing up, maturing, gaining experience, or other things that happen naturally with age. We are talking about choices today that allow for different choices in the future. Game theory for an individual is quite literally the fork in the

road. Every choice you make today can preclude or allow a future choice. As Granny would say: “when one door closes, another one opens.”

Consider the curious situation that you are in when you play games with your future self. You think that of all the people on earth, you most assuredly know *your* preferences and payoffs. In fact, you do not. You know yourself **today**; you do not know yourself 20 or 50 years from now. You don’t know what will matter to you, your biases, your constraints. You don’t know what will give you jollies. You don’t really know yourself in the future in any meaningful way.

The proof of that can be seen if you look back over your life a decade or two. Why is it easy to give your teenage son advice about education drugs or sex? Because, as Granny might have said, you’ve “been there and done that.” You can (hopefully) remember your feelings, what gave you jollies, and what mattered to you when you were his age. You know how your choices at his age changed outcomes for you, and you can give some sage advice. “Hindsight,” Granny said, “is 20/20.”

It is easier to predict player preferences and accurately specify a payoff matrix if you’re already past the time in life your players are experiencing. Giving your kids advice is easier than dealing with your aging parents. You simply haven’t gotten there yet.

Too Tired to Think

If acting on emotions is (sort of) the opposite of doing game theory, then being “too tired to think,” might mean you are too tired to figure out the structure of the game that you’re involved in and what your best choices are. It happens. We’re human.

Why does fatigue make decision-making more difficult?

Emotions and tiredness change information processing. In game theory, they affect your ability to accurately assess a game before you even play it. Game theory is a rational, logical process as long as we recall that everyone can have different ways of being rational and logical. It is really a brain-only thing, and the brain works better at some times than it does at others.

Leadership training often includes recommendations like getting enough sleep, avoiding multitasking and distractions, and controlling your emotions. These are good habits all for effective leaders, but they are especially important when you make strategic decisions. Why? Accurate and measured informational processing is key to evaluating all options and choosing the optimal one.

When you're "not thinking straight," you tend to do things that are mentally *easier*. Strategic decision-making depends on analysis and evaluation. So, tiredness is strategically problematic because you might mis-specify the players, choices, and payoffs. For example, you might think that everyone you're dealing with is just like you (has the same payoff matrix). You might also fail to recognize there are more than two choices or that there's another motivation for the other player. A reduced ability to accurately evaluate and set up a game often leads to bad outcomes. It's why Granny told you to "sleep on it." Don't try to make difficult decisions when you're too tired to think!

Physiological factors can affect decision-making in a couple of ways. Pain, for example, can make people less able to process things—since it reduces concentration—and thus less able to make correct strategic decisions. Inebriation reduces the ability to process potential consequences in the short- and long-term, also leading to suboptimal outcomes. When we focus on the human aspects of game theory, we can't ignore the effects of physical health.

But, going down this path needs to be done with great care. Hormones are often associated with things like overactive sex drive and rage. Just because the science is valid doesn't mean we can use biological excuses for bad decisions. When the correct outcomes for all players are included in a payoff matrix, game theory specifically never allows for a "hormonal justification" for making bad decisions. On the contrary, every punishment in a game exists to take away jollies from bad choices and reduce the incentive to harm the other player. Correctly specified jollies de-incentivize bad behavior. Game theory does not allow for biology as an excuse.

Revenge

There's a well-known, rather cryptic, expression (usually ascribed as an Italian proverb): "revenge is a dish that's best served cold." What does this mean?

One interpretation of this proverb is that reactions to other players choices should not be rushed. A player made a choice that gave you a bad outcome. Revenge is a reaction to that choice. The nature of the word revenge implies a sequence of games with the same player. The expression could mean exacting your revenge quickly (and probably emotionally) will not be as fulfilling as waiting until the situation or the payoffs from the game allow you to punish the other player effectively.

All of this sounds pretty ominous, but it doesn't have to be. Revenge can be silly. If you're building a snowman with your sibling and out of the blue, they hit you with a snowball and snow goes down your back (usually the worst outcome of a snowball sneak attack), you aren't likely to get your best revenge if you respond immediately; their guard is up. So, you laugh it off and wait until they relax, maybe after they turn their back on you. Revenge, literally, is best served cold.

Granny's expression, often mis-interpreted, that has the same meaning is "he who laughs last, laughs best," or simply "getting the last laugh." This has nothing to do with humor. This one is about repeated games. If you're making choices with the same player over and over, the expression tells you to exact your revenge in the last game. Laughing is the jollies you get by taking revenge for something, knowing the other player can never get back at you. There's no chance for them to take revenge on the revenge. We'll talk more about this motivation when we examine repeated games.

In Hot Pursuit of an Irrelevant Success

There is an interesting complication of the payoff matrix and the concept of rationality. Rationality is just consistency, an ordering of outcomes that is applicable on a regular basis. That's all well and good if the payoff matrix itself is well-ordered. However, payoffs are personal things. We often find ourselves wanting something for a reason that we can't (or don't want to) explain.

If your company's softball team has never won the league championship and they are in the finals for the 10th time, players might care a lot about the outcome of that last game. They might care about it an awful lot. They might care about it more than work or even home. The potential payoff, the value of the "win," might seem irrational to people who aren't on the team (spouses, other employees, etc....). When team members skip work meetings or home duties to attend softball practice, they look like they are making bad decisions. This is best understood as a misunderstanding of the payoffs that they perceive. To the softball team members, winning the championship is an affirmation. To others, it is an irrelevant success.

That problem carries over into many more spaces. Rankings, awards, and rules for promotion can be forms of irrelevant success. Organizations spend millions of dollars enforcing historic rules that might well be irrelevant for

actual profitability. If you get stuck trying to figure out another player's rationality, think about the issue of relevance. What matters to you may not matter to others (even a large group of others), and vice versa.

Suppose there are two restaurants close to each other with equally good food. One focuses its resources on talent, ingredients, and ambiance that earn it a Michelin star. The other spends the same number of resources on marketing. Is it obvious which one will survive? Do the restaurants have the same goal? (hint: do they have the same audience?)

Or, how about this one: success, to academics like college professors, includes getting tenure. Roughly speaking, that means you have a job for life at a college or university as long as the school and your department survive. At many schools, getting tenure requires publishing several research papers in academic journals that are peer-reviewed and highly ranked. The rules for getting tenure were created many years ago when creative and research freedoms needed formal protection.

Who's the customer for a school? Students. Do students care about any of the metrics that lead to tenure? Do they even know what they are?

What happens when the priorities of two players (in this case, a customer and a producer) aren't aligned?

Rules, rationality, and the payoff matrix are all interrelated. You will find yourself in games where you are told the rules and you have to decide if you even want to play the game. The choice can be difficult and problematic. For better or worse, the rules were written by the people who got there first. You can see something that looks like a lot of fun—looks like it will give you a lot of jollies (like teaching), but the rules involve educational attainment (a PhD in many cases) and other seemingly arcane metrics (published research) that make the goal *less* fun and reduce your jollies. Knowing this, you might become a Marketing Director and write books instead.

Historically, rules are set to establish and maintain some level of quality. Teaching college means teaching concepts and approaches that are more than your opinion. Aha! Light Bulb Moment: the irrelevance of your success depends on its impact. If you teach people well and they live a better life, or if your research solves a problem and it changes the world for the better, or if the work team wins the Softball Championship, it may be worth it. If you can trace the outcomes of your accomplishments to something beyond you, then you did not spend your life in hot pursuit of an irrelevant success.

If mortality is the finish line for accomplishments, then, to paraphrase Da Vinci: "a life well-spent brings happy death." When all is said and done (thanks, Granny), getting the last laugh might be the best outcome.

The Cover of the Rolling Stone

The hot pursuit of an irrelevant success was actually the point of a popular song in the 1970's. *Dr. Hook and the Medicine Show* was a band that achieved moderate success during the seventies when it released the song "The Cover of the Rolling Stone." Written by poet Shel Silverstein, the song laments that the group won't feel like they are a success until they have appeared on the cover of the then-iconic music magazine *Rolling Stone*.

At the time of the song's release, that cover was considered the height of success. But, the long-term relevance of that achievement more or less died with the demise of print magazines. *Rolling Stone* is still around, but it's mostly an online publication. "Cover of the Rolling Stone" is a great song with diminished cultural impact.

Sincerity

There are humanist things that affect game outcomes that can't be easily taught, but they are well-understood. When you're trying to communicate your motivations, what makes *you* tick, to the other player, you're often put in the position of saying things like "money doesn't matter to me" or "I'm not in this for myself." Roughly speaking, you're trying to get the other player to *believe* that you want to make cooperative choices—as opposed to self-interested choices.

When the other player believes you, they likely feel that you are sincere or genuine. While these words have definitions, their application, and real-world use is usually informal. People "pick up on things" when they talk to you. Executive training consultants say it's eye contact or facial expressions or body language—all of which can be associated with sincerity. (But, those things can be taught, learned, and therefore mimicked. They may not be the most reliable indicators of sincerity.)

"How you come across" is a critical part of the outcome of all games that you play. As we have more interactions, play more games, and see "what works and what doesn't," we are ultimately testing signals and learning. Hopefully, many subtle signals of things like sincerity "work for us." And that's the key: what works for one person doesn't always work for others. Communicating sincerity is often a matter of reinforcing some aspects of your personality and softening or removing other things that make you seem self-interested.

A Sense of Fairness

Using jollies can help you put a lot of things into a payoff matrix and incorporate more of the human element into a game. There are concepts out there that are more like feelings; they are things that are easy to say but hard to define. One such concept is fairness. Being fair might be as generic as “everyone gets to play by the same rules,” or it might be as personal as “things turned out a lot better for me than they did for you and that makes me feel bad.” The latter approach can be incorporated in the payoff matrix of a game.

Suppose the payoffs for a simultaneous (everyone chooses at the same time), noncooperative (players can’t collaborate) game with full information (everyone can see the payoff matrix) looks like Fig. 3.3:

To make this game easier, let’s assume that these players each get one jolly for every dollar that they end up with. This game will end up with Person A getting 3 and Person B getting 10. Person B has a dominant strategy to pick Up; Person A does not have a dominant strategy, but they can see that Person B will pick Up, so they pick Left.

Now suppose person B feels really bad—as in, loses jollies—whenever he ends up a *lot* better off than person A. Specifically, B loses 6 jollies if his winnings are more than twice A’s winnings. B is guilty.

That sense of fairness on the part of B will change where the game ends up. To see that, let’s look at the payoff matrix, now *only* with jollies. We can

		A	
		LEFT	RIGHT
B	UP	A 3 JOLLIES B 10 JOLLIES	A 2 JOLLIES B 5 JOLLIES
	DOWN	A 3 JOLLIES B 5 JOLLIES	A 4 JOLLIES B 4 JOLLIES

Fig. 3.3 Joystick Choices

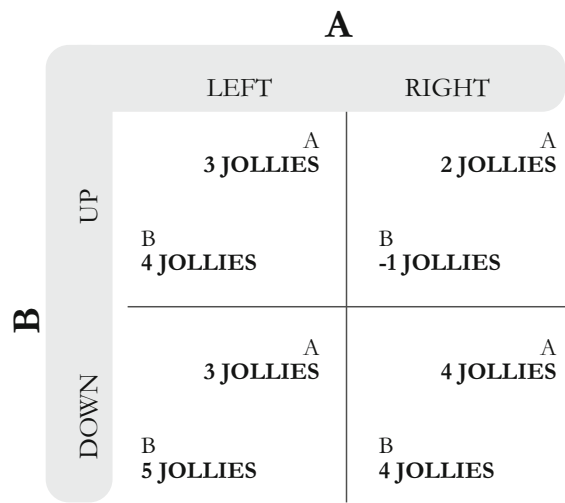


Fig. 3.4 Joystick choices—with guilt

include this self-imposed punishment in the upper right corner to see how guilt affects B’s jollies, as in Fig. 3.4:

This change doesn’t affect Person A; he still doesn’t have a dominant strategy (we didn’t change his payoffs). However, it does affect B. He now has a dominant strategy to choose down, and the game will now end up with both players getting 4. B’s guilt changed the payoff matrix and therefore the outcome of the game.

There are people who would say B is illogical or irrational for not acting in his own self-interest, but there’s nothing particularly wrong with basing your jollies on things like how the other player might feel. We do this all the time in healthy relationships. The choice was still self-interested: B chose the option that didn’t make him feel guilty. When we use happiness, we have to remember that it isn’t measured the same way as money, revenue, house size, or other metrics. More/bigger isn’t necessarily “better” in everyone’s mind.

Conclusion

We run the risk of “losing” every game, making incorrect or inappropriate choices, when we don’t understand what matters to the other players. The easiest thing to do is assume that everyone is just like you. But game theory isn’t easy (sorry). The very nature of doing game theory correctly involves open-mindedness, a worldly perspective, and a lack of prejudice.

(We'll see in Chapter 7 that prejudice can dramatically alter the outcome of a game.)

In classroom game theory, there are usually straightforward games (the traditional version of the Prisoner's Dilemma), and difficult, complicated, heavily mathematical games. Straightforward games are perceived as easier. In the classroom, that's correct. In the real world, where you have to figure out "what makes the other player tick," specify their payoffs, and balance your own preferences, you realize all game-theoretic situations are tricky.

Players in important games are more than just other ages or other genders. They are other countries, other cultures, other belief structures. It's very easy to be judgmental when you do game theory out in the real world. Feeling superior because you always do what you perceive as the cooperative thing might give you jollies, but it doesn't always give you enough (extra) jollies to compensate for playing a game naively or mis-specifying a game. Understanding the player is step one toward strategic success.

As it turns out, what you don't know *can* hurt you. Ignorance, in game theory, is rarely bliss.

Questions

- 3.1. Two people meet while partying in Las Vegas. Over the course of an alcohol-soaked weekend, they get married at the Chapel of Love (yes, that's a real place). One year, 2 months, 3 weeks, and 4 days later, they filed for divorce. Another couple is set up by their families in an arranged marriage. Despite never dating before their wedding, they are married for the rest of their lives. What game-theoretic information is different between the two couples?
- 3.2. The point of learning game theory is to better understand and navigate situations where you and another person are making choices that together determine the outcome for each of you. Given that, it's pointless to study game theory if you assume everyone is just like you. Consider this situation: a 65-year-old grandfather is taking care of his 14-year-old grandson. They decide to watch TV together. Grandpa is only interested in a fishing show. Grandson only wants to watch a newly released science fiction show. Neither of them has any interest in the other's show. How will they be able to decide which of these shows to watch?

- 3.3. There's a well-known debate in many corners of society about the primary determinants of a person's outcome—nature vs. nurture. Are we the product of biology and genetics or are we the product of our upbringing? We're not biologists, so let's restate the question in a game theory context: does nature or nurture determine a person's payoff matrix? Put differently, are the sources of jollies learned or intrinsic?
- 3.4. Good long-term relationships (such as long, happy marriages) are the product of well-played, repeated games. Learning what's in your partner's payoff matrix is a critical part of playing those games well. Mel and Nel have been living together happily for 30 years even though Mel loves baseball and watches or listens to many games during the year and Nel likes to crochet. How then could they have stayed together, happily, for so long?
- 3.5. You're going to break up with your boyfriend and you decide to do it in a public place. Why? The outcome is the same. Why does "where" you do it matter?



4

Fool Me Once: The Value of Information

Granny used to say: “a little knowledge is a dangerous thing.” No, she wasn’t telling you to drop out of school. The key to this cautionary expression is the word “little.” Granny was saying that *too little* knowledge about something might cause you to misstep in the future. You think you know enough to make an informed decision, but you don’t.

There are probably a dozen expressions extolling the value of knowledge and information. In game theory, “who knows what” and how information is used almost always determines the outcome of games. Information removes uncertainty. Full information brings you closer to a full understanding of game setup: the other players, your choices, and your outcomes.

Uncertainty happens in the absence of reliable information, and it creates a problem in any decision-making process. As we saw in the last chapter, the underlying challenge in game theory is figuring out what someone else’s payoff matrix really includes—what they really value. In the real world, it is nearly impossible to understand other people’s jollies, what makes them tick, fully. Information (in many forms) becomes an important tool in getting closer to clarity.

Let’s take a look at the types of information that can be helpful, how and when we use them, and where you can leverage information across multiple types of games.

Information is a broad term, and as we stated before, we have to be specific about what we mean. In a game theory context, information is anything that helps one player better specify the potential outcomes of a game. It could be related to the preferences or motivations of the other player, the type and frequency of the game at hand, and the choices available to all players.

Because information is so impactful, game theorists use a lot of in-house jargon and new terminology when talking about information-based issues and problems. Because of that, we'll be spending a fair amount of time in this chapter restating Granny's wisdom in terms of game-theory-speak.

In classrooms and game theory books, you are given all information up front. It's right there in the setup. There's the game, up on the board, crystal clear. Each player can see everything plain as day: payoffs, preferences, players, and all available choices. In daily interactions, well... it's not so clear. Most games we play have some degree of asymmetric information, where one player has more (and more relevant) information than the other. Information asymmetry is at the heart of every job interview, used-car transaction, and first date. Each of these games is an exercise in gleaning (or hiding) relevant information from the other player and, in doing so, getting the game to turn out better from your perspective.

The Fine Art of Information Sharing

"Never tell anyone outside the family what you're thinking," was The Godfather, Vito Corlene's warning to his son Sonny after he blurted out his opinion to an outsider. Side note: if your Granny said this too, you may want to look into your family history.

The default assumption in most games is that information known only to you is more valuable because it can help you attain the best outcomes. If information is private, it's perceived as having more impact on the player. The Godfather's advice is to share information only with people that you trust, like family. Otherwise, you run the risk of another player using information against you.

Information is certainly a tool that can be used to manipulate, and in the context of The Godfather, it was a matter of life and death. Hopefully, we're not engaging in daily dealings on this level, but Vito's point is well-taken: once information is out there, it can't be "taken back." Each piece of information is part of a puzzle to how you, as a player, "tick." Once it's available for all to see, it's difficult to put it back in the bottle. Granny advises us to "play it close to the vest," and keep valuable information from other players until you are certain of their intentions.

Information can be withheld or revealed depending on how that information can be used by another player. The act of revealing information to someone else is called a signal. If it makes you better off, you can signal the information you have. If it doesn't, strategic theory says you should

keep it private. (Spoiler Alert: there's a whole subdiscipline of game theory called Signaling Theory. Everybody loves a good subdiscipline). Signaling by a more-informed player can grant a lot of power in negotiation games, especially when the information can impact the other player's jollies in a significant way.

If you're the *less*-informed player, signaling has value as well. The less-informed player will usually try to glean information to help reach *their* best outcome. When a less-informed player asks questions or attempts to gather baseline information to better her position, we call this a screen. You screen the other, better-informed player to glean useful information. Every time you fill out an application for a job, loan, or credit card, or you apply for college or graduate school, or you fill out an application to rent an apartment, you are being screened by the entity that you are trying to do these dealings with. You know more than they do about yourself, so they're trying to fill in informational holes to ensure their player position is stronger.

Aha, did that confuse you? We bet you thought that the hiring manager or company was the more informed player, right? Wrong! In hiring decisions, the candidate knows much more than the company. Why? You know you best. The company tries to screen you to get some information to tell them if they should hire you. Keep in mind, information affects power, but the two are not the same. Power is an ability to drastically alter the other player's outcome with your single choice. We're not saying a less-informed player doesn't have power to influence the game. Simply, from an information standpoint, a job candidate always has more knowledge about what they bring to the table than the company looking to hire.

Here's a fun tactic: bluffing. Bluffing is a signal that's not true. It is false information intentionally given to alter the game in your favor—and this is problematic for both players. From the perspective of the less-informed player receiving the bluff, there's no way to confirm how true a bluff might be. The player might not even know the true impact of the bluff, which means they can't accurately specify their *own* jollies anymore. If you think of the signal as the total amount of information that you're getting, a bluff is noise. It's part of that information that's hard to verify, and the ability of a player to bluff implies there's a lot of potential noise in these signals. In game theory and acoustics, this is sometimes called a high signal-to-noise ratio, where a ratio of one (1) means the signal is all noise, unverifiable, and dangerous to rely on.

In some circumstances, we are able to walk away from (not play) games with a significant signal-to-noise ratio. If calling the bluff (asking for evidence) costs you a relationship, for example, you might not play the game

at all. On the other hand, we regularly play games that are based on bluffing, like poker. Great poker players assume bluffing is happening, and their skill lays not only in betting on a great hand but also leveraging bluffing players against each other. It takes a keen eye to see a pattern (and therefore information) in a bluffing signal. Some would say it's the whole point of the game.

Signals are given to change the behavior of a player. If a person sending a signal is not bluffing and she wants the signal to change the behavior of the recipient, she will have to decrease or eliminate the noise to increase the validity (trustworthiness) of the signal. This can be done by including some form of proof to back up the claim. In day-to-day interactions, proof is more than "I swear" or "I promise." As Granny said, "talk is cheap." When you apply for a loan, you can swear that you'll repay it, but the lender will probably look at your credit score or another metric to know with reasonable certainty whether the loan will be repaid. You provide a signal, and your credit report verifies it's not a bluff.

Can bluffs happen on screens? Absolutely, we do it all the time. A bluff can inject noise into a screen easily, since the less-informed player doesn't know whether the screened information is correct. It could be that he doesn't know the other player well enough to evaluate the answers for truth. It could also be a processing problem: did you ask the right question? The person trying to get information might also face a cognitive bias in his interpretation of the screened information, based on many factors (do I like how is the response worded? Is it cold in here? How long have I been screening candidates?)

"You can't judge a book by its cover" is one of Granny's cautions about what your screened information is actually worth. She means you can't tell how good a book is by simply looking at the front; the cover doesn't give you enough information to accurately assess how good the book is. Granny is cautioning you against an unreliable screen, lest you buy the book and hate the content.

If the game is on a board in a classroom: yes, you have all the information you need to solve a game. In reality, information gathering takes time, skill, patience, and the ability to see through many human factors.

"Don't judge a book..." is talking about the misuse of spurious visual screens—making judgments about people based on their appearance. Because of this, screening has become rightfully regulated or outlawed in many cases. You aren't allowed to use screens like ethnicity or gender to make decisions about most things. Historically, screening on those sorts of things has been used inappropriately to make decisions on hiring, housing, compensation and promotion, mental toughness, and college admissions. Regulations have

helped stop non-data-driven biases from creeping into these life-changing decisions that require much better information to assess accurately.

Of course, as a human, you can put any screens in place you like. For example, if you're an astrophysicist attending an international conference and your personal screen is to only talk to people who look like serious, tie-wearing, strait-laced scientists, you might avoid other people... like Dr. Brian May, award-winning astrophysicist, published author, and lead guitarist of the rock group Queen.

Consider the Source

When information is being shared openly and freely in an obvious strategic situation, the first thing you should do is think about the *type* of game you're playing. For example, if you're in a "winner take all," what we call zero-sum game and the other player tells you what they intend to do, you should ignore them. Why? In this case, the other player has no incentive to help you; it's zero-sum, win or lose. They are telling you information to mess with your head, skew things to their favor, or otherwise influence the game to *their* advantage. In zero-sum, any information conveyed by your opponent is a setup to try to make you lose.

Consider tennis. Suppose your opponent yells, "I'm going to hit the ball to the left side next time." Why would they tell you that if it was true? They wouldn't. Why would they bother lying to you at all? Because they want you to deviate from your current strategy that has you ahead 5-0 in the final set.

First and foremost, information sharing only makes sense if the game allows the information to actually make a difference. In poker, it takes a trained player to filter bluffs and signals that come their way. In zero-sum, it may take one or two bad outcomes before you stop trusting all signals altogether. Even in games that might be win-win, information can be used to move the outcome in your favor or against you.

Separating the Wheat from the Chaff

Granny was all about signaling. She knew that it's "hard to tell a book by its cover" and "actions speak louder than words." She also told you that "you will be known by the company you keep." Sure, she might have told you this to get you to think twice about your dear friends. The expression is great for game theory too.

Suppose a young, highly talented basketball prospect is trying to choose between two colleges that are equal in most ways. Each college's Athletic Directors (AD's) will likely tell her that she'll get a lot of playing time, the team will promote her in the press, and she'll start each game. Assume both teams kept most of their starters from last year, and the colleges finished their conference in first and second place, respectively.

How can the prospect choose which school to attend? Both coaches are signaling what they think is useful information, and it's virtually the same for each school. Both AD's said the same things about opportunities to get playing time and the likelihood of wins. The team makeups are almost identical, records are close, and the facilities available are state-of-the-art. The schools are equidistant from the player and each program is offering her the same scholarship.

What other information could the candidate use to break this tie?

What about the personality of the coach? Suppose Coach X has more charisma, is more energetic, and seems more excited about the prospect of getting the student on board. He calls all the time and makes a big show of trying to recruit her.

The other coach, Coach Y, is a close friend and frequent confidant of one of the most successful professional coaches in the country. This is not something he talks about openly, but everyone, including the prospect, knows about it.

If the player believes that friends are a signal, and if she believes Coach Y's friendship is indicative of his commitment to basketball, she might choose Coach Y's college. The friendship is a signal, a new piece of information, and it influences her decision and the outcomes of the game. The player is judging the choice before her by the company each coach keeps.

Signaling and interpreting signals is complicated and personal.

We said that both coaches offered the same amount of playing time, but an experienced player knows that outcomes—both individual and team—depend on the amount of effort that all players put forth. Being a good friend of someone successful is a positive signal, but so is being a charismatic, inspirational coach. As it turns out, many of us "follow our gut" when it

comes to choices like this. We could have just as easily said the player values enthusiasm as a force in leadership and chooses Coach X. Interpretation is extremely personal, dependent on the player's priorities and preferences.

There are infamous examples of charismatic, inspirational leaders who led followers down a path to very bad outcomes. Pick your favorite repressive dictator, for example. This is not a reason to ignore charisma, but it's a reason to put these two signals (charisma and friends) together. Most charismatic leaders have friends—previous followers. Before you follow a charismatic leader, you might want to talk to, or do a little research on, other people who have made the choice that you are about to make. Granny's advice is even better now: you'll be known by the company you keep.

What's the Worst That Could Happen?

You're back in high school (ew) and you want to ask a classmate to go out on a date. You're shy and afraid of rejection. Granny says, "there's no harm in asking; the worst thing is they could say no."

This and other similar expressions sound completely innocuous. "There's no harm" makes it seem like nothing bad could happen by simply asking a question. True?

Let's try another example: you get pulled over by a police officer because you were speeding. You're asked for your license and registration. As you're fumbling around trying to find them, you ask in a friendly tone, "have you ever had to shoot someone?"

Ooh, not the time, friend! Not the time. Why does it feel like such an ill-advised question? It shouldn't hurt to ask, right? Well, take a step back and put yourself in the officer's shoes. She was just doing a routine traffic stop, but now she's playing a different game. The question's timing and circumstance changed the perceived risk in the traffic stop. Asking a friend out at school also changes the game, assuming you have to continue seeing the person in class—but the risk to personal safety is much less.

What about asking out someone that you like at work (assuming this is allowed)? There's a colleague that you get along with and you're thinking about asking if he wants to go to dinner. You remember the advice from long ago, "there's no harm in asking, all they can say is no." So, you ask and, politely, and he says no. Is it business-as-usual on Monday?

Nope.

By asking the question, you changed the relationship forever. Maybe not in a bad way, but all future interactions, all games, are now re-informed. Both

players know about your feelings and your willingness to act on them. The question changed the game.

That all future games are changed doesn't always mean there's harm in asking a question. A question is a type of information, and it always helps players respecify a future game.

How do we address questions in terms of signaling? Maybe you don't ask the question directly; maybe you just "feel out" the situation. You could ask *other* questions (maybe of other people) the information you need. This is a strategy to screen the other person's position indirectly—and more cautiously. When stakes are higher, like in a workplace or getting pulled over, you might start with a smaller version of the game. In High School, it's "do you want to go on a date Friday night?" At work, it's "let's go grab coffee sometime." On the side of the road, it's "traffic stops are so stressful; here's my license and registration."

In many game-theoretic situations, making the game smaller can often make the game turn out better for all players. Recall the Maximin Game in Chapter 3: human nature (jollies) often makes us shy away from choices that could lead to a very bad outcome—even if the outcome isn't likely. When you break a larger game into smaller games, the potential negative payoffs become smaller. As humans, we do this all the time: in personal relationships, financial investing, traveling, and gambling. You don't ask her to marry you on the first date; you don't throw millions into an unknown hedge fund, you don't make your first trip overseas a 90-day transatlantic cruise, and you don't go all-in on your first poker hand. It's what Granny would have called "pacing yourself."

Diluted Signals

The most straightforward signal is to communicate directly: say it, write it, create a PowerPoint for it, or otherwise reveal it completely. This ensures the information is received. However, direct communication doesn't guarantee the information is *understood*. When dealing with signals between humans, we must be cautious that any signal communicates what we are *hoping* to communicate.

Once upon a time, telling someone that you loved them was a very powerful signal. These days, when "love ya" is something people toss around mindlessly, the word doesn't necessarily convey a clear message from person to person. Along the same lines, when you teach your child manners you teach them to say please and thank you. These are good habits to be sure, but saying

please and thank you isn't supposed to be a substitute for good behavior. Those words do not allow you to ask too much of people. Well-mannered words are not the same as good manners.

Granny cautioned us well to be wary of verbal signals. She said, "actions speak louder than words," and "talk is cheap." Game-theoretically, these expressions are absolutely true. Notice: she isn't saying words don't mean anything, just that direct communication of some things can be interpreted in many ways. People absorb and interpret information through their own, individual lens, with their own biases, processing capabilities, and previous experiences. Even direct communication—the best form of signaling and information sharing—can become diluted and misinterpreted.

Curiosity: Educational Attainment

Formal education is important to build hard and soft skills for future career and achievement. However, there are those who argue that educational attainment is actually a signal of something more than formal skills. Being able to say that you have an associate's, certificate, or bachelor's degree doesn't necessarily mean you're *smarter* than those who do not. It really means you were able to get into, stay in, and graduate from a college program. The degree provides information about your privilege, your work ethic, your commitment, or your ability to learn. A potential employer or client takes the degree as additional screened information, assuming the values of the college align with what they're looking for.

Like other signals, a formal degree has lost much of its informational power over recent decades. Policies that make it easier to get into college and make it harder to flunk out have skewed the information a degree provides. Advocates for these policies are (rightfully) trying to improve life outcomes for people based on data that show college graduates have higher incomes and better achievement throughout life. There's nothing wrong with wanting to give everyone an equal chance. Unfortunately, these policies can also take away part of the value of the signal, depending on how they are implemented. If a potential employer no longer *believes* a degree is a major indicator of potential and performance, she may not prioritize an applicant with the degree. Perhaps she moves on to other indicators, like work experience or longevity with a prior company. In this case, the policy intended to help actually removed some opportunities, since it diluted what the degree represents. It has the opposite effect of what policymakers may have wanted.

Mixed Signals

“You’re sending mixed signals!” said Granny. In game theory, a mixed signal is both a big deal and a real problem.

When players face asymmetrical information, a signal conveys information that the signaler wants the recipient to believe and use to solve the game in the signaler’s favor. We talked about bluffing, purposely signaling something untrue to try to change the outcome. It’s a noisy signal, where the interpretation depends on the believability of the *signaler*.

Mixed signals are a bit different. A mixed signal is a signal that has more than one possible interpretation. And that’s key; how the *recipient* interprets the signal determines whether it is mixed. Here’s an example:

A very smart high-school senior is trying to decide where to go to college. She was accepted to many schools including the University of Chi-town—a famously exclusive, academically challenging school, and Hoity-toity University—a famously elite (Ivy League), academically excellent school. The student asked the opinion of a learned family friend. The student said “I have to go to Hoity-toity, right? No one turns them down, and if I go there, I’m set for life.”

The sage, wise, family friend responded, “If you choose Hoity-toity, a future employer will think one of two things: either you’re smart or you’re from a well-connected, affluent family. If you choose Chi-town, the only signal you send is that you’re smart.”

These are just one person’s interpretations of the college-choice signal. Others could certainly interpret them differently. *That is the problem*. Mixed signals are subject to interpretation. The choice of a college has to be made early in life, but what it signals lasts, more or less, forever and will be subject to interpretation by people the student doesn’t even know yet.

Mixed signals also permeate the world of high-price, high-quality goods. When you consider buying an ultra-expensive, high-quality watch, are you communicating your sense of style or just your ability to afford something extravagant? When you buy designer clothes, are you buying their style, fit, and other attributes, or just the right to wear a brand name? Does the brand convey that you are stylish or snobby? Or both?

Signals are subject to interpretation, and mixed signals depend solely on the parameters of the *receiver*. Specifying a game depends on one player’s ability to accurately identify what matters to another player, so they don’t send a signal that can be interpreted negatively. When decisions are made without knowing all future players, mixed signals can be risky.

K.I.S.S. Strategy

Granny didn't mince words a lot. Sometimes, she said "come on, keep it simple." Some of us had a Granny that embellished it: "keep it simple, stupid." She was a peach.

We said the most straightforward signalling strategy is direct communication, where you simply tell the other player what you want them to know. This is the K.I.S.S. (keep it simple, stupid) strategy. If you tell the other player exactly what you want them to know, there's no way it can be subject to interpretation, right?

Look at it from the other side. If a player tells you something in a game-theoretic setting, why do you believe them? To use information correctly, you have to know the game and its payoffs. If you're at a four-way stop perpendicular to a stranger and they wave for you to go first, you don't usually question that information. You don't even think about this very long because, well, why would they lie? Mind you, there are possibilities; they might work for an auto body shop, or they might get jollies from the sound of crashing cars, but far more likely, they just want you to go first.

Now, go back to playing tennis. You've been hitting the ball back and forth with the other player a few times. It's a competitive game; you both want to win, and it's zero-sum. Just before she hits the ball back to you, your opponent yells "I'm going to hit this one to your left." Of course you don't listen! It might be true, sure, but this time, the other player has every reason to lie. Your opponent is trying to "get inside your head"; change your strategy by giving you completely useless information. That's the nature of that game.

The K.I.S.S. strategy—direct provision of information—only works when there are no reasons for any player to bend the truth. Weirdly enough, in game theory and in life, it's rare to find that type of game. Information is almost always shared to sway a player. Think of most communication that comes from a parent of a three-year-old right before bedtime. Information—even directly—is intended to promote better choices.

The Winner's Curse

You're wandering the streets of Paris, and you happen upon a little art gallery, Galerie De Frèd. As you're perusing the paintings in a bin, you find an unsigned piece that you think might be a Van Gogh. You offer the proprietor 100 Euros, and he happily accepts your offer and hands you the painting. Did you buy a Van Gogh? Nope...you probably just bought a "Frèd."

When a transaction involves asymmetric information, specifically when a good's value is known to a *seller* but not a buyer, a buyer will often overpay for the good. In the case of the painting, any offer that the gallery owner accepts will be greater than or equal to the true value of the artwork. The gallery owner—the seller—is the only one who knows the true value. It's usually illegal for the seller to outright lie about the quality of a good, but he can certainly hold back on key information, forcing the buyer to guess (aka, lie by omission).

Some states or municipalities have tried to reduce information asymmetries by forcing sellers and buyers to disclose information. In California, for example, all job listings must include the salary range for the position. Revealing such an important factor ensures that the "seller," the potential candidate, does not accept an undercut salary. Incidentally, the policy also helps the "buyer," the recruiting firm, screen candidates who want more than the intended range.

Asymmetry and Acquisition

In our gallery example, a key piece of information is that the painting was not, in fact, an original Van Gogh and the seller knew this. How? Well, while it's true that the "true value" of art is not quantifiable, there is a bottom-line value that's usually easy to assess. A Van Gogh original is going to be worth millions on the open art market, and it is unlikely a painting in a bin in a corner store in Paris is in possession of such treasure.

There are other situations where a seller knows what the asset is worth *to him*, but not to everyone else. Corporate acquisitions, where one company buys another, are particularly susceptible to information asymmetries. Even with mandated financial disclosures and legal discovery periods, the true assets purchased are unknown until management assumes control.

Let's say you develop and patent a new form of artificial intelligence (AI). It's new, it's efficient, it's cutting edge. You nurtured the project from start-up to innovation to a successful first year, and now Microsoft wants to buy it. You value your firm at \$1 billion. Microsoft offers you \$2 billion. You think you've hit the jackpot, you sell, and Microsoft overpays for your company.

Not necessarily.

Your valuation of \$1 billion is your assessment. It is based on your firm's ability to use your new technology (directly, for a particular project; or indirectly, marketing it to other companies). A large firm with better connections, more applications, and deeper marketing experience might be able to use your technology *better*. The value you created with your technology is \$1 billion for you, but it may be in the \$ trillions for Microsoft. Did they really overpay? The question will surely keep you up at night on your new yacht in the Bahamas.

In practice, companies often do overpay for acquisitions, because the ability to gauge value from the outside is difficult. It's hard to judge how useful something is before you've bought it. Novelty, news, even international pressure can skew the perception of value, pushing the purchase price up. In fact, for publicly traded firms, stock prices usually go down for a while after they acquire another company. The market almost always assumes they've made a bad choice and overpaid.

Let's take one more example of mixed signal information that might hit closer to home. You're in the market for a car and you decide to peruse Craigslist for some options. You see a great option, contact the seller, and offer twice what the seller is asking. The seller is the only person who knows how the car really runs, and it's not that great. The seller laughs all the way to the bank—what a sucker you were!

But wait, there's more. You're an expert in auto restoration, the daughter of a mechanic, and a lifelong lover of cars. You transform that beat-up car and sell it for five times what you paid. The information asymmetry, in this case, put the *seller* at a disadvantage. The buyer knew more about the "true value" after all.

The winner's curse is a cautionary tale about the value of information in a game. Information asymmetries can be used by either player in a game, and they can result in a variety of outcomes. Remember, payoffs are uniquely personal and to specify them correctly takes a lot of understanding on the part of both players. Granny knew this. It's why she said, even though you prepare and think you know everything, "be careful what you wish for, you might get it."

Negotiations

One unique area of strategic decision-making is negotiations. Every negotiation is a well-choreographed dance of signals and screens. Like many other things we've discussed, the situation and setup of the game is rather specific, and classic. You are involved in a discussion when you and another player are trying to divvy up the end result of a cooperative solution. Your choices affect the other player, and theirs affect you.

You and your potential employer are negotiating your salary. Salary represents two things: for the firm, it's the *value* that you will be able to create if you work for this company; for you, it's the *value* of your time spent away from doing other things. Both your resources (skills) and those of the company (assets) are required to create this value (some strategists call this "synergy"). Given that, how much of the value do you get, and how much does the company get?

A negotiation is a series of choices that help set up a game. The actual game is your final choice: to accept or decline an offer. Through negotiation, you reach a choice set where each player sees a specified outcome. If a mutually acceptable agreement can't be reached, you won't take the job and all of the value disappears. Both parties move on and try to find an acceptable agreement somewhere else.

From here, things get tricky. If—and this is a big if—both parties agree on the amount of value that's being created, deciding who gets what often comes down to negotiating from what's called a "position of strength." That usually means you both have a Plan B, something you can or will do if you don't get an acceptable outcome from the negotiation. Negotiations coaches would call this a fallback, the best minimum, a BATNA (Best Alternative to a Negotiated Agreement, for our negotiations friends).

It's usually easier to negotiate a good starting salary with a new employer if you're already employed somewhere else. Your Plan B, your backup plan, is to stay where you are. This minimizes negative jollies, especially if income is your major motivator. Similarly, your potential new employer will usually have more than one candidate for a job opening. They're minimizing their potential downside as well. There's lots of gamesmanship in negotiating, and some of it needs to be used carefully.

Suppose you know the other candidate and you know something bad about her (we'll let you make the bad thing up). Do you tell the potential employer? Do you say something in passing like, "I know you're also talking to Gladys, and I think it's wonderful that you're not holding what happened at her last firm against her." That might damage the company's backup, but

it might damage you as well. You look like a gossip. The employer might not want to hire either of you.

When a labor union goes on strike because its members want higher wages, it sometimes forms a picket line around the company entrance saying “this company is unfair” or “this company abuses its workers.” That might convince the company to give in to the union’s demands, but it will also hurt the company’s business if customers don’t want to shop there anymore. Even if the Union is paid more eventually, it might get what’s called a Pyrrhic Victory: a win in which everyone ends up losing.

Information sharing in negotiations is particularly tricky, because one form of mixed signal or bluff can derail the choreography. The important take-away is this: a negotiation gets two parties to the point of making a decision. All information shared during negotiation brings players one step closer to specifying payoffs for everyone involved.

Information, Education, and Patriarchy

In Chapter 3, we said that patriarchy and other discriminatory systems limit a group’s ability to make a choice, to play a game. It’s the macro-societal version of the control freak. When societal rules only allow one type of player to make choices, it is taking what could (and should) have been a game-theoretic situation and making it into a unilateral choice that benefits one player.

The justification for this often revolves around information or education. Players in power often say the oppressed group is not capable of deciding for some reason. The implication, sometimes reinforced by other structural factors, is that the disempowered people aren’t well-enough informed to be allowed to make choices.

Socially, personally, politically, emotionally—there are many ways this approach doesn’t make any sense. But let’s indulge the insanity for a moment. If a society is removing choice on the basis that the repressed group can’t possibly know how to make a good choice, this is still not a long-term solution. Economists sometimes refer to this as a second-best solution, meaning it does help solve the game, but the outcomes (payoffs) were not as good as they could have been if both players acted with full information. The better way—the way to improve society over time—is to fix the information problem: allow all players to get an education and become better informed, and then stop making choices for them. This approach allows everyone to play the game for themselves and should maximize jollies for all players.

Social structure is usually discussed in areas like political science and history, but things like racism, sexism, and policies on education (information) fit very neatly into game theory. After all, games are just decisions where two players determine a joint outcome. Most, if not all, of society's interactions (within and between cultures) depend on another player of some sort.

History has shown us that having hard and fast rules that are based on *cooperative* principles are more likely to have better outcomes, even longer survival rates. Dictatorships rarely thrive beyond a certain point because a lack of information stunts payoffs and growth.

Our Modern Society: Fake News

One of the more problematic aspects of AI is the *ambiguity* it creates about the accuracy of information. Maximizing your outcome in a game requires accurate information, something not skewed by lies, personal interpretation, or other biases. Easily created—and easily disseminated—false information, through the use of AI, chat, media manipulation, etc., makes verification difficult. Even with direct communication, doubt can cause misinterpretation of available information.

When faced with fake news, decision-makers turn to alternative approaches to help inform their choices. We start questioning facts and discrediting sources based on various factors. Personality traits like charisma, charm, and other subjective criteria are used to correct a perception of noise—even if the facts are proven. Indeed, the reliability of the messenger becomes a substitute for the reliability of the message. This has long been a tenet of public relations and politics, but it has emerged as a leading force in everyday life.

Strictly speaking, there's nothing wrong with making a choice based on personality; we do it all the time. But when personality trumps all other sources of information, a game may be mis-specified and choices might turn out poorly.

The Pitfalls of “Reply All”

We've all done it. You hit Reply All, write your message, hit send, and *immediately* regret your entire life.

Sometimes, we signal information inadvertently. The convenience of modern communications is undeniable, but there's a clear danger of revealing things to the wrong people at the wrong time. How many people look at an entire email or text stream before adding to it and sending it to the entire family or the entire company? In this day and age, do we realize that every email and text is permanent?

In the next chapter, we'll contextualize the expression "what you don't know can't hurt you" when we talk about threats. As we know from the Reply All Conundrum, what someone does know, in particular about your feelings and opinions, *can* hurt you. It's inadvertent because you didn't mean to share the information, but you did. It's out there. And, as we know with information, you can't take it back.

Once a sibling or your boss knows what you really think about them, you can't play any future game the same way. Keeping some things private is critical to achieving certain outcomes. This isn't a lie by omission. It's keeping information that would *not* lead to optimal outcomes out of the game.

Conclusion

Every game-theoretic situation depends on information. You can't begin to play unless you know who you're playing against, what your choices are, and what the payoffs could be. Furthermore, and usually trickier: you need to have some sense of the other players' choices and payoffs so you can leverage information to your advantage.

The list of expressions that relate to information is long. "You can't kid a kidder," means you'll have a harder time successfully manipulating information set if the other player is more experienced at doing that very thing. "Backstabbing" conjures images of people saying one thing but doing another. Doing things "behind your back" means the same thing; you can't or don't see it coming. Maybe it's because of the player, maybe it's their payoffs, but these strategies are the result of manipulating an information asymmetry to your advantage.

The discussion of information in game theory also includes many of the problems that we associate with prejudice and intolerance. When we make bad assumptions about people based on appearance, culture, age, gender, or demographic, we find that outcomes are often worse for everyone involved. This decision-making problem is one that often has people aligning attributes with different motivations: if someone doesn't look like me they probably don't get jollies the same way as me. This can cause you to mis-specify a

game in a bad way. Maybe you and your much-older aunt like donuts the same way.

If you play the Prisoner's Dilemma game with a stranger who looks different from you, your choice might be the self-interested choice because you've assumed that's what they will choose. You don't know them, and they don't know you. Human nature says to defend yourself first. When it turns out they weren't as prejudice as you were and they made the cooperative choice, they got the worst possible outcome. Going forward, how do you think *they* will play the game the next time they encounter someone like *you*!

Questions

- 4.1. Some fables fundamentally change with technological innovation. In the "Boy who Cried Wolf," the kid was ultimately eaten because he summoned help too many times when there was no threat. What could the boy do with today's technology to reinforce his cries that there really is a wolf?
- 4.2. Let's look at regulation. Asymmetric information always gives the better-informed player an ability to steer the outcome of a game in his favor. In an unregulated world, a really good art forger could get really rich. She can sell well-done forgeries of famous artists' works for millions of dollars. In the real world, how do we stop this from happening?
- 4.3. "Walk a mile in my shoes" is a good reminder that information changes outcomes in ways that both players—especially the less-informed player—should consider. Another popular expression, lifted from football, is when a leader says, "we need to be more focused on blocking and tackling." Why might that suggestion not bring for desired outcomes? (no one likes doing that!)
- 4.4. Bluffing is part of the game when you're playing poker. It's part of the strategy and part of the fun. Lying in other games is much less acceptable and definitely not fun. Why, for example, can you lose your job if you lie about your qualifications? (Hint: think about the surgeon who is about to remove your appendix).
- 4.5. Many human attributes are difficult to signal, but knowing could relieve some information asymmetries. Can you think of establishments, organizations, or groups that exist to solve this problem?

5

Do Unto Others: Cooperation in Games

We've invoked the concept of cooperation quite a bit. Cooperation, in its truest form, is something that works only in certain games and under certain circumstances. This will seem strange, but you can get cooperative outcomes in games that, strictly speaking, we don't *think* of as cooperative games. We know that because we just saw it. There's a way to get a cooperative result in the Prisoner's Dilemma game, and that is most decidedly not set up to be cooperative.

Zero-sum games, which we described in Chapter 2, are games that have a winner and a loser. Most competitive sports, board games, job competitions, and Tic-Tac-Toe fall in that category. Cooperation doesn't align with the outcomes of zero-sum games because, well... they're zero-sum. There has to be a winner and a loser, and that doesn't incentivize players to be cooperative.

Cooperative games are ones in which cooperation is possible. Players have an ability and opportunity to observe everyone's choices and identify a mutually beneficial outcome (what we sometime call "win-win" games).

Cooperation is a term that we should unpack. In the modern lexicon, it refers to interactions that focus on mutual benefit, maybe even synergistic benefit (where the cooperative solution is better than anything any one person could have accomplished on his own). In game theory, we don't make any assumptions about the level of the outcomes. We don't say that a cooperative game has a better outcome than a noncooperative game, because these are two completely different types of games. Rather, we say that cooperative games have outcomes that are non-binary: there can be two sets of positive jollies, two sets of negative jollies, and/or a mix of each, depending on the setup of the game.

Learning and understanding game theory is all about figuring out what's possible in the realm of human interaction. Arguably, we are born mostly self-interested. We've never met a baby who decided to stop crying just because mama was a little tired. In practice, we have learned that cooperative behavior usually leads to better outcomes for everybody. In a game-theoretic sense, cooperative behavior means players end up better than they would have playing the game in a self-interested way.

The Game Within the Game

It never failed: you were ready to head out to show off your new bicycle to your neighbor. She was going to be so jealous! You pack up, head out the door, only to hear Granny call out, "play nice!" So much for being super competitive. Looks like bragging will have to wait.

Let's say there's a basketball coach attempting to create a "team mentality" in her players. She's trying to raise the level of cooperation among teammates. The goal: to ensure the team collaborates toward the same goal (winning), that they share responsibility, and that they capitalize on each other's strengths to get the win.

But, wait... Basketball is a zero-sum game, right? One team wins and the other team loses. How can cooperation enter the picture? True, the basketball game itself is zero-sum, but the *team* makes choices of how to play, and these choices vastly affect their payoffs. This is what we call a subgame, or a game within a game.

Is each player in the right place at the right time? Are they making the right choices (staying within their role on the team)? Is winning the most important thing to every player? Is everyone doing their job, or are some covering for others that are falling short? These are all elements of cooperation that affect the outcome of the basketball game, even though it's technically zero-sum. They are also things that could affect the outcome of future games. If the team wins this game but team members cooperate badly, it could affect future outcomes and the overall season.

Suppose there is an exceptionally talented player on the team, and she wants to be noticed by the people (often called scouts) who are choosing players for the next level (a college player looking to go pro). She knows she'll have to appear to cooperate, or she won't get any playing time. She also knows scouts are looking at the individual. Ultimately, she needs to show her individual skills to fulfill her personal goal of playing at the next level. This will affect how she "cooperates" with her team members and the outcome of

the game. If she's looking to be recruited, but her teammates are not, she is really playing a different subgame.

Cooperation is a trickier concept in game theory than it is when we invoke the word “cooperate” every day. Making a game-theoretic cooperative choice is making a choice that usually goes *against* any single player's best interest unless other players also make cooperative choices. Put another way, cooperation only leads to good outcomes if everybody does it. To that end, cooperative games need a fully observable set of payoffs, so everyone is in the know. It also helps to have a way to enforce cooperation or de-incentivize noncooperative behavior in some way.

Cold War Cooperation

A game's structure has to allow cooperation to exist and work.

Nuclear disarmament is a nice, uplifting example. Let's start there.

Suppose there are two countries—Us and Them—and both countries have nuclear weapons sufficient to obliterate the other. If each country knows about the other's capabilities, this can lead to a stalemate: neither side can use its weapons knowing obliteration is likely.

Why does a stalemate matter? Well, in a classroom game, we don't account for silly things like costs of maintenance or upgrades. The game's choice set becomes “fire, don't fire,” and the payoffs are usually specified to show a first-mover advantage. In the real world, maintaining a functional and state-of-the-art nuclear arsenal is quite costly, even if you never fire a rocket. A stalemate is rather pointless and very expensive. So, how do you do you reverse costs? Simply disarm! Aha, that's another game entirely.

Let's formalize this game with some made up numbers so we can visualize the situation. Our units are jollies but on a national scale. Remember, jollies are relative units of happiness for each player; in this case, each nation.

Let's break down the payoffs shown in Fig. 5.1:

- If one country disarms and the other does not, the remaining country becomes a world superpower, militarily. The disarmed country's jollies are lower, since the dominant country can basically control the rest of the world.
- If neither country disarms, they are equal superpowers. However, each has to spend resources to maintain vast nuclear arsenals, and those costs are reflected in the payoffs.

		US	
		DISARM	DON'T DISARM
THEM	DISARM	US 8 TRILLION THEM 8 TRILLION	US 10 TRILLION THEM 1 TRILLION
	DON'T DISARM	US 1 TRILLION THEM 10 TRILLION	US 5 TRILLION THEM 5 TRILLION

Fig. 5.1 The disarmament game

- If both countries disarm, they again end up with equal jollies. PLUS, they're no longer spending on armaments and can put those resources to greater economic effect. As the payoffs reflect, both nations are better off than they were when they had a lot of nukes.

If we play this game simultaneously and without cooperation, both countries will keep their weapons. They both have a dominant strategy to not disarm (run through the analysis in Fig. 5.1 to confirm). This is a Prisoner's Dilemma-style game, and both countries can see that the cooperative solution is better. Question is: can they get there?

If Us and Them sign a treaty in which they both agree to disarm, they are both better off. Signing the treaty is their attempt at cooperating. It's a good signal, but is it enough? The treaty itself isn't enough, even with the punishments and impartial observers and ongoing monitoring. Both countries once had an incentive to engage in nuclear armament—it's how they got to the current position. Assuming that incentive hasn't gone away, both countries now have an incentive to cheat and to keep their weapons.

We're circling around a word here: trust. In order for cooperation to work, there must be innate trust OR a mechanism built in to ensure trust in some way. Us and Them need a way to keep each other accountable. Like Butch and Sundance, it would help if these players could observe each other.

The solution in the 1960s Cold War was spy planes. Famously, for the US, the high-speed, high-altitude camera with wings known as the U2 helped document disarmament by the USSR. (Turns out, they weren't high-speed

or high-altitude enough and the USSR shot one down. Not exactly cooperative behavior on both sides, but we digress). Perfect observation mechanisms weren't technologically possible in the 1960s, and neither was a cooperative solution to the Cold War.

Until...

The US and USSR needed something that couldn't be shot down; something that was not in a country's protected air space. Something like... a spy satellite: an orbiting camera with the ability to see anything and everything that looked like a nuclear weapon within a country, but not be shot down. More importantly, observation could be done by both sides, but at unknown times. With satellites, each side would assume it was being watched. By the 1980s, both countries had this capability, treaties were signed, and the Cold War was done. Observation reinforced the trust necessary for the cooperative solution—the treaty—to work.

The structure of most games is not carved in stone. Rules and other aspects of the structure can be changed to allow for things like cooperation if and when that leads to better outcomes. In the case of the Cold War, the game was restructured so that bluffing wasn't possible. Once each side could observe the other, trust was in the cards.

Recall that bluffing is one way to take advantage of an information asymmetry, for the player who knows more. For cooperation to be possible, information needs to be shared, transparent, consistent, and available to all parties.

The Prisoner's Dilemma, as normally structured, isn't a cooperative game. Either player can *say* that they'll choose Don't Confess, but their self-interest is better served by choosing to confess and they always choose that way. But, we said this Cold War game was a Prisoner's Dilemma style game. What's the difference? Why doesn't the ability to observe get Butch and Sundance to the cooperative solution?

Both Prisoner's Dilemma and the Cold War games are structured as simultaneous games—both players make their choices at the same time. The nature of the Cold War game in the real world is different because it is a series of subgames over time. This creates a timing difference that makes long-term cooperation possible. There's nothing permanent about the choices Us and Them make.

If a cooperative treaty were signed that said both countries would stop building new weapons, it would require years of cheating before the cheater could take advantage of their additional (secret) weapons. With modern surveillance technology, that's (hopefully) impossible to achieve. Even if they did, there would be a next iteration of the game, a next decision to make and

another after that. Because the game is repeated, it allows players to iterate to a solution that's better for everyone. This is consistent with the real world. We might not trust somebody enough when we first play a game, but by the tenth round, we are willing to hear them out.

In its usual form, the Prisoner's Dilemma is a bit more immediate. Imagine the players are able to observe each other but they still have to choose to confess or not simultaneously. Not knowing what the other will choose still leads both of them to confess. The immediacy and finality of the decision make cooperation difficult.

The Nash Equilibrium

Both life and game theory are easier if at least one player has a dominant strategy. Ah, to be so lucky. Most times, in many games, finding one optimal solution isn't possible. When there's more than one possible outcome, we try to find other ways to approach and if possible, solve the game. One such approach is called a Nash Equilibrium. Named for Nobel Prize-winning economist (and subject of the movie "A Beautiful Mind") John Nash, the Nash Equilibrium is one type of stopping point for a game.

Let's back up. An equilibrium is a set of choices that solves a game. It is a form of stasis for that game, where neither player wants to move. In our games, we've reached equilibrium by dominant strategies, dominated strategies, and other methods. In some games, there were even two equilibria, and we needed a tie breaker. Ultimately, an equilibrium is a solution to a game where no player wants to move to another choice.

A Nash Equilibrium is one such solution, where a game could end up. In a Nash, you are doing the best you can given what the other player has chosen. Put differently, in a Nash, no player can *unilaterally* make himself better off. In these games, you don't have a choice that's always best for you the way you did in a dominant strategy; you have a choice that's best for you *given* the other players choice.

Let's see how this works in a familiar situation: the Dropped Call Game. You and your significant other are talking on your cellphones when suddenly the call drops. One of you loses cell service and the call is terminated. Who calls whom back?

Take a look at the payoff matrix in Fig. 5.2. Let's assume that you were in the middle of a great conversation, and "Talk More" gives both of you the most jollies.

		CALLER 1	
		CALLS BACK	WAITS
CALLER 2	CALLS BACK	BOTH GET VOICEMAIL	TALK MORE
	WAITS	TALK MORE	DON'T TALK AGAIN

Fig. 5.2 The dropped call game

The game has two Nash Equilibria: one player calls while the other player waits. It doesn't matter who waits and who calls as long as they don't do the same thing. So, the game has two possible solutions. How do we get to one of them? How do we get to finish our conversation??

When games have multiple Nash Equilibria but no dominant strategies, you need to do more to solve the game—to get to one of the equilibria.

Games like these are often called coordination games, and the hardest thing about them is to find a way to get to the mutually beneficial outcome. They need coordination. In many cases, people will revert to an understanding or a rule for solving the game. Something like “the person who originally made the call should always call back.” That certainly works as long as everyone knows the rule and everyone follows it.

These rules emerge when people have what game theorists have called a focal point. This sounds fancy and theoretical and at some level, it is but the idea is straightforward. Focal, like focus, means we've found a way of cutting through the noise and randomness of the game. We have found an informal rule that works for the players most of the time, the rule is easy to remember, and the rule often springs from what we call common sense. Fancy strategists will formalize rules into heuristics or other modes of operation. The bottom line is this: humans draw on their own preferences and ways of doing things to get out of a Nash situation. It is usually something intangible—some unstated rule—that allows the conversation to continue.

That's a phrase we say a lot and don't really think about, but “common sense” is two really powerful words. *Common*, meaning shared by most

people, well-known, or typical, and *sense* meaning intuitive, sometimes even automatic. A coordinating mechanism that comes from common sense should work most of the time and should be easy to apply to multiple players.

We needed a coordination mechanism like “whoever made the call, calls back” in the dropped-call game because of the specifics of that game. If the situation allows, the easiest coordination mechanism is direct communication. Just say what you’re going to do. That always works in these games because the only real problem is coordination. How many times in life have we said: “I’m heading into a tunnel/elevator/building; if I lose you, I’ll call you back.” In one direct statement, you pre-solved the game.

We’ve talked about games that have outcomes in which one player is better informed than the other and uses that advantage to improve their outcome. Those games will bring in things like bluffing and lying. Those asymmetric information strategies are of no value in a coordination game, however, since players can only coordinate with full information.

When a coordination game has a rule that everyone knows and uses, like “I’ll call you back,” game theorists call it an assurance. When there are game-theoretic situations in which players need to coordinate in order to reach a mutually beneficial outcome, finding and using that assurance guides toward the cooperative solution. It helps players know that things will turn out okay.

Punishment

One way to restructure the single-game Prisoner’s Dilemma to make it a cooperative game is to incorporate a form of punishment. Doing that means that we have to make explicit changes to the payoff matrix to punish a player if they confess. Recall the original Prisoner’s Dilemma game in Fig. 5.3:

Suppose both players agree that if one of them confesses and the other doesn’t, the snitch will do something to the no-snitch in jail that would effectively ruin their day (break something that takes a long time to heal). The punishment would add 5 years to the snitch, because...you know, violence is bad. The payoff matrix would then become Fig. 5.4.

This changes the game. Now, neither player has a dominant strategy. The game could end up at either “both confess” or “neither confess.” Both of those outcomes are Nash Equilibria; neither player can unilaterally make himself better off.

Is the game now cooperative? Yes... with one caveat.

Suppose Butch says he won’t confess. If he does what he says and Sundance also doesn’t confess, they both get 3 years. If Sundance confesses despite what

		BUTCH	
		CONFESS	DO NOT CONFESS
SUNDANCE	CONFESS	BUTCH 5 YEARS SUNDANCE 5 YEARS	BUTCH 10 YEARS SUNDANCE 1 YEAR
	DO NOT CONFESS	BUTCH 1 YEAR SUNDANCE 10 YEARS	BUTCH 3 YEARS SUNDANCE 3 YEARS

Fig. 5.3 The Prisoner’s Dilemma, redux

		BUTCH	
		CONFESS	DO NOT CONFESS
SUNDANCE	CONFESS	BUTCH 5 YEARS SUNDANCE 5 YEARS	BUTCH 10 YEARS SUNDANCE 6 YEARS
	DO NOT CONFESS	BUTCH 6 YEARS SUNDANCE 10 YEARS	BUTCH 3 YEARS SUNDANCE 3 YEARS

Fig. 5.4 The Prisoner’s Dilemma, with punishment

Butch said, Butch is worse off—but more importantly, so is Sundance. If Sundance ignores what Butch says and does, and confesses, Sundance’s prison time goes from 5 to 6 years. So, Sundance has no reason not to believe Butch. The punishment—reflected in the payoffs—seems to be working.

If Sundance does not confess, would Butch have any reason to not follow through and confess? Nope. Butch will get 6 years if he does not follow through, instead of 3 years if he follows through.

Both players can now get to the cooperative solution by agreeing, upfront and explicitly, to choose “don’t confess.” The game doesn’t have the observability that we normally require for a cooperative game, but we’ve restructured the payoff matrix to make it a coordination game in which direct communication (just say what you’re going to do) will get both players to the cooperative outcome.

Punishment, Mafia Style

There are game-theoretic setups in which punishment is metered out by someone other than the players in a game. Members of organized crime families and other, less nefarious organizations, like the military, often have an honor code that requires all members to monitor and enforce cooperative choices for the entire group. If that were the case, the payoff matrix would have both Butch and Sundance being punished for confessing by a third party. Let’s call him Mickie.

We’ll make two changes to the payoff matrix, if Mickie is in the picture. First, the “punishment” for confessing is eight extra years (Mickie’s, like, 6’5” and used to be a prize fighter) and second, both of them would get punished if both confessed (Fig. 5.5).

The presence of a third party doing the punishing means both or neither man could be punished. This gives even more incentive to cooperate, since

		BUTCH	
		CONFESS	DO NOT CONFESS
SUNDANCE	CONFESS	BUTCH 13 YEARS SUNDANCE 13 YEARS	BUTCH 10 YEARS SUNDANCE 9 YEARS
	DO NOT CONFESS	BUTCH 9 YEARS SUNDANCE 10 YEARS	BUTCH 3 YEARS SUNDANCE 3 YEARS

Fig. 5.5 The Prisoner’s Dilemma, mafia style

there's an opportunity to both "win," or at least survive. Mickie gives both players a dominant strategy to not confess, and the two get the game to the cooperative solution.

We can't assume that this will work consistently. Those third-party enforcers have to be "described." How do they work? How are *they* getting jollies? Why should they be trusted to do the enforcing? What if Mickie's bluffing?

Let's leap from mobsters to international relations to show how the approach is imperfect. Both the League of Nations (circa 1920) and the United Nations (circa 1948) were intended to eliminate aggression between countries. They were supposed to make sure war never happened again. They were created to get the world to an international cooperative solution.

While the latter organization has, so far, prevented World War III, it has not in any way prevented all war. The governing structure of the UN shows us why. While most countries are members (part of what's called the General Council), there are 15 countries on the Security Council, five of which are permanent members. Any of the permanent members can unilaterally veto a UN choice.

That veto power means that those five countries can, more or less, do anything that they want to do. Unfortunately, that often extends to countries that are "friends" with the five permanent members. These organizations were certainly founded with a meaningful charter, but the structure limits its ability to serve its charter. As Granny said, "the road to hell is paved with the best intentions."

Game Theory Goes Back to the Movies

There are many movie quotes and situations that are game-theoretic. Let's talk about one that comes from *The Godfather Part II*. Now head of the family, Michael Corleone tells a top lieutenant: "keep your friends close, but your enemies closer." Michael is providing a rule to address the problem of decision-making when the other player is not likely to act in your best interest.

If the downside of a situation is dire—possibly fatal—then you need to know if, when, and why the other player might make the choices that will lead to your demise. The implication of the expression is that your enemies are much more likely to make those choices than your friends. Keeping them close = you can keep an eye on them. Keeping them close = you have more information about their intentions at any given time.

Automatic Punishment

Punishment is important in reaching and maintaining the cooperative outcome, but dealing out said punishment can be hard to do. Causing pain, for most people, is itself painful and thus unpleasant. Did Granny say it? “This is going to hurt me more than it hurts you” implies punishment negatively affects a person’s jollies, even if you’re on the receiving end.

Given this reality, we sometimes try to make the punishment automatic. We remove the hesitation from the player who needs to punish because the so-called “human element” might interfere with follow-through. Making punishment automatic means it is no longer a choice being made by a person who might be unwilling to “do what has to be done.” It also removes punishment from the punisher’s jollies, since they are absolved from having to personally make that decision.

In the classic movie *Dr. Strangelove*, the Soviet Union builds a computer-controlled, unstoppable “doomsday machine” that would destroy the entire world if anyone (but at the time, the US) ever launched a nuclear attack against it. The idea: this device would end the arms race because a preemptive nuclear attack would kill both sides with certainty.

SPOILER: unfortunately for the world, Soviet leaders didn’t tell anyone about the doomsday machine. Nobody knew until a deranged US general launched an attack and, as it turns out, the machine automatically retaliates and destroys the world (*apologies to anyone who hasn’t seen this truly great comedy—definitely go see it if you can*).

In effect, automatic punishment is what some dog owners might think of as a shock collar; a device that gives a dog an electric shock every time the critter barks. The idea is that the dog will learn not to bark as a way of avoiding the automatic shock. Veterinarians and animal trainers generally agree that these devices don’t work and might be counterproductive; instead of controlling a dog, they might make the dog more aggressive. That and *Dr. Strangelove* should be a cautionary tale; automatic punishments are difficult elements of a game since they directly affect the outcome but aren’t in the control of either player.

Removing the human decision-maker from the punishment phase of a game can backfire. Use of a gun in the commission of a crime (robbing people at gunpoint) adds mandatory extra prison time in many places. That makes sense if you’re trying to deter the use of guns, but some people believe that if that extra time is excessive, it creates an incentive to use the gun which means the criminal might be able to eliminate a lone witness and avoid being caught.

While we won't wade into this gun control controversy, it's easy to see how automatic enforcement is imperfect. Here's another thought: some countries have automatic penalties for crimes like murder to avoid even the appearance of selectively applying harsher punishments to different groups of people. That is another aspect of humanity that makes punishment harder than it appears. For punishment to work, to change the decisions of all players and reach the cooperative outcome, punishment cannot be applied selectively.

Let the Punishment Fit the Crime

We've talked about punishment, but what about forgiveness? De-incentivizing noncooperative actions might not require punishment *per se*. Sometimes, pressure or negative perceptions that could take effect in the *next* round of games is enough to deter that kind of behavior. In repeated games, where we play in an iterative way, can we bring a noncooperative player back to the cooperative solution?

Punishment is a funny concept. It can be formal, as in jail time. It can also be more esoteric, as in giving somebody the cold shoulder or outright ghosting them. In legal terms, most societies have put limits on punitive judgments. Capital punishment is just not an option if the crime is, for example, grand larceny. In game theory terms, this means jollies need to be reasonable and punishment should be in line with the jollies that the bad behavior created. Anything out of line with that metric would give players the sense that the person doling out the punishment is doing so with other payouts in mind (perhaps they have an agenda, a vendetta against you, or some other reason for blowing things out of proportion).

What can go wrong in a game-theoretic situation if the punishment is excessive? Among other things, people won't want to play with you. Overbearing punishments often direct people to a maximin strategy. If the difference is large enough, they might decide not to play, or not to confess mistakes or indiscretions at all. With large punishments, we often see a reduction of information, and therefore fewer games that land on an optimal solution for all players.

Punishments are an enormous problem in many organizations. If the threat of punishment is severe, meaning it significantly alters jollies beyond a certain amount, the incentive to play honestly is removed. Now, jollies are based on avoiding punishment instead of cooperating to find the best outcome. Mistakes happen in any repeated game. So, if the intention of the threat of

severe punishment is to stop people from making bad choices, the result could be more cover ups and inefficiency rather than higher performance.

Bad behavior sometimes has implications that the decision-maker doesn't understand. The tricky thing about jollies is that they come from a lot of different parts of our psyche. "An eye for an eye," which dates back thousands of years (Hammurabi 1750 BC), is an early attempt to align the punishment with the crime. But, as subsequent scholars have said, you end up with a lot of people without eyes. Explicit and pre-stated punishment is interestingly missing from many of the most famous rules of societal games. The Ten Commandments and the US Constitution have lots of do's and don'ts—but no specific punishments. That's left to the players to decide.

Realistically, punishment and forgiveness are sometimes not possible, and making some bad choices even once changes all future aspects of a relationship. Let's say a professional basketball player snaps at practice 1 day and attacks the head coach. The League may punish him with a suspension or a fine, but that doesn't mend the relationship. He will probably be traded the next season.

In the News: Nuclear Disarmament vs. Climate Policy

When the US and (then) USSR agreed to reverse their build-up of nuclear missiles, it marked a dramatic, positive change in the direction of international relations. But as much as we'd like to think this was the result of pragmatic, practical leaders seeing the advantages for the world in cooperative behavior, it was facilitated by the innovations such as surveillance satellites and other technology that let both countries see each other's missile silos.

Let's pivot to another crisis: climate change.

Policymakers believe that we need cooperative agreements among the largest countries in the world to control greenhouse gas emissions caused by carbon-based fuels used to power most cars, factories, and electricity generating plants. Once these international agreements are signed, there will be a reduction in emissions, right?

Well, not automatically.

Agreements are in place to control output and impose other restrictions that cost short-run money to improve long-run outcomes. No problem there, except countries are short-sighted. Large economies will take a

larger hit, so the agreement is there to make sure everyone is incentivized to participate. Like nuclear disarmament, though, observation is key. Without a reliable and transparent way to monitor emissions (and control cheating), an agreement is just a piece of paper.

Do we also need punishment? This is where nuclear war and greenhouse emissions are a little different. In the long run, sure, they're both pretty bad, don't get us wrong. But, cheating on building nuclear weapons is different than maintaining fossil-fuel consumption. The punishment for going against a nuclear treaty might be nuclear war. This is a big deterrent. Punishments for going against emission standards should be equally powerful: significant economic sanctions, fines, or other costs. They also need to be larger than the possible gains from deviating from the treaty (lest the payoff matrix incentivize noncooperative behavior).

A viable greenhouse gas limitation treaty needs both observation and punishment to succeed.

While cooperation is a useful area of discussion in game theory and one of the most real-world forces we see, one could argue that cooperation defeats the purpose of game theory. Most games, most of the time, are noncooperative. Having to make a choice without knowing what the other player is doing is the whole point of game theory, right?

It turns out, some of the biggest, most impactful games need bi-lateral cooperation to ensure a good outcome.

Noncooperative games are not malicious. As we have seen, they are not the same as self-interested, selfish behavior, nor are they unilateral. You're still playing with somebody else. So, we think of cooperation as a restructuring of a game, something with commitments, promises, threats, or other things that help us approach a cooperative solution in the real world. We'll examine these three things more in Chapter 7.

Too Many Cooks Spoil the Broth

A common complaint about academic game theory is that most of the mathematical models and examples have only two players. Even worse, most games are what's called 2×2 —two players and two choices. In the real world, there are usually more players and more choices. When we teach and learn game theory, we do these smaller games because they are easier to solve, understand, visualize. Remember, we aren't trying to be literal in most cases; we're

trying to learn technique and grow an approach to analyzing an environment and specifying a game.

Granny has a zinger for this one too: “too many cooks spoil the broth.” If we have a game with a small number of players and choices, the game is easier to solve. It’s also easier to find ways to cooperate and thus reach the good-for-everyone outcome that we have called the cooperative solution.

When you have too many cooks (or players), the cooperative solution becomes less available. Beyond a certain number of players, observation becomes more difficult, and we revert to more self-interested behavior. If and when this happens and we end up at the noncooperative solution, most people think that the broth has been spoiled. This is why we teach socialization early in childhood education. Learning how to navigate more complex choices with more players and still reach a cooperative solution is part of growing up.

The Golden Rule: Words Well Chosen

“Do unto others as you would have others do unto you.”

Granny didn’t say this one, but she might have quoted it a lot.

While a good expression, as we’ve pointed out, is often catchy and cute, you have to think about the specific words that were chosen. In this case (sorry, everyone), grammar matters.

Look at the tense of the verbs.

If you have a repeated Prisoner’s Dilemma game (or any noncooperative game), one heuristic, MO, or way of deciding is a strategy known as “*tit-for-tat*.” I do to you next time what you did to me this time. Tit-for-tat is a great deterrent to noncooperative behavior, and it’s especially humbling in each game. After all, you don’t want something to come back around in the next game that could have been avoided, and you control that choice.

The expression that would fit that solution would be “do unto others as others *did* unto you.” Tit-for-tat is a response rule. When everyone follows that rule, and everyone has full information, it should lead to the cooperative solution.

The Golden Rule, on the other hand, is more prospective and forward thinking. “Do unto others as you *would have* others do unto you.” Note the choice of the verb “would have.” For this rule to help you reach the cooperative solution, you need more detail and more structure. In particular, everyone would have to be “golden rulers.” The downside of cooperating with a noncooperator is large, and you will abandon that strategy the first time somebody acts in a self-interested way.

Mind you, you wouldn’t necessarily become completely self-interested. You would try to sort the “others” in an attempt to discern who you could form an

alliance with. This might explain the human tendency to join clubs, churches, and other groups with shared values. Familiarity becomes your method of sorting and determining who you should trust to act cooperatively.

The Simple Logic of Rumspringa

Business leaders sometime call it “buying in” to the corporate culture. The Marine Corps calls it “Semper Fi.” Granny called it “rowing in the same direction.” One requirement for any successful, cooperative social structure: people have to want to be part of it. Outcomes depend on a voluntary and complete “buy in” to get to cooperative solutions time and time again.

The more rigid the cooperative structure, the more this is true. The Amish lifestyle, for example, is based on cooperative tenets, where community supersedes individualism. To choose this lifestyle, members must believe that this approach will make them at least as happy as other lifestyle choices. It's one thing to be told the lifestyle is beneficial; it's another to make the choice voluntarily given available information.

Many Amish sects encourage individuals remove themselves from the local community and experience the non-Amish world prior to committing to the way of life. Called the Rumspringa, the experience allows younger members of the community to take residence elsewhere and live under different rules. If the individual returns, they are actively choosing to be a part of the institution, with full information about other available choices.

That the Amish religion is still around speaks to the validity of this approach. Very few religions take this approach. Most have a rite of passage, formal acceptance, or ceremony of indoctrination—sometimes taken at a young age before informed choice is even possible. They don't encourage young members to test drive other religions.

Cooperation, sacrifice, and other words that mean selflessness, trust, and even faith are associated with group outcomes. Indeed, a lot of our upbringing—across almost all modern religions and societies—is intended to help us avoid the noncooperative outcomes we observe in the Prisoner's Dilemma.

Conclusion

Learning to share, to wait your turn, and play nice with others are all preschool lessons that train you to act in a way in which society is better off. These skills are meant to improve the collective; they are for the benefit of the group, not only the individual. Sometimes, the downside of noncooperative behavior doesn't occur to people until they are on the wrong side of that behavior.

When someone butts in front of you in line, or hogs all of the toys, or hits and pushes during playtime, you (hopefully) realize that when everyone does these things, no one is better off. Good parenting is trying to get that lesson ingrained without the (probably repeated) bad behavior punishments. Bad behavior in the real world does sometimes win; bullies and crooks don't always get punished. But, are those individuals supported by society?

In legal parlance, the goal is to maintain individuals as "functioning members of society." Lessons are small, constant, and repeated by many people. They are also reinforced by many social structures (corporate codes of conduct, school bylaws, organized religion, legal codes). Reinforcement is both the presence of a rule and the punishment for not following it. How well they work, however, depends on the individual.

Granny always had something to say, especially when people evaded capture or punishment: "you'll get what's coming to you." When we see noncooperative behavior, sometimes we *wish* authority were there to meter out punishment (where's a cop when you need one?). Granny was convinced judgment would happen eventually, and this makes sense in a Game-theoretic sense. In our heads, the bad guys can't win in the long run. Without controls, we iterate to smaller and smaller payoffs. Society becomes unpleasant and unstable, and we are guided to suboptimal solutions.

Questions

- 5.1. We made it five chapters before bringing up COVID-19. Yay! Countries attempted to slow its spread with cooperative behavior, specifically masks and accelerated (publicly sponsored) mass-vaccination programs. Mitigating communicable diseases is a good example of the upside of cooperation. Our recent experience also shows us how difficult that is. Discuss the recent debate between mask mandates and personal choice using a game-theoretic approach. What is different between masks and vaccinations? Why might one reinforce cooperative behavior better than the other?

- 5.2. Many behaviors that we would think of as cooperative also fall into the category of politeness. There can be, unfortunately, a bit of a culture clash when we think about polite behavior. Culture clash is another word for personal interpretation. Take opening a door, for example.

A man holds a door open for a woman. This was taught as polite behavior once upon a time, but it might now be considered a bit dated—especially if he *only* holds doors open for individuals that he perceives are female. What if someone who identifies as male wants to maintain the custom for politeness, without judgment? What could he do? Think about the information problem(s) with this action, depending on the perspective of the person he's holding the door for.

- 5.3. The ability to observe the actions and decisions of all players is an important factor in getting cooperative outcomes. Today, observability is at an all-time high. Cars, houses (doorbells), stoplights, and retailers are among the many places where cameras see everything. This new ability to observe all behavior, cooperative and noncooperative, hasn't become sufficient to decrease self-interested or illegal behavior, though. What else—besides the ability to observe players' choices—is required to stop noncooperative social behavior?
- 5.4. Waiting in line is a classic, yet frustrating cooperative behavior. At places where wait times are notoriously long (ex, Disneyland, a ballpark restroom, the school pickup line), some organizations have devised ways to have customers pay to skip the line. How is that extra fee similar to a fine for noncooperative behavior?
- 5.5. Cooperative behavior is taught from an early age. Sharing and patience are not innate; they are learned. Are there limits to cooperative behavior? Are there games in which cooperation does not help the game reach the best outcome?



6

A Stitch in Time...: Sequential Games

Granny may have grown up on a farm, so she had the best expressions ever. Take the short, simple saying “you reap what you sow.” Granny was saying that your future crop is connected to the seeds you plant today.

Well... duh, of course. It turns out this classic bit of cautionary advice defines *sequential games*. Good outcome or bad, what you get in the future will depend on your choices and actions today.

Sequential games have the same features as simultaneous games: two people are making choices and that, together, determine their outcomes. Sequential games simply have a different structure: rather than choose at the same time, one player chooses before the other. Checkers, chess, Tic-Tac-Toe, poker, and many other games are sequential. Unlike rock-paper-scissors or the Prisoner’s Dilemma, sequential games are fully observable: each player can see the first player’s decision and then make his own. As we’ll see, this temporal nature of sequential games can give some players a unique advantage or disadvantage.

Sequential games occur as part of the natural order. You spend your entire life playing games with your future self. The “present you” makes choices that the “future you” has to reckon with. While we don’t want to make it seem like every choice you make is in need of a game theory deep-dive, having the tools and approaches of game setup could help you sort out complex decisions.

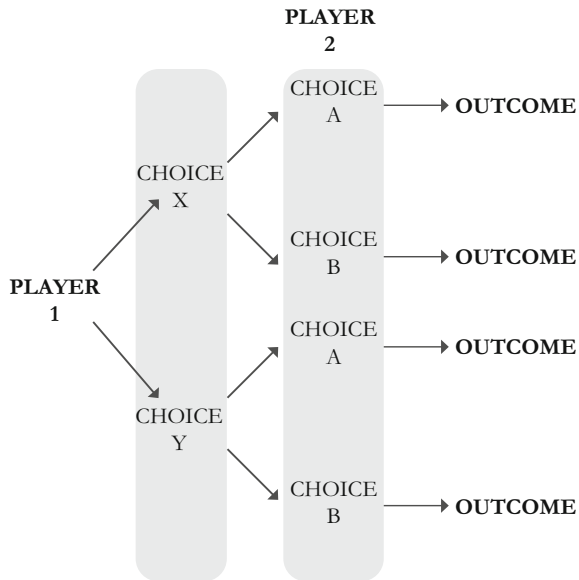


Fig. 6.1 A tree!

Sequential Games: The Setup and the Approach

The easiest way to see how these games differ from ones we've already done is by rethinking the payoff matrix. You can certainly try to keep our payoff matrix and impose timing on it, but there's an easier way. For sequential games, we use what's called a decision tree, or as game theorists call it, the extensive form (Fig. 6.1):

Player 1 makes her choice. *Then*, Player 2 chooses. Both choices have an effect on the outcome, so Player 1 has to look ahead to Player 2's possible choices to see what might happen. Sound complicated? It can be, but we do it all the time.

The Nancy Reagan Game

This is a book on game theory, not statistics and probabilities, so we aren't going to spend any time talking about the likelihood of future outcomes. Rather, for sequential games, we want to focus on the choice *process* and the impact of today's choices on future outcomes.

If you decide, when you are eighteen years old, to try smoking cigarettes, you might become addicted to them and smoke for the rest of your life

(a possibly shorter life). If you never try cigarettes, you really can't become addicted to them. The choices of the two players, the young you and the grown-up you, are inexorably connected. You *could* start talking about probabilities—you'll *probably* be able to quit, don't have an addictive personality, etc. Yet, even from a probabilistic standpoint, one thing is true: if you don't start smoking, you can't become addicted to cigarettes.

With that in mind, we can look at a different extensive-form game where both players are you. As simplistic as it sounds, looking into your own future and seeing outcomes that you really don't want to happen, you can give yourself a straightforward approach to decisions you have to make today. Let's see smoking in extensive form (Fig. 6.2):

Former First Lady Nancy Reagan's famous, oft-mocked advice about using drugs, "just say no," applies very well to this game. The idea is simple: you really can't become addicted to anything if you never try it in the first place. Note, the outcomes for both versions of "try smoking" are worse than the *only* outcome possible from not trying. This is similar to other game setups where we show one or more very bad outcomes that we simply want to try to avoid.

In the case of the game above, we're not including any *positive* aspects of smoking. Smoking might make you feel good (chemically), or sophisticated,

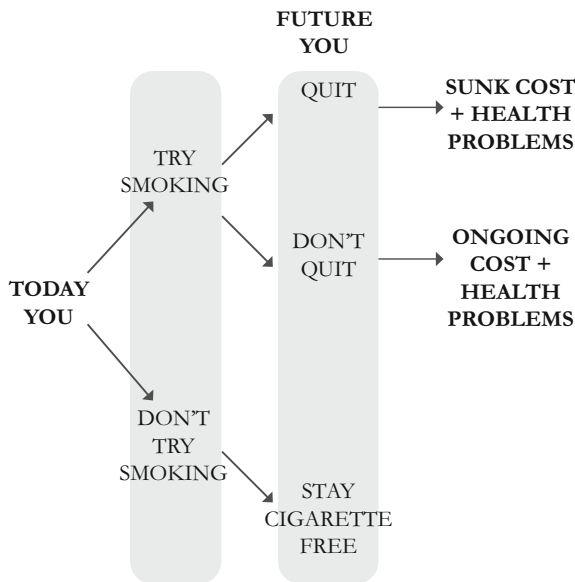


Fig. 6.2 Smoking or non?

or less stressed, and these feelings increase your jollies. This isn't a simultaneous game, so we have to remember there's a temporal element. These feelings increase your jollies *in the short run*. In sequence, we are able to specify a portion of the payoffs (in this case health outcomes and money) and still navigate the game.

Sequential games are both useful and a bit misleading. Suppose we're doing the game that Nancy Reagan was talking about, but instead of smoking legal substances, we're choosing whether to start illegal drugs like heroin or crack cocaine. We can specify the *entire* sequential game that you play with your future self when you consider trying drugs, in terms of another outcome: careers (Fig. 6.3).

There are better and worse futures in either case. However, there is still one, unambiguous bad outcome—being a drug addict—that only happens if you try drugs today. At this point, we aren't interested in the likelihood or probability of these outcomes. The point of game theory is to look at payoffs and, in this case, walk back a path to avoid the worst possible outcome.

From this maximin perspective, it makes sense to avoid the worst possible outcome by never starting drugs. This reality is loosened by those who consider the first step “experimenting.” Whatever the rationale behind the choice to try the drug, it is a choice that changes all possible future outcomes. An outcome that you almost certainly do not want is now possible. If you never “experiment” with cocaine, heroin, or crack, you can't possibly die of an overdose from the addictive use of those drugs. From a sequential game

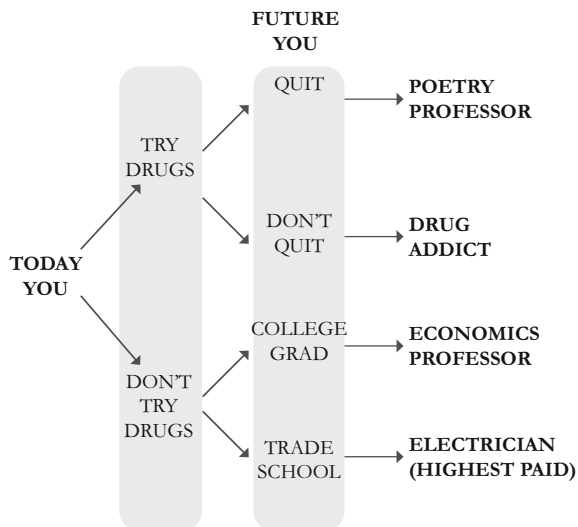


Fig. 6.3 Future careers

perspective, “just say no” is good advice, simplistic though it is. It removes the possibility of the worst possible outcome, showing a maximin strategy in an extensive form.

The First Credit Card

One of the hardest things for a person to learn is the difference between a loan and a gift. In some countries, kids grow up getting many gifts and not enough loans. Later in life, the ability to do things like buy a car, rent an apartment, and buy a home will depend on credit history—a signal to lenders that you know the difference between a gift and a loan and that you have the wherewithal to pay something back.

As a parent, you want to help your offspring navigate these tricky waters. That’s easier said than done.

Your teenager wants his own credit card, and you agree to co-sign—become the default payer of the card—which allows your son to get his first credit card. This thing is magic. He can swipe or tap to his heart’s content and quickly spend more than his allowance or after-school job is capable of paying. What do you, the parent and co-signer, do next? If you let your kid start missing payments, his credit history will be damaged, and he might be living in your basement a lot longer than you had planned. If you pay off the card, he might not learn the difference between a loan and a gift, and he’ll overspend repeatedly.

So, his choice and yours can affect his future outcomes. The fun part of this game—and all games in the parenting space—is that you’ll both have different outcomes, *and* you have to navigate the difference between money and jollies.

Spoiler Alert: we’re not solving this game for you! Everyone plays it game differently, and there isn’t a “right” answer. Indeed, later in life, your kid will be playing games like this with you. Adult children trying to deal with adult parents have just as many complex conflicts. Maybe Granny didn’t say it, but “it’s hard to raise your parents.”

Rollback Solutions and First-Mover Advantages

Sequential games with your future self highlight the importance of sequencing in decision-making. Throughout your life, you make contemporaneous choices that affect all future choices and outcomes. Game theory has approaches and perspectives that are different for sequential games because of this temporal element. When outcomes are a factor of choices and time, there are specific considerations we, as humans, may use to make sense of choices.

Playing a sequential game well means looking ahead at how another player will respond to your move. You'll never win at checkers or chess if you can't look ahead and figure out your move based on the other player's response. Chess Grand Masters (champions) often see 10–30 moves ahead in any given game (and some can play multiple boards at a time)!

The solitaire game Pyramid is an excellent, one-person sequential game. You are dealt a deck of cards in a large triangle starting with one card, then two, then three, etc. until you have seven rows; the cards should overlap a bit to “block” the cards in rows above. You will have twenty-eight cards when you are finished. The remaining twenty-four cards form a draw pile.

You match up two cards that equal 13 to eliminate them and reveal the cards underneath. Every number card equals its sequence: kings = 13, queens = 12, jacks = 11, and aces = 1. If you have no available combinations that equal 13, you draw a card and look for a combination with that card.

When played according to the rules, you can combine 6 and 7 (ok, here's some math: $6 + 7 = 13$). Let's say you deal your cards, and you have two possible combinations of 6 and 7: one doesn't free any new cards and the game will be over; the other allows for more choices and the game keeps going.

Looking ahead before you make your next choice is critical for winning this game. It also means you're looking at future choices you *will* want to make but can't make yet. This approach to sequential games is called a rollback solution. A player determines an ultimate outcome of winning the game and works backward through choices to get there.

Like simultaneous games, you can also use the temporal aspect of sequential games to determine whether or not you should actually play. You could look at the cards and see a layout that makes it impossible to win at all. That means there's no rollback solution available to you. What do you do in that case? End that game immediately and play a new game. Granny might have been a Kenny Rogers fan, singing “you've got to know when to hold 'em/know when to fold 'em...” It is a poker version of the Pyramid game,

but he lays out a clear choice. There are times we have to walk away from a no-win situation.

Sequential games do get trickier when you're playing with another person, in part because the final outcome is not as simple as you win or you lose. In sequential games, potential outcomes are dependent on both players' choices. Depending on the circumstances, it may depend more on who goes first. Playing Tic-Tac-Toe? We prefer to go first. This is less strategic than it is numerical. Turns out, the source of the first-mover advantage in Tic-Tac-Toe is actually mathematical. There are nine squares, and the first mover will get to put their mark in five of them. We've heard it said that you have to be really bad at sequential games to go first in Tic-Tac-Toe and lose. Try it, let's see.

If there's a distinct advantage to go first in a sequential game, it has a first-mover advantage. If there's benefit in waiting and seeing what happens first, there's a second-mover advantage. Let's take a look at a few examples.

First-Mover Advantage: The Coke and Pepsi Game

Suppose The Coca-Cola Company and PepsiCo, Inc. are considering adding a new drink to their respective product lines. Working with their research and development departments, each company has narrowed the choice to a new diet cola or a new flavored soda. The companies have done some market research, and they know that there won't be enough new markets for them to both choose the same thing. Ideally, Pepsi would launch one and Coke the other. Their marketing departments do a lot of research and determine the potential profits under all scenarios. The payoff matrix they develop, in revenue, is in Fig. 6.4.

Ok, let's go through the usual ideas. Neither player has a dominant strategy. If Coke chooses Diet, Pepsi wants Flavored, and vice versa. These are the two Nash Equilibria: outcomes where, if they can get there, neither company can unilaterally make itself better off.

How do we get to one of these results? We didn't specify anything about how this game would be played—simultaneously or sequentially. If it's simultaneous, the best we can say in game theory is that a heuristic or other form of advantage will push the game to a particular solution. This is largely unhelpful and not reflective of the real world. It's appropriate in this case to use a sequential game to see what happens when one company goes first.

THE COCA-COLA COMPANY

		DIET SODA	FLAVORED SODA
PEPSI CO.	DIET SODA	COKE -\$1M PEPSI -\$3M	COKE \$10M PEPSI \$6M
	FLAVORED SODA	COKE \$15M PEPSI \$5M	COKE -\$2M PEPSI -\$4M

Fig. 6.4 A sweet choice (in revenue)

Company 1 chooses one innovation, then Company 2 chooses the other, and voila, they get to one of the solutions.

Here's the problem with that simple approach: Coke and Pepsi both want to go first. This game has what's called a first-mover advantage—and you can see it in the payoff matrix. Both companies prefer diet soda over a new flavored soda, as long as the other company chooses the opposite. Likely, it's because the market for diet sodas is much larger.

Let's show the first-move advantage with two sequential decision trees. If Coke goes first, it looks like Fig. 6.5.

If Pepsi goes first, it looks like Fig. 6.6.

But... the outcomes are the same, right? Not really. Remember, a sequential game's first-mover advantage has each player looking at the final outcomes and then the possible paths to get there.

If Coke goes first, it can see its best outcome is \$15 M, and this is achievable if Coke chooses Diet first. Pepsi sees that its only possible outcomes given Coke's choice are \$5 M or - \$3 M. It will choose flavored, and that gets Coke its \$15 M.

If Pepsi goes first, its best possible outcome is \$6 M, which would be possible only if it chooses diet. After that first move, Coke's only possible outcomes would be \$10 M or - \$1 M. Coke would choose Flavored, and Pepsi gets its \$6 M.

If Coke goes first, it ends up with \$15 M and Pepsi ends up with \$5 M. If Pepsi goes first, it ends up with \$6 M and Coke gets \$10 M. In this

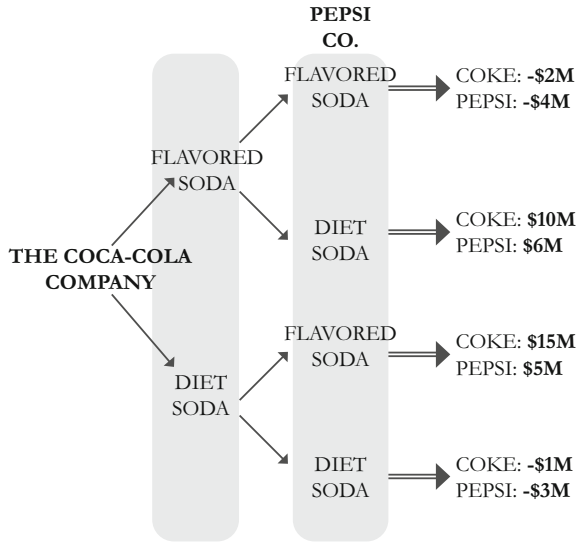


Fig. 6.5 Sweet choice, Coke as first mover

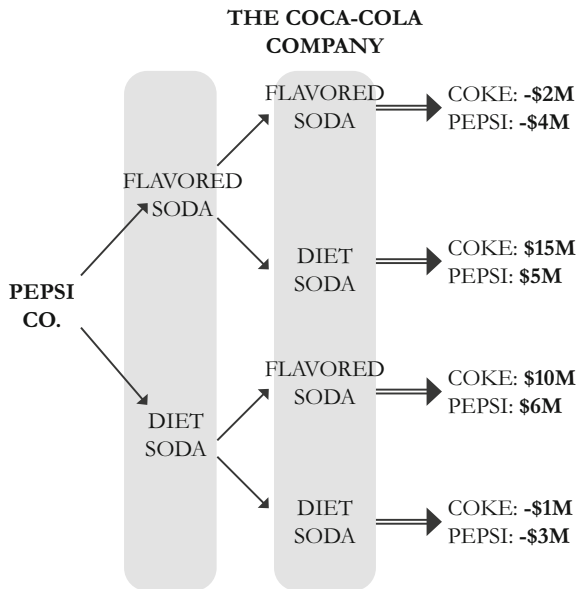


Fig. 6.6 Sweet choice, Pepsi as first mover

game, with these payoffs, the player that goes first can steer the outcome to its advantage. Coincidentally, this lowers the payoff of the other company, which is consistent with competition in the soft drink market.

When the rules of a simultaneous game are not pre-established or very clear, we often experiment with timing. “What do you want for dinner?”... “You go first!” We try to play the game sequentially instead. In some cases, it takes another rule or requirement, like who goes first, to make this a useful way to get to a solution.

Second-Mover Advantages

It doesn't surprise people, based on experience with Tic-Tac-Toe or many real-world situations, that there are first-mover advantages in sequential games. Are there times when it's better not to go first? We could simply make up a sequential game that has a second-mover advantage and prove it theoretically, but the point of this book is to find real-world examples of game theory.

It's 1989. You live in the Midwest. It's a SNOW DAY, and you're oh so happy. Reading Rainbow is done, and now you're in front of the TV for some midday programming. A familiar face appears and says to the audience: “come on down!” Life is good.

A familiar second-mover advantage (in this case, last-mover advantage) from US television is in the game show “The Price is Right.” Contestants will win a prize (a new jet ski! Because that's so useful!) if they correctly guess the price of the showcase prize. Whoever is closest without going over wins. There are three contestants. You can hardly contain your excitement as you pour another bowl of Fruity Pebbles.

The first person guesses \$500, which you're sure is too low. The second person guesses \$900, and you're sure that's also too low, but getting closer. Here comes the third person's guess, and it's... yep, \$901. The last contestant has an advantage because they can simply guess \$1 higher than the highest previous guess. They get all the benefit of the second person's guess, but now contestant 2 can't possibly win.

A few things about this game. First, it's zero-sum. There is one winner and two losers. When there is uncertainty in a game, going last allows you to decide with a little more information than the first person had available to them. The first mover, roughly speaking, has no idea what range to set, how high to guess, or even the relative value of a Jet Ski. The last mover now has a clear range, and he knows exactly how to beat at least one of the other players.

Business strategists, especially in innovation and marketing, have explored the issue of first- vs. second-mover advantages for decades. The consensus: a first-mover may reap initial benefits, but they will also lead in flaws and mistakes. Second movers, or any subsequent followers, have a chance to see these mistakes and improve on the product before launch. As an iterative, sequential game, it is often the second mover who ends up with most market share.

Second Mover Advantage: The Joke

Two people walking through the woods are suddenly set upon by an angry bear. As they try to run away one of them says “we’ll never be able to run faster than that bear!” The other responds “I don’t have to run faster than the bear. I only have to run faster than you!”

Home Field: A First-Mover Advantage?

There is a well-accepted perception that playing a team sport in one’s home stadium gives the home team an advantage. Baseball is zero-sum, and it has a sequential aspect to it; the visiting team bats first (has the first opportunity to score), and the home team plays the field. The teams reverse roles after the batting team gets three outs. That pattern repeats in every inning for nine innings, or until there’s a clear winner. So, in every inning, the home team knows how many runs the visiting team has scored when they come up to bat.

This structure gives the home team an advantage; they always know how many runs they need to score to tie the game, take the lead, or win. They can devise a strategy with more information than the visitors.

Home field advantage isn’t just the ability to observe and react in the game. It also includes human factors: players can sleep in their own beds, eat in their own kitchens, and use their own locker rooms. Fans in the stands are more supportive—perhaps quieter—when the home team is trying to score (and they can be disruptive when the visiting team is trying to score).

Home field advantage has been shown, statistically, to be a real thing. Is it game theory? After all, sports are supposed to be outside of the traditional game structure, right? Well, some games—including some sports—have a distinct advantage based on location or timing.

We know that our choices and decisions require some thought process; an ability to work through all of the choices, payoffs, and iterations that lead to your best possible outcome. If you're making choices in your home, generally relaxed, you probably make better choices. It's not just that you make more cooperative or better-informed choices with more information, it's that you process things better in a comfortable situation.

When you're in public, at a gathering of professionals who are mostly strangers, or in an art museum, or at a concert, you are, quite literally, a different person. You are subject to peer pressure, noise, discomfort, and other external forces. You make choices that often, in retrospect (or "when you get home") can be seen as incorrect or inconsistent. These choices might not have been optimal, or they didn't fit the model of your typical choice. Peer pressure, in particular, might be thought of as a jolly killer. You find yourself doing things because the people around you are doing them, not because you actually want to do them. This is best thought of as a "jollies conflict." The desire to be, or feel, accepted is real, but it's also often in conflict with other sources of happiness. One could imagine that a person makes different, perhaps even better choices when they are able to make their choices on "home turf."

A good example of this difficult situation can be seen when a person goes away—as in, away from their hometown—to college. The short-run desire to feel accepted frequently blurs the long-run consequences of a student's choices. Sports fans think of home field advantage as the impact of home comfort on players. Parents think of home field advantage as a reason to keep kids close to home: things are more familiar and easier to control, with more information and a shorter processing time.

Order Matters

The sequential reality of life, what we might think of as historical realities, make us realize that order matters. Here's a colorful example: The Beatles' first number one hit in the US was "I Want to Hold your Hand," released in 1964. It's a bouncy, catchy, non-threatening song in every way and arguably, it helped to cement rock and roll as a musical form. Jump ahead thirty years to Sir Mix-a-Lot's 1992 hip hop hit "Baby Got Back," and you're tempted to say popular music has come a long way. Granny might have some strong opinions as well.

The bands that came before and pushed the limits of censorship paved the way for today's artists. Question: would it be possible to reverse the order? Imagine the moral outrage, the parental protests, radio stations being shut down, etc. if "Baby Got Back" had come first. As it is, there were protests against some songs and some performers, but "I Want to Hold your Hand" being sung on the Ed Sullivan Show allowed us to, three decades later, hear "Baby Got Back." In sequential games, even esoteric ones, order matters.

The Farm System

All games have an inherent uncertainty in them. You can structure the payoff matrix any way you like—dollars, jollies, time in jail, whatever—but you can't be 100% certain you'll receive what you should, what the payoff matrix tells you. In the introduction, we said that most of those issues are for another book—a book on probabilities and statistics.

Sequential games give us another way to link uncertainty and game theory. It's easy to tell a child "stay in school" and "don't use drugs," with the justification that they'll be a lot better off in the long run. What if the child has no examples of the long run? What if they don't see a lot of people around them with a happily-ever-after? Long life is something a young child can't fathom, and therefore it may not be a major factor in sequential decisions.

Let's go back to baseball. Baseball, and many other professional sports, use a farm system, where the major league club gets some of its players from lower-level teams—often called the minor leagues. The minor leagues are where players are honing their skills, training for the big time. They grow, thrive, and eventually move up, if asked. While this personal growth is happening, the players aren't paid as much as they will be if/when they become stars. So, making it to the majors is a major achievement. For the major league teams, the process seems a lot like farming.

There's a catch (ba-dum-tiss): major league teams have different financial resources. Clubs located in big cities (ex, New York City and Los Angeles) have a lot of fans and thus a lot of revenue with which to incentivize star players to join. Teams in smaller cities will often lose their star players in what's called free agency, a time when a player can sell their services to the highest bidder.

A team in a small city usually builds its pool of talent by building a good farm system, but they have to decide when to bring their up-and-coming

stars to the major league level. The choice: when to bring a young talent to the big league. The outcomes: winning a championship and building for the future.

Inject some human uncertainty into the problem. A team's decision-maker thinks "if we call up all of our best prospects and try to win today, some of them could get hurt, our strategy will fail, and we will have lost the season in the short and the long run." Another could say "bring up our best prospects today, train them up, and have them ready for the World Series." A third coach could say "bring the prospects in today, but don't use them in the championship." Whatever the approach, there is a connection between the choices today and the long-term success of the team.

Economist John Maynard Keynes once said, "In the long run, we're all dead." He was saying that economic policies whose payback was expected in the *very* long run wouldn't be beneficial, because it was just too far in the future to matter to everyday people. His statement was made in the context of government policy during the Great Depression, but the idea is important for all sequential games.

There is a distinction between repeated games, which could go on forever, and sequential games, which we think of as having a last period. The approach that we used for solving a sequential game assumed an endpoint. Chess, checkers, Tic-Tac-Toe, and other sequential games end. You can say that the end is far in the future, or even uncertain. Still, there is an end—and its presence changes how you play the game.

Unpacking: A Stitch in Time

Granny's advice tended to be short and sweet. It didn't include a lot of explanation and middle ground. Here's a fun one: "a stitch in time saves nine."

Sewing is something that was far more common when Granny learned this expression from her Granny. If you mend something small now, you won't have to mend something big later. A tear in your pants or shirt or dress tends to get bigger over time if you don't mend it.

There's a game in there (Fig. 6.7):

Similar to the expression "never put off until tomorrow something you can do today," "a stitch in time" warns you that little problems usually get bigger. When we try to put real life into formal games, we have to make a few assumptions:

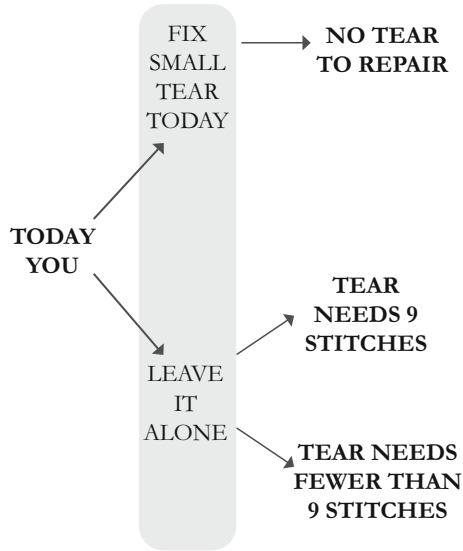


Fig. 6.7 A stitch in time

- The only reference to the future is the number “nine.” The rest we have to infer.
- The only way to measure the magnitude of the downside is to compare the number of stitches—one versus nine.
- How long is “long?” The expression doesn’t give us a way to know how long this process takes. Will the larger tear happen in hours, days, months, or longer?

The quote is consistent with simple, sequential, game-theoretic advice. No one knows the future, but we can connect potential future outcomes with the actions we take today.

By now, you’ve realized that a lot of Granny’s expressions are short and memorable. The “stitch in time” expression leaves out a lot of detail so that it can be easy to remember and easy to repeat. What’s the significance of “nine?” It rhymes. That’s it. Memorable expressions often have a rhyme, cadence, or are catchy in some other way. In that sense, these are usually best thought of as mnemonics; they aren’t meant to be literal.

One more thing: we noted that the outcome of every game, even with your future self, is jollies. Jollies are the happiness you get from a decision. The “Stitch in Time” game assumes that having a much bigger tear in the future costs is bad, and it costs you a lot of jollies. If the person involved in this problem is a fan of tears and distressed clothing, then the stitch doesn’t

really apply. If the person runs a sewing business, they might not like it either. Jollies, preferences, can change over time, making the other player (even your future self) harder to understand.

Moving a Big Ship/Building the Plane While Flying It

Granny might have been a pilot. When we make decisions that are parts of sequential games, we often find ourselves immersed in a process with many steps, many decision-makers, and a great many unknowns. Uncertainty—outcomes that are based on probabilities—is not the focus of this book. In that sense, we are separating game theory from Statistics, since we want to focus on game setup and patterns of choice.

In the real world, though, probabilities of choices weigh heavily in daily decisions. Specifically, we make decisions at a point in time when we have made past choices and will make future choices (what we call, well, “life”). Real games never happen in a vacuum, so we need to be aware of the realities of change over time.

Legally, organizations and their officers can be held liable for decisions that lead to bad outcomes. Not all bad outcomes, though. The legal system has determined that you can hold a firm accountable for things that could have been avoided, or *predictable* bad outcomes.

Consider the literal application of “building the plane while flying it.” If the people “flying it” understand the risks and are paid to take the risks (like test pilots), and the incomplete plane crashes, the company involved probably hasn’t exposed itself to undue liability. If, on the other hand, you are still perfecting the plane when you fill it with passengers, you’ve got a very different situation.

Here’s where game theory and statistics shouldn’t (can’t) be separated. When a player makes choices in a sequential game that he knows could lead to very bad outcomes for others and “hopes for the best,” other players can blame him (even sue him) for exposing them to high risk. Risk is just another word for probability.

Legal systems, to varying degrees, don’t allow people to make unfettered use of uncertainty when making decisions. When something bad happens, you could sue the decision-maker if the uncertainty posed too high a risk. Legally, the affected party has to show actual harm caused by the first player. If an unbuilt plane’s cargo hold opens and millions of dollars in luggage and goods are lost, that harm is easy to see (and quantify, financially).

Some situations aren't so direct. Let's take the expression "moving a big ship" or "righting a big ship." Companies use this to illustrate (to shareholders and employees or other stakeholders) why change is taking such a long time. The implication is that although small changes may be imperceptible, they *should* add up to big changes in the long run. The big ship will course-correct slowly, but effectively.

Because the future is so long run, and because the changes are unbelievably small, it is hard to observe whether the decisions are right or wrong. Some of them may actually be perceived as "the wrong thing to do" in the short run. Because decisions are happening frequently and their results compound, the process of "righting a big ship" is inherently uncertain and risky. In the case of corporate development, it's the reason stakeholders want to see a strategic plan. The plan helps everyone see the final direction of the ship, removing some uncertainty and anxiety around incremental decisions.

In the US, legal codes include regulations that limit a decision-makers ability to cause harm with uncertainty. The obvious reason is to save lives and prevent injuries. The less obvious one is that suing a company, or even a person, for causing harm that they should have been able to avoid assumes that the victims (or their survivors) will get monetary compensation for the harm. Companies and people have an escape clause from paying damages: they can declare bankruptcy. Even without that, the decision-maker that caused the harm might not have enough money to pay for the damages they caused. As with other rules, societies can attempt to control decision-making. Without adequate enforcement, these rules aren't always as effective as people choosing to be responsible.

We'll Cross That Bridge When We Come To It

Chess Grand Masters, the best of the best, are often thought of as some of the smartest people in the world. There are a lot of ways a person can be considered smart. In the case of chess, the perception is that a Grand Master can process a large volume of information—all available decisions on a chess board—at a single time and use that information to chart a path to a win.

Suppose you have a sequential game with ten or twenty or fifty decisions in it. Achieving the rollback solution, winning the game, is unfathomable. I do this, then you do that, then I do this, then you do that... many, many times and with each player trying to out-think or sometimes fool the other player.

Granny wasn't especially good at chess, but she was aware of the daunting task that decision-makers encounter in real-world situations. Complexity can be demoralizing (or inspiring, depending on who you are).

When you told Granny about a complicated, multi-step, sequential problem in which you were focusing too much on a decision that's down the road, she would say, "just cross that bridge when you come to it." She's telling you to focus more on the immediate choices and their outcomes, the short-run decisions. The hope is that payoffs of later games—and therefore their correct choices—become clearer after each step.

In a complicated, chess-like game, you are seeing a path to your desired outcome which has twenty steps or choices over time. One of those, step seventeen, requires you to cross a rickety old bridge that might collapse when you're halfway across it. That fear of Choice 17 might stop you from making Choice 1.

Here's where Granny will really bake your noodle: focusing on a problematic choice seventeen steps down the road is a mistake because you might never have to make it. Somewhere in choices 1–16, the possibility of crossing the rickety bridge could go away. You should start playing the game, make your best choices, and deal with the bridge problem when you're more certain that you have to cross it, IF you have to cross it at all.

Suppose when you start seeing things more clearly, your desired outcome takes you down a path that means you will have to cross that bridge and you are certain the bridge will collapse. Given all of that certainty, you know you won't make it to the end if you start down that path. This is when you follow Plan B; look for the *next* best outcome and find your new path. Reaching the end of the game alive is more important than certain death.

Granny had a term for this too. She said, "that's a fool's errand," or maybe something about snowflakes in hell. The decision-making problem involving the rickety bridge comes down to your level of certainty about your ability to ever make it down that path and come out on the other side.

Conclusion

Life is a sequential game; there...we've said it.

While there are many sequential games involving more than one player, games like poker, chess, and Tic-Tac-Toe, you will play sequential games with your future self more than you play with anyone else. The ability to think ahead is a lot of what Granny was trying to get you to do for your own sake.

Somewhat ominous proverbs like "as the twig is bent, so grows the tree" are warning us that actions today will change our future in ways that we might not like. Sequential games are not simply the outcome of experience,

learning, or other actions. They are forks in the road; situations where one choice changes future choices permanently. In a sequential game, each choice eliminates some possible futures and adds others.

Staying in school and learning how to do math allows you to become an accountant or an engineer. It doesn't eliminate the possibility of becoming a poet. Never learning how to do math (somewhere) does pretty much mean you can't be an accountant, an engineer, and a lot of other things. Game Theorists have attempted to formalize, mathematically, the Sequential Game called Life.

Questions

- 6.1. Sequential games depend on time and order. In that sense, they are different from repeated plays of simultaneous games. We've said that rules (like who goes first) are imposed by players and can usually be changed if it leads to better outcomes. Discuss how making a simultaneous game sequential is like making a commitment in a simultaneous game.
- 6.2. Societal and cultural "norms" are often just rules or approaches to games that were determined once upon a time and have become ingrained in daily life. They are expectations, *modus operandi*, "accepted" ways of doing things. Life is a sequential game. Discuss the "rule-writing" part of a society as a sequential game. Can you show how some injustices (unfavorable outcomes) or progress (favorable outcomes) are the result of who got to have a say first?
- 6.3. Going first in Tic-Tac-Toe is considered an advantage because there are nine possible places to put your X, and the first player will have a chance to occupy five of the nine spaces. This would lead one to believe that anytime the first player has more total choices than the second, there is a first-mover advantage. If you always go first, are you always going to win? Not necessarily... When is having the opportunity to make more choices *not* an advantage?
- 6.4. When a sequential game is thought to have a first-mover advantage, we sometimes randomize the choice of who gets to go first. For example, in the game of (American) football, one team will start the game by kicking the ball to the other team and the "receiving" team will therefore have the first opportunity to score points. This switches in the second half. The decision about which team will kick and which will receive is determined by a coin flip where the winner of that random event gets to decide if they will kick or receive.

- Frequently, the winner of the coin flip decides to kick the ball and defer the opportunity to receive the ball until the second half. Why would that team give up the first-mover advantage of being able to score first? Might there be a different first-mover advantage in the second half?
- 6.5. The rollback approach to decision-making in sequential games is an important differentiator from simultaneous games. You still can't guarantee your best possible outcome; the other player's choice might preclude it. In most cases, you can make choices that avoid the worst possible outcomes when the sequential game is with your future self. Sequential games allow someone to make a Maximin-type decision without having to convert the game into jollies. Given that, could a person make a Maximin-type decision in a simultaneous game that's played with her future self?

Appendix 6.1: The "Marxism Versus Capitalism" Game

Should an outcome be fair? Or large?

In game theory, we classify outcomes as balanced, where every player has a relatively similar payoff, or imbalanced, where one player has a much larger benefit than the other. Consider the game in Fig. 6.8 between two populations in the same country, specified in jollies.

This game has players, choices, and payoffs, but no rules yet.

		A	
		X	Y
B	X	$\frac{A}{2}$ B 1000	$\frac{A}{1}$ B 1
	Y	$\frac{A}{5}$ B 5	$\frac{A}{2000}$ B 3

Fig. 6.8 Two populations, A&B

At a quick glance, the game has two Nash equilibria. If the game is played simultaneously without cooperation and with each player only interested in maximizing their own outcome: B gets 1000 and A gets 2, OR B gets 3 and A gets 2000.

What can we infer from the payoff matrix? For starters, the country is Capitalist. Outcomes in any quadrant are lacking a lot in terms of “fairness,” or what some people call a balanced income distribution. There are very rich people and very poor people. In addition, there is no way to solve this game beyond identifying the two possible Nash equilibria.

Can we impose a rule to guide us to a single outcome? Well, we could let somebody choose first, make the game sequential. Turns out, the game has a first-mover advantage. The group in society that chooses first always ends up better off.

Suppose the prevailing culture says whenever a game with multiple Nash equilibria can't be solved straightforwardly, both players revert to a Maximin strategy. A and B should avoid the choice that could give them their worst outcome. B would avoid X and A would avoid Y. The Maximin solution would lead to the outcome B gets 5 and A gets 3. Let's call this the Marxist outcome. This is the second-best outcome for both players: it is fair and balanced—and also worse than each player's best outcome.

How could we structure the game, make rules, that would allow for something that feels fair, but doesn't give up so much of our potential payoff?

What if the game is sequential, but repeated? The first-choice alternates between the two players. This would allow each player to get their best outcome every *other* play. Over the course of time, several iterations, that would increase the average payoff considerably over the maximin result.

This should lead every society to take this approach, right? Maybe. The win-win result depends on two critical things. First, the alternating first move needs to be immutable and enforceable. Second, the amount of time between plays can't be too long. If the time between each round is, say, a hundred years, the first-mover advantage will return. Human beings aren't incentivized to make decisions that make them worse off in the short run for a long run they might never see.

We leave it to those interested in politics and economics to use this result to explain why long-term income imbalance tends to be the norm in many capitalist countries. The shorthand is, being “first” usually means controlling the means of production (once upon a time, land), and no society has figured out a way to “alternate” that advantage given the human lifespan.



7

I Solemnly Swear...: Promises, Commitments, and Threats

Commitments, promises, and threats feature prominently throughout our life. If you think about it, these three are among the first game theory-related experiences that people can remember. When we're young children, we aren't aware that we are in a game. At some point, the usefulness of these and other tactics becomes evident.

We make commitments, promises, or threats all the time not knowing that these are actually manipulations of games called strategic moves: things that one player does to change the other player's choice, thus changing the outcome of a game.

Game theorists have tried to differentiate these tactics, but the line is a little blurry. When do we call something a promise versus a commitment? Most of the time, a promise has an unstated threat built into it. At the same time, built into any threat is an implicit promise.

It's easy to get lost in the weeds in this area of game theory, and we don't want to miss the forest for the trees. So, let's focus on the big picture, keep our eyes on the prize, and filter out the noise. (Did we miss a saying?).

To understand promises, commitments, and threats, we'll draw on what we've learned about game structure in simultaneous and sequential games and how we operate in each scenario. This chapter covers perhaps the most human of everything in game theory: the ability of a player to change the outcome of a game via communication. Why? Well, the efficacy of promises, commitments, or threats comes down to very human things like reputation, credibility, and who someone perceives a player really is.

Commitments

We usually associate commitments with simultaneous games and promises with sequential games. This often means a lot more to academics than it does to normal people, so let's break it down logically. In the context of decision-making, a commitment is an information we share *before* players make choices. A commitment happens in what we call the “pre-game” of a simultaneous game. It's information, known to both parties, that changes how players choose at the same time.

A commitment is made to push the other player to change their perceived choice. If all goes well, the person making the commitment ends up better off than they would have been without the commitment. Why does this matter? A commitment needs to be credible to affect change. Think of it this way: if a person makes a commitment to something that clearly makes them worse off, you'd second guess them. Why would they do it? A commitment is credible if and only if it changes the outcome of the game so the person making the commitment gets more jollies. It should convince the other player to make a choice that they would not otherwise have made.

Let's look at a game involving two stores: Jane's store, the larger of the two, and John's store. Both stores are trying to determine if they should cut their prices to capture a larger share of the market. While getting more customers is nice, they both would be better off if they both kept their prices (and thus their profit margins) high. Here's the payoff matrix for this game:

If you play the game in Fig. 7.1 simultaneously, without cooperation, and with full information (everyone knows everything), Jane's and John's both cut their prices.

The folks at John's Store will see that if the owner of Jane's Store cuts prices, John's Store will get either \$3 M or \$1 M. \$3 M is better, so John's owner will cut prices. If Jane's owner doesn't cut prices, John's owner will get either \$4 M or \$5 M. \$5 M is better, so the owner of John's Store won't cut prices. The owner of John's Store does not have a dominant strategy.

Jane's owner sees that if John's owner cuts prices, Jane's Store will get either \$30 M or \$25 M. \$30 M is better so Jane's owner cuts prices. If John's owner doesn't cut prices, Jane's owner will get either \$54 M or \$50 M. \$54 M is better so again, the folks at Jane's store will cut prices. Jane's owner *has* a dominant strategy to cut prices.

Jane's has a dominant strategy to cut prices. John's does not. However, with full information, the folks at John's know that the owner of Jane's will cut their store's prices, so John's owner simply has to make the choice that's best in that case. John's too will cut prices.

		JOHN'S STORE	
		CUT PRICES	DON'T CUT PRICES
JANE'S STORE	CUT PRICES	JOHN'S \$3 MILLION JANE'S \$30 MILLION	JOHN'S \$1 MILLION JANE'S \$54 MILLION
	DON'T CUT PRICES	JOHN'S \$4 MILLION JANE'S \$25 MILLION	JOHN'S \$5 MILLION JANE'S \$50 MILLION

Fig. 7.1 A tale of two stores

As with other games, we see that this one ends up somewhere we could call suboptimal: both players could be better off, but the equilibrium didn't land there. In previous games, we've fixed this by allowing somebody to go first (making a game sequential). We could also employ a commitment.

Suppose, before the game is played, the owner of Jane's makes a commitment to not cut their prices. Maybe they call the owner of John's and state it outright, or maybe they imply it at a community meeting somehow. Given the setup of this game, the owner of John's store is effectively playing "follow the leader," so Jane's owner's commitment would make John's owner switch choices; they would not cut prices at John's store. The game ends with neither of store cutting prices and both of us are better off.

Our first criteria for a commitment is met: the commitment in this case is credible. So is the second: the player making the commitment is better off because that strategic move convinced the other player to change their choice. As we mentioned earlier, there are some important details that need to be worked out. For example, there needs to be a way for the owner of Jane's Store to communicate this commitment in an effective way.

Students of business law are ready to jump up at this point. Legally, Jane's store can't announce publicly that they won't cut their prices. In the US, a formal announcement of price setting is considered collusive or anti-trust behavior. So, stores get around it by being more subtle: "we are holding the line on prices." This is an important part of the story, and it leads us to a third criteria: a commitment must also be observable. The point of a commitment is to change the behavior of the other player in a simultaneous game. There

has to be a way to communicate the commitment and observe the intent of the committer.

So, Jane's store commits to hold prices and John changes his choice.

What if Jane's store... bluffs?

Imagine Jane's commitment is credible, and the owner of John's store chooses not to cut prices. The folks at Jane's could renege and cut their store's prices anyway. This could happen in a simultaneous game. Eek.

The outcome for Jane's store is actually better in this case; indeed, they get the best possible result from bluffing the commitment. In fact, using a bluff, Jane's store ends up with profits of \$54 M—the best outcome in the payoff matrix.

To quote the Bard: here's the rub. We said both players have full information, so this possibility—the possibility that the commitment is a bluff—is known to the owners of John's store. Furthermore, if this is a bluff and the folks at John's fall for it, they will get their worst possible outcome. So, the owner of John's store won't believe a commitment unless they have some kind of guarantee. The people at Jane's store will need to do something more than say that they won't cut prices. As Granny said, "actions speak (a lot!) louder than words."

A commitment must be credible, observable, and irreversible. No take-backsies. A commitment must be permanent once made. Problem: this isn't *possible* in a simultaneous game. Any way you slice it, the owner of Jane's store could find a way to convince the owner of John's store that this commitment isn't a bluff, and they could make any choice when the time comes.

One more detail should be mentioned here. Since commitments are made before the game is played, *either* player can make a commitment. In this game, the owner of John's store isn't likely to make a commitment because they know that Jane's has a dominant strategy (a choice that Jane's should make regardless of what the owner of John's does. A commitment on John's owner's part wouldn't change Jane's choice or the outcome of the game because of this dominant strategy. There is no commitment John's could make.

The John's and Jane's game shows how a commitment might change the outcome of a game if only one player has a dominant strategy. It also showed how that player has a clear incentive to bluff. There are many other games in which commitments lead to unambiguously better outcomes. Supposed you had a game with no dominant strategies, for either player, and a mutually beneficial Nash Equilibrium (where no player can make things better unilaterally). In this case, a commitment by either player would easily solve the game to one of the equilibria with no downside risk of bluffing.

What Happens in Vegas, Stays in Vegas?

This catchy advertisement would seem to promote fun and merriment in a city with fewer rules and regulations than most. One could also argue that it wants people to feel less inhibited—and even forget their commitments. If you work for a company that expressly forbids gambling as a condition of employment, you certainly wouldn't organize a Saturday poker game at your home with everyone in your department. On the other hand, if you're at a conference in Vegas, what might the slogan suggest you do?.

Commitments with Threats

When is a commitment not a commitment?

The transfer portal is, roughly speaking, a commitment combined with a threat. I'll "commit" to playing for your team (commitment). If I don't like how things went after any year, I'll go play somewhere else (threat). Signing with a college team used to be a commitment that lasted for your entire college career. Now, with the transfer portal, the commitment is taken 1 year at a time. In game theory, when a commitment is combined with a threat, the threat wins: if they give you an out, a threat supersedes the commitment every time.

A commitment works because it allows the person receiving the commitment to believe that the other person's choice is already made. The receiver can then act accordingly with this new information. How does a threat in a commitment differ from a bluff in a commitment? A threat shortens the lifespan of the commitment. In the example above, what once was a long-term, 4-year commitment is now 1 year at best. The choice has to be re-made each year. Both the player and the college have lost the ability to plan (reliably) for the long term, and the game structure has changed.

Does the Transfer Portal hurt school sports? Let's look at motivation of the players (the players of the game, not just the athletes...): a coach's goal is to develop players. Development takes more than a year. A player's goal *might* be to get playing time quickly, regardless of adequate training, since their commitment horizon is much shorter. The Transfer Portal misaligns each player's jollies and makes a cooperative solution at the end of each season harder to reach.

Threats

On its own, a threat is also a tactic employed before the first round of a game. Where commitments are made *before* a simultaneous game, threats work best in a sequential game. Before a game begins, threats are made by the *second* mover to change the *first* player's choice: "If you choose x, I'll choose y."

Let's look at a bigger example:

While this isn't the most fun game in the book, let's see how a threat can help solve what we call the Mugger Game. Suppose you're walking through the park and stranger walks up to you and demands your iPod Shuffle. Also, it's 2009. Let's call you, owner of said Shuffle, the "muggee." Perhaps to minimize his chance of escaping, the mugger says, "if you scream, I'll hurt you." This is a threat intended to change your behavior. The mugger's idea is simple: you don't want to get hurt, so threaten to hurt and get the Shuffle.

The formal version of this game is shown in Fig. 7.2.

This game, by its nature, is sequential. The "muggee" will choose whether to scream first, and then the mugger responds. And, yes, you can analyze the game with a traditional payoff matrix instead of in tree form, the way we usually show sequential games. The matrix actually has advantages in this case.

In this game, neither player has a dominant strategy, and the game has two Nash equilibria: scream and get hurt (upper left quadrant), or don't scream and don't get hurt (bottom right quadrant). When we discussed games like

		MUGGER	
		HURT	DON'T HURT
MUGGEE	SCREAM	MUGGER 30 MUGGEE -6	MUGGER 10 MUGGEE -4
	DON'T SCREAM	MUGGER 20 MUGGEE -8	MUGGER 50 MUGGEE -3

Fig. 7.2 | I scream, you scream

this earlier, we said there usually needs to be a way to get to one of the equilibria—and the mugger’s threat might be just the ticket. If the threat works, the mugger—the threat issuer—*and* the muggee are better off! Like the commitments that we talked about earlier, a threat is said to be credible if it leads to an outcome that makes the player issuing the threat better off. In this case, it’s a win–win.

Well, not really. Because... you know... felony theft.

We still have things to establish, like how can you know the payoffs of the other player (described in the setup as a stranger) in a game like this. How can you know that the threat is, in fact, credible? If all goes well, the threat gets the game to the best of the Nash equilibria for both players. Not coincidentally, this threat can be issued as a promise, “if you *don’t* scream, I *won’t* hurt you,” with the same result. In most cases, a threat has an implicit promise associated with it and a promise, as we’ll see later in this chapter, has an implicit threat tied to it.

Credibility and Leniency

There’s a lot more to issuing a threat than you might think. Human nature, empathy, and what’s sometimes called a “lack of resolve” can all affect the outcome. The first thing that must be true is that the threat changes the actions of the other player. Second, the threatener needs to be better off if the other player chooses differently. These tests of the effectiveness of a threat are another form of credibility.

Many laws in our society are effectively threats. If you’re caught after robbing a bank, you’re going to jail for 10 years. Losing 10 years of freedom will certainly deter the bank-robbing tendencies of most of the population, so the threatener—generally, society at large—is better off because the money put in the bank will stay safe.

A bank robber might have a compelling story, though... a good reason for undertaking the heist. If that story convinces the judge to be lenient, to lessen the sentence to, say, 5 years, prospective bank robbers everywhere will question the credibility of the threat. Hang on... credibility was established when we saw that the threatener was better off...right?

Here’s where payoffs in jollies get very tricky. Before the game is played, the threat of 10 years would, by our definition, be credible. However, if the robber is caught and his sob story “tugs on the heart strings” of the judge who then changes the punishment to 5 years, we see a problem with agency. The jollies of society (an anonymous, amorphous concept) and the jollies of the threat’s

enforcer (a human judge with a family and a career putting careers in jail) were not aligned.

Every parent knows about this problem. An effective threat requires an understanding of how exacting the punishment will feel on the punisher. In coparent households, for example, you never know who will meter out the punishment in each case. As soon as a kid learns she can get ice cream from Daddy even when Mommy says no, all bets are off.

If You Give an Inch, They'll Take a Mile

Threats work when they are credible, unambiguous, and, to a large extent, immutable. If a threat is really going to change the other player's choice, possibly leading to a better long-run outcome for both of you, Granny says: "don't give an inch."

You tell junior to turn off the computer and do his homework or else you'll take away his computer time completely. He might well test the voracity of that threat with a small pushback: "Can I just finish this one game?" or "Just five more minutes, I'm almost done." What Granny warned you was, "If you give an inch, they'll take a mile." Not enforcing a threat even a little bit shows the other player that you aren't (entirely) credible.

Threats, like rules, have exceptions. The cautionary expression about inches and miles is really about threats and credibility. Credibility isn't always lost all at once. Sometimes, it's chipped away over time. Little Junior is surprised when he gets five extra minutes the first time; but after awhile, he won't even take off his headphones to hear your threat.

Threats, Information, and Prejudice

Let's do the Mugger game again (because it was so fun and non-polarizing the first time), but this time, you're prejudiced. You have a negative connotation to people who wear jeans. Yes, jeans.

Suppose there are two types of muggers in the world: the earlier one, we'll call him Type A, and an eviler one, Type B. Their payoffs are in Fig. 7.3 and 7.4, respectively. You can see that the Type B mugger has a dominant strategy to hurt you, so the threat attempting to get you not to scream isn't really

credible; this mugger is going to hurt you either way. So, if you encounter a Type B mugger, you should scream.

There is no way to know which type of mugger you'll encounter. There is no accurate screen. Statistically speaking, let's say 99% of muggers are the friendly, nice Type A.

		MUGGER	
		HURT	DON'T HURT
MUGGEE	SCREAM	MUGGER 30 MUGGEE -6	MUGGER 10 MUGGEE -4
	DON'T SCREAM	MUGGER 20 MUGGEE -8	MUGGER 50 MUGGEE -3

Fig. 7.3 A "Type A" mugger

		MUGGER	
		HURT	DON'T HURT
MUGGEE	SCREAM	MUGGER 30 MUGGEE -6	MUGGER 10 MUGGEE -4
	DON'T SCREAM	MUGGER 60 MUGGEE -8	MUGGER 50 MUGGEE -3

Fig. 7.4 A "Type B" mugger

Unfortunately for you, you believe that any mugger that wears blue jeans is a Type B. To you, jeans signal somebody mean, cruel, and an overall worse human being. Half of all muggers wear blue jeans. So, your prejudice will make you scream half the time and lower your jollies, for no good reason. Based on reality—there is no good screen for a Type B mugger. Since they only comprise 1% of the mugger population, the numbers say you should never scream, and you will almost never get hurt. Your prejudice led you to more negative jollies.

All this said, if you find yourself repeating this game, we should talk.

Be Careful What You Wish For

The credibility of a threat requires both sides to have a lot of information.

Consider the problem of a “high-skilled jerk.”

You’re the boss at a small technology company that has ten employees. All of them are good at what they do, but their outcomes depend critically on their ability to collaborate. The most skilled person in the group is smart, but he’s just not a team player. Actually, it’s even worse than that: he’s a total jerk.

One day, high-functioning jerk walks into your office and says he wants a big raise because he’s the smartest person at the firm. He says he’s gotten an offer from another firm across the country that pays 20% more than he’s currently making. “If I don’t get a 20% raise, I’m quitting and taking that other job,” he threatens.

Threats can be bluffs too. The first part of credibility is easy to see: he would indeed be better off if he got his 20% raise. So, you ask: does he really have the offer? It turns out he does. He shows you an offer letter that’s legitimate.

The part of the threat that he couldn’t know for sure is how *you* determine the company’s jollies. He thought you prioritized smarts, intellectual labor, and individual achievement. Those were his prized factors. However, your company rewards an ability to collaborate and mentor a team. Your values are not aligned, and you probably take some pleasure in using a really great Granny expression: “don’t let the door hit you on your way out.”

Jerk mis-specified the game, and you called his bluff. In the parlance of poker, he guessed wrong at your hand, at what determines your jollies. He assumed you needed him more than you actually do, and that your interests and value propositions were aligned. Now he’s picking up bubble wrap on the way home.

Funnily enough, the night before he came to see you, your (now former) employee's Granny listened to his plan and said, "be careful what you ask for, you might get it." She knew that he should delve a little deeper before issuing a threat he couldn't take back.

Threats and Future Games

Remember how simply asking a question can change a game? Threats can change games too. These implications are subtle and cautionary.

You're in a bad mood 1 day at work and you blurt out "if I don't get my raise, I'm going to quit!" The boss calms you down and the day proceeds. Does your boss think the same of you before the outburst? Maybe. Or, maybe there's new uncertainty around your commitment to the company.

What if you say "if I don't get that promotion, I'm going to sue?" Like many such threats, your boss knows that employees sometimes sue when they feel they were wronged. He wants to keep you, but he's an officer of the firm. Saying "I'm going to sue" changes your relationship with your boss—and your company—forever. In some companies, once you threaten legal action, your boss has to escalate the discussion to HR, legal, and other parties. Your threat has irrevocably changed your game.

This isn't always a bad thing. When you're trying to reign in a spoiled child, a threat can "get her attention" and make some future games work out better. Here's where credibility and the payoff matrix become intertwined. You don't like making your 3-year-old unhappy; that decreases your jollies. So, threatening to take away her favorite toy for a day to get her to go to bed on time seems like a non-credible threat; you just don't want to do it. But, bedtime (and all parenting) is a repeated game. The "payoff" comes on future nights when her behavior is modified, and bedtime becomes less of a problem. As parents, you may choose to live through your discomfort with the punishment in the short run to raise a better child in the long run.

Promises, Promises

Most people think that the difference between a promise and a threat is just the tone of your voice. It's more like they're two sides of the same coin. Game theorists try to differentiate promises and threats more explicitly by focusing on the person *receiving* the strategic move. Threats are trying to *deter*

a person from making a choice. Promises are trying to *compel* the person to do something. The difference can be subtle.

You threaten your teenager by saying “if you don’t clean up your room, you can’t go out with your friends today.” The promise would be “if you clean up your room, you can go out with your friends today.” Tone of voice notwithstanding, we see the mirror image aspect here. When you say “if you don’t clean up your room you can’t go out with your friends,” you are implicitly promising that you can go out with your friends if you do clean up your room.

Similarly, if you promise your 6-year-old, “if you eat your vegetables, I’ll take you to get ice cream,” you are implicitly threatening “if you don’t eat your vegetables, I won’t take you to the get ice cream.” Aha, here’s a fun different: a promise allows the player to *suggest* future outcomes that the player receiving the promise didn’t know about or didn’t think about. A promise can give more information. A promise expands the potential jollies of a game.

Strike While the Iron’s Hot

The advice in the expression “strike while the iron’s hot” is related to metallurgy. Hot iron is much more malleable than cold iron (which... isn’t really malleable at all). So, if you’re trying to change the shape of a piece of iron, as any blacksmith would tell you (if you could find a blacksmith), you heat the iron until its red hot, and then you strike-shape it with your hammer.

In game theory, threats and promises exert influence on decision makers. The level of influence depends on the timing. Whether you’re using commitments, promises, or threats, you have to use them before the other player has made a choice. That doesn’t mean that the game is already over, it means that the other player has decided what they’re going to do. Even if we loosen that a bit—the other player is “pretty sure” what they’re going to do, we can glean some wisdom from the local blacksmith. It’ll be easier to influence the other player’s choice when they haven’t thought much about it. Threats, Promises, Commitments are most effective with the player is malleable. Strike while the iron is hot.

Reputation Redux

Reputation becomes a big part of promises and threats in repeated games. Like the parenting example earlier, if you show an ability to do something that makes you worse off in the short run, you gain credibility and can get better results in the long run.

Putting a sign on the door of your restaurant that says “Restroom is for Customers Only—No Exceptions” might seem harsh. It deters non-customers from asking. If the restaurant owner is, by nature, kind and considerate, she doesn’t like saying “no” to anyone. The problem is, her customers are paying for that restroom, and there’s no way of limiting non-customer use to just a few select people. Signals and strategic moves combine here in an interesting way. The commitment—the harsh sign—isn’t because the restaurant owner is mean. It’s because she is too nice and needs the sign to reinforce the policy.

Game Structure and Strategic Moves

Promises, threats, and commitments are made all the time. Like so many aspects of game theory, they are often misused, overused, and misunderstood. Not all games are structured to allow these strategic moves to work. Sometimes, a strategic move isn’t able to change the outcome of the game. Sometimes, you can’t declare the commitment or make the promise or threat openly in the first place.

When playing a sequential game with your future self, we sometimes say we’re making a commitment to self-improvement (a gym membership, spin class, giving up chocolate, etc....). That sounds great, but by the nature of sequential games, you’re really just making a repeated, sequential choice. Whether you actually improve yourself will happen over time and depend on many more choices. Making the commitment is step 1 in the right direction, but it is not the choice of the game.

Similarly, if the situation that you’re dealing with is one period only and a simultaneous game, promises and threats can’t work at all. Commitments may help guide choices prior to the game being played, but the payoffs of a promise or a threat have to be built into the matrix before it starts. Again, the strategic move in this case is simply another choice.

Dr. Strangelove, Teddy Roosevelt, and the Biggest “To-Do” for a Viable Threat

Question: When is a threat never going to work?

Answer: When no one knows about it.

The point of a threat is to change the actions of the other player by communicating your intention to do something they won't like. Key word: communicating. A threat never works if you keep it to yourself.

In the earlier-referenced classic movie *Dr. Strangelove*, the title character (emphatically) tells the Russian ambassador that the whole point of their Doomsday weapon (an earth-destroying nuclear bomb that will wipe out everyone if Russia is ever attacked by a nuclear strike) is lost if they don't tell anyone about it. The goal was deterrence. Nobody was deterred because, well, they didn't know about it.

Similarly, President Teddy Roosevelt's famous line “speak softly and carry a big stick” might work in a jungle, or a zoo. He was trying to deter international aggression by saying diplomacy should be backed up by military capabilities. That said, you don't want to speak too quietly lest your threat (aka, the stick) not be recognized.

Successfully communicating your desire for the other player to change actions is key. What's less important is the use of specific, verbal communications. Parents and economics professors alike will tell you that a stern glare will work just as well as an actual threat.

All strategic moves—promises, threats, and commitments—need to be communicated if they are to change the outcome of a game. Communicating a commitment means you are telling the other player what your choice is before a simultaneous game starts. Communicating a promise or threat means you are telling the other player about some action you will take depending on their choices in the second round of a sequential game. In all cases, communication is the key.

Bribes

Consider the game in Fig. 7.5.

If both players make their decision simultaneously, without cooperation, acting in their own best interest, and knowing the entire payout matrix, they

		PLAYER 1	
		A	B
PLAYER 2	A	PLAYER 1 11 PLAYER 2 15	PLAYER 1 9 PLAYER 2 21
	B	PLAYER 1 9 PLAYER 2 10	PLAYER 1 10 PLAYER 2 12

Fig. 7.5 Bribe time

will end up with Player 1 getting 11 and Player 2 getting 15. That's because Player 2 has a dominant strategy to choose A and, knowing that, Player 1 (who has no dominant strategy) would choose X.

Now, suppose Player 2 offers to pay Player 1 a bribe of 3 (a promise of sorts) to choose Y instead. Well, Player 2 is trying to get her highest possible outcome, 21. That would require Player 1 to choose Y instead of X. Would the bribe change the outcome?

You can look at the numbers and tell yourself it would; if Player 1 did choose Y and got the bribe, they would get 12 instead of 11 and assuming Player 2 paid the bribe out of the 21, they would end up with 18 instead of 15. It worked, right?

Sorry, trick question. The bribe won't change the game as it's currently structured. No player has an incentive (if they are entirely self-interested) to follow through on the bribe. To see why, ask yourself *when* the bribe would be paid. If it's paid before the game is played, Player 1's best strategy is to stick with X and get a total of 14 ($11 + 3$). If the bribe is to be paid after everyone chooses, why would a self-interested Player 2 ever deliver? Knowing this (remember, full information!), Player 1 won't choose Y if the bribe will be paid after the game is played. The bribe doesn't help us solve the game.

As Granny would say, "if an offer seems too good to be true, it probably is." Granny often worried about what we call the "strings attached" to a gift. In the case of the bribe, the challenge is in the implementation. It looks like it could work, and feels like a bribe *should* work, but the structure of the game makes it hard to get to a great outcome.

Conclusion

We make promises, threats, and commitments all the time. Little did we know that these things are always game-theoretic in nature. Strategic moves are attempts to influence another player in a situation where two or more decision-makers will determine the outcome.

Commitments depend heavily on a lot of information being easily available all of the time. Commitments always operate by moving one player away from a dominant strategy toward a choice that is better for both players. Dominant strategies don't change. The player making the commitment is always better making that choice. That's because commitments work when they convince the other player to change his choice simultaneously.

Among the interesting things that we discovered is that strategic moves, especially threats, sometimes change all future games between those players. A threat reveals information. It tells the other player "where your head is." So, in addition to the direct outcome of the threat, we usually have a change in the information set for all games between these people—including future interactions. Unlike in classroom game theory, commitments, promises, and threats have real, long-term implications for those of us who go through life interacting with others. Strategic moves can alter your reputation dramatically in all walks of life.

Questions

- 7.1. Commitments and information go hand in hand in any real-world situation. A successful commitment includes observability, or monitoring. Information can come from many sources. One of the most important commitments people make is marriage. Explain how a big wedding—one that includes hundreds of friends and family members—is actually an example of a commitment with lots of information.
- 7.2. Breaking a promise doesn't make much sense if the promise really was credible. Credibility means that the player making the promise should be better off. We hear about breaking promises all the time, so there must be something missing from the analysis. One possibility is uncertainty about the future outcomes.

Suppose you promise your child that you will take her to Disneyland for her fourth birthday. She's elated. You buy tickets, make arrangements for the pets, and start a preemptive ibuprofen regimen, but when the

time comes, you have to break your promise because of a family emergency. She's devastated. When you try this again for her fifth birthday, what should you do differently?

- 7.3. Death threats are interesting. Ignoring everything related to the after-life, and assuming the threatener is making you choose between your money or your life, how would a credible death threat dramatically change future payoffs? What is the implicit promise behind this threat that might change your choices?
- 7.4. Suppose a bully says, "give me all of your money or I'll beat you up" and you give him your money. On seeing this, someone says, "you have to stand up for yourself." Aside from the fact that they weren't the one who was going to be beaten up, how is that advice-giver seeing your situation differently? (hint: think about sequential games.)
- 7.5. Based on what we've discussed about game structure, why can't you make promises and threats in simultaneous games under normal circumstances? What about the fulfillment of a promise makes it impossible?



8

Life Imitating Art: Repetition and Reputation

Some common expressions are literally advice on playing a game: “you’re going to get a reputation,” “once a cheater, always a cheater,” and “never say never” are commonplace. We say them almost without thinking of the implication. Namely: life is a series of repeated games, and behaviors today greatly affect your choices and payoffs tomorrow.

The frequency of a game is how often you play it, and this is an important determinant of a game’s outcome. In daily life, you certainly prefer to know if an interaction is only once or repeated. It helps to know how many sets your friends plan on bowling; you might pace yourself better. You might ask your blind date if they’re staying local or moving away, since this will determine how often you interact. You might treat a 1-on-1 with your professor differently if you know you have to take three of his courses in the coming semesters.

Formally, we need to define a few things to see how and when frequency matters in the setup of a game.

As you increase the number of times you repeat a game, your outcomes can change. In fact, change is the basis of the Granny favorite “if at first you don’t succeed...” She was telling you that outcomes might be different the next time, so you shouldn’t give up. What exactly changes game-to-game over time? In this book, we have specified most games as having full information, which means both players know exactly what possible choices and payoffs are. We don’t allow for many unknowns, so there isn’t much more information time could give you, right?

Wrong.

First, repetition allows us—as humans—to develop things like trust. In repeated games with the same individual, we incorporate these considerations into choices and payoffs and develop our feelings about the other player: their reputation. Based on reputation, gameplay can evolve. Players develop a cadence about the game: a historical record of how the other person acts and a new set of strategic actions to benefit both players.

Second, repetition allows players to find a path to cooperative outcomes—a possibility that might never exist in single-play games. If a game is repeated infinitely (what Game Theorists call “at the limit”), repetition can shift a game from fully self-interested to fully cooperative decisions. In turn, this cooperation can lead to higher payoffs for all parties.

Sure, it’s impossible to play a game to infinity (at least, for us mere humans), but the time horizon doesn’t have to be too long to develop a reputation and learn how to interact better with the other player. Let’s use a few examples to show how core social structures, family values, and even religious beliefs evolve as repeated games.

Prisoner’s Dilemma Redux

Can repetition help Butch and Sundance out of their Dilemma? We hope the guys won’t be arrested an infinite number of times, so let’s play the Prisoner’s Dilemma, as we described it in Chapter 2, just twice. Each robber finishes his prison sentence, then they both rob another bank the day they are released from jail. They’re caught and interrogated again. Butch and Sundance are... well... tenacious, and really bad robbers.

Recall the basic setup: full information, two self-interested players, two choices, and a simultaneous game. The payoff matrix is shown in Fig. 8.1.

We showed that both players have a dominant strategy to confess, and the game ends with both players going to jail for 5 years.

To play the game twice, we need to add all possible combinations of choices and punishments together. The easiest way to do this is in a decision tree. Note: a tree can make it look like a sequential game, where one player goes first. It is not. It’s just the most straightforward way to show a multi-play, simultaneous game.

If both players get the same prison sentence, they can meet again after they are released, rob a bank again, and go to jail together again. Note 2: If they are in jail for **different** amounts of time, we need to rethink the structure of the repetition. The easiest way to do that is the player that gets out first

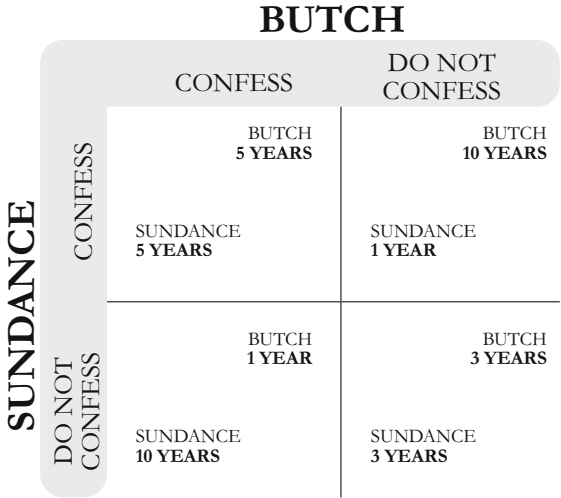


Fig. 8.1 Butch and Sundance are recaptured

will wait for the other one before they play (rob a bank) again. Butch and Sundance are besties.)

If both players are simply trying to get the shortest combined prison sentence, each would confess every time the game is played. Confirm that choice by working backwards through Fig. 8.2. Look at what each player would do in the second choice and work backwards.

Once you have payoffs at the second and first choice, you can combine them and put everything back in a new payoff matrix. Here’s what we mean:

Look at the endpoints of the tree. There are four cases in which Butch confesses in the second game. When he does, Sundance has the following potential payoffs:

C → 10	C → 15	C → 6	C → 8
Sundance	Sundance	Sundance	Sundance
N → 15	N → 20	N → 11	N → 13

In each case, Sundance will get a lower *combined* sentence if he chooses to confess. If Butch does not confess, Sundance’s possible outcomes are:

C → 6	C → 11	C → 2	C → 4
Sundance	Sundance	Sundance	Sundance
N → 8	N → 13	N → 4	N → 6

Once again, Sundance has the incentive to confess in all cases. So, Sundance has a dominant strategy to confess in the second round of the game.

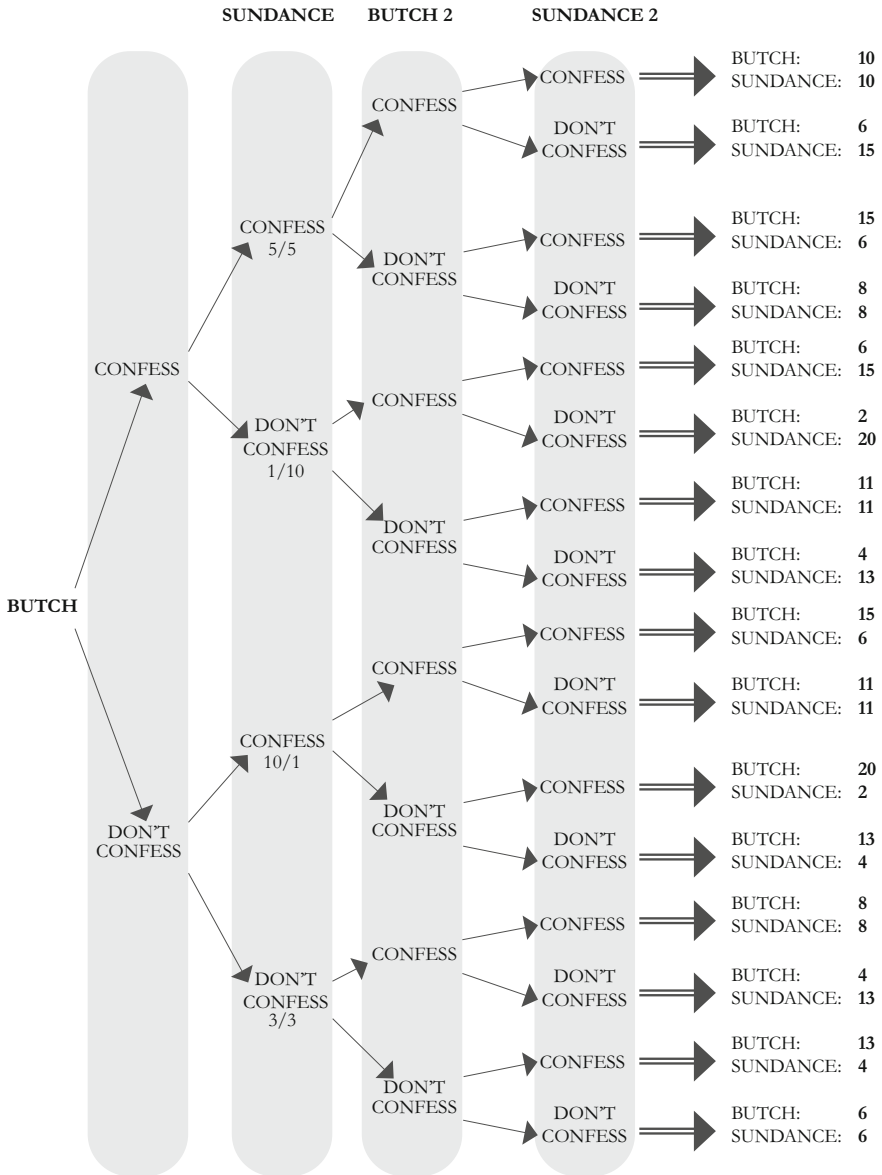


Fig. 8.2 ... Here we go again: Butch and Sundance, sequential

Similarly, there are four cases in which Sundance confesses in the second game. When he does, Butch's possible payoffs are:

C → 10	C → 6	C → 15	C → 8
Butch	Butch	Butch	Butch

(continued)

(continued)

N → 15		N → 11		N → 20		N → 13	
And, Butch's potential payoffs if Sundance does not confess:							
C → 6		C → 2		C → 11		C → 4	
Butch		Butch		Butch		Butch	
N → 8		N → 4		N → 13		N → 6	

In each case, Butch chooses to confess in the second round. He too has a dominant strategy to confess.

Now, work backward. Since Butch and Sundance both have a dominant strategy to confess in the second round, that limits the possible outcomes for the first round. Of the sixteen possible outcomes in round two, only four of them could happen.

After the second game, the only possible outcomes are:

Butch → 10		Butch → 6		Butch → 15		Butch → 8	
Sundance → 10		Sundance → 15		Sundance → 6		Sundance → 8	

Since they are added together, these four outcomes can become a new payoff matrix for the two games (Fig. 8.3):

When we combine the outcomes from both periods, we see once again, that both Sundance and Butch have a dominant strategy to confess. Knowing

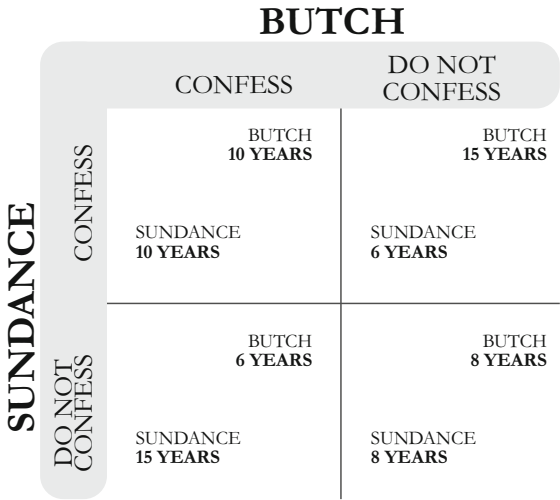


Fig. 8.3 Butch and Sundance, combined sentences

they will confess in the second round means they will **also** both confess in the first round. Playing the game twice with self-interested players did not change anything for Butch and Sundance.

We're sorry to have put you through that, but here's the thing: if the setup doesn't change in any way, playing the game five times or ten or fifteen times wouldn't change the choices or the outcomes. Why? The Last Game Matters: each player always has an incentive to confess in the last period. Working backward, that means they would confess in every period that comes before, no matter how many games they play.

In game theory, we call this a last-period problem: self-interested players are always self-interested in the last period, and that strategy feeds back to earlier games.

No End in Sight

We're all comfortable with the Prisoner's Dilemma at this point. However, if the payoffs are based on years in jail, the game doesn't lend itself to a sensible discussion about repetition.

Let's go back to our two stores, Jane's Store and John's Store. For this new discussion, assume the stores are the same size. Recall, each store is trying to decide if it should cut prices and capture more market share. Both store owners know the payoff matrix (aka, complete information). Figure 8.4 illustrates each store's profits in all possible outcomes.

As with the Prisoner's Dilemma, if you play the game once, the game will end up with both stores getting profits of \$3 M when they *both* could have gotten \$5 M. Not ideal. Let's confirm that and also add some details about the game.

Setup: Jill's and John's choose simultaneously, *without* cooperation. Each store wants to maximize profits.

The folks at John's store see the game like this:

- If the people at Jane's store cut their prices, profits at our store will be either \$3 M or \$1 M. \$3 M is better and that happens if we also cut prices.
- If Jane's owner doesn't cut prices, our profits will be either \$10 M or \$5 M. \$10 M is better, so again, we'll cut prices.

Regardless of what Jane's owner does, John's owner has a dominant strategy to cut prices.

		JOHN'S STORE	
		CUT PRICES	DON'T CUT PRICES
JANE'S STORE	CUT PRICES	JOHN'S \$3 MILLION JANE'S \$30 MILLION	JOHN'S \$1 MILLION JANE'S \$10 MILLION
	DON'T CUT PRICES	JOHN'S \$10 MILLION JANE'S \$1 MILLION	JOHN'S \$5 MILLION JANE'S \$5 MILLION

Fig. 8.4 Jane's & John's play the odds

Spoiler alert: the owner of Jane's store has the exact same numbers in the payoff matrix (just remember to look up and down in the columns), so the folks at Jane's also have a dominant strategy to cut their prices.

If this game is played once, both stores cut prices and they both get profits of \$3 M. If neither of them cut prices, they could have \$5 M each. How can we get there?

Maybe they should arrange a meeting and agree not to cut prices? Unfortunately, in many places, that's illegal. It's called price collusion, and you can see why it's banned: collusion usually keeps prices higher, and that hurts consumers.

Given that reality (same as the Prisoner's Dilemma), the result doesn't change if you play the game two, five, or ten times with this setup. Each store should/would always cut prices in the last period; they have nothing to lose. That choice "feeds back" to every previous round.

What if you never had a last period? That makes a huge difference. Knowing there's no last period means you don't have a last-period problem. The same mechanism that perpetuates the self-interested choice (cut prices), also works for a cooperative outcome. If the incentive to cut prices is *never* there (in a last period), the cooperative choice prevails—forever.

If John's and Jane's play forever, there are only two possible scenarios that could occur every time they review prices: both stores cut forever or both stores don't cut forever. In repeated play, one player may deviate occasionally, but look at the payoffs. With that kind of decline in profits after a single game, neither player would make the same choice more than once. There

may be momentary lapses, but in aggregate, each player would make the cooperative choice every time.

True, we can't play the game to infinity, but we can approximate it. We can play the game a thousand times, and our outcomes would be in billions of dollars. The payoff matrix in Fig. 8.5 is what we call an infinity payoff matrix. It combines "forever" numbers and "one-time" numbers to illustrate a repeated game. Because the only options that could occur every period are both store owners cut prices (forever) or both store owners don't cut (forever), each store can approach the game with those numbers.

The stores can cut prices forever and get \$3B or not cut prices forever and get \$5B. The game now has two Nash Equilibria; if you don't cut (forever); you'll be better off by \$2B. Each store's owner sees if they're ever tempted to be self-interested and cut their prices just to get *one period* of higher profits, they risk losing \$2B forever!

This aggregated, "infinity" payoff matrix shows how the short-run gains from being self-interested are (far) less than the long-run loss. The key to this result is imagining the game will be played forever and seeing any one-time noncooperative behavior as a switch, of sorts. Multiplying the outcomes by a thousand is a way of magnifying the results for either of the only two outcomes that could happen forever. This is a useful abstraction.

As we said, "forever" doesn't have to be forever. Dominant competitors in the industry play this game quarterly, sometimes weekly—and any large number of games can achieve this aggregated outcome. Here's an interesting

		JOHN'S STORE	
		CUT PRICES	DON'T CUT PRICES
JANE'S STORE	CUT PRICES	JOHN'S \$3 <u>BILLION</u> JANE'S \$3 <u>BILLION</u>	JOHN'S \$1 MILLION JANE'S \$10 MILLION
	DON'T CUT PRICES	JOHN'S \$10 MILLION JANE'S \$1 MILLION	JOHN'S \$5 <u>BILLION</u> JANE'S \$5 <u>BILLION</u>

Fig. 8.5 Jane's and John's forever

tidbit of human nature: if we have a game that is perceived as forever (like this one), and one of the players falls on bad times and believes that this will likely be their last period (they're facing bankruptcy, for example), they will cut prices or in self-interest. This is the nature of decision-making when one player feels that they have "nothing to lose." This reality makes people revert to self-interest and reinstates the "last-period" problem.

Similarly, if the payoff for being self-interested even once is a **lot** bigger, say \$2B, then both players will try to cut prices. The cost of cutting prices and benefit of cutting prices would be the same.

So, if the short-run benefits of self-interest are really big, or if there's a change that creates a last period, noncooperation is likely for both players. Fundamentally, cooperative behavior comes from the belief that the short-run benefits of noncooperation are smaller than the long-run costs.

Repeated games have all kinds of tricks for getting people to see this long-run versus short-run problem. One we referenced earlier is "tit-for-tat"; a sort of repeated play version of the golden rule. Each player knows that the other player will mimic his choice in the next period. If you don't cut your prices, neither will they; if you cut, they'll cut too.

Price Cooperation

Legally, stores can't cooperate to fix prices. The setup of Jane's and John's is realistic in that sense: companies need to choose independently and simultaneously. Still, companies the world over collude all the time, since there are other ways to signal understanding of a game and its outcomes. Jane's and John's can't directly say "here's where we're setting prices," but they can put signs in store windows that read "Price Match Guarantee."

This is a form of tit-for-tat known as "price cooperation." It is not literally (or legally) collusion. It is just a signal that tells your competitor something like "don't bother cutting your price; we'll match it, and you won't get any of our customers anyway." Fundamentally, getting to the forever cooperative solution becomes a matter of signaling that you're on board with that plan without actually colluding.

This works really well unless, and until, there is a last period. If one of the stores is in distress, perhaps facing bankruptcy or closure, they give deep discounts, knowing they won't be around to play another day.

What Goes Around Comes Around

If stores play a Prisoner's Dilemma style game forever, they can reach the cooperative outcome. Can price cooperation apply to individuals?

We invoke reputation frequently in this book because it is, fundamentally, a game-theoretic concept. Once you have a reputation—good or bad—it determines how people interact with you; how they play games with you.

Granny warned you at some point, when you were being too self-interested or otherwise doing something that was potentially damaging to your reputation, “what goes around comes around.” She’s reminding you that games are played again and again. People talk; people say things about you. They say you’re a “good egg” or a “bad seed.” Some of these expressions made more sense, as far as cautions are concerned, when we lived our entire life in a small town where everyone knew everything about you. Social media and other modern forms of interaction have connected us in ways we never thought possible—and re-upped the importance of things like reputation.

Cheaters Never Prosper

When you’re trying to convince a child that noncooperative behavior doesn’t work out well in the long run, you can’t just say “cheating never pays off.” In the short run, it can and often does. By the time she reaches an age when lifetime-improving advice begins to resonate, she will have had personal experiences in which a noncooperator (a bully, a teacher on a power trip, her naggy younger brother) got the best outcome by cheating with no consequences. Someone cut in front of the line; someone hogged the best toy in the playroom; someone ate all of the cookies.

You didn’t tell her “cheaters never succeed.” You said: “cheaters never prosper.”

In the John/Jane game, either of them could do better in one period by cheating (cutting their price) but by doing so, they definitely don’t prosper. The cheater loses \$2B in the long run!

Prospering, by definition, happens over time. It’s not a one-time return, like success. It represents repeated successes, or an individual who succeeds more than he fails. Cheaters aren’t better off in the *long* run.

Remember to clearly unpack the expressions you hear. It doesn’t say if any one person cheats, no one prospers. It is focused on the cheater and the cheater’s long-run outcome: Cheaters *never* prosper.

While this view might be more aspirational, the expression implies that cheaters will be noticed and singled out in a world where most people do not cheat. Being rude, being a bully, being a jerk, anything that might be described as an obvious noncooperator will make other players change their behavior when they play games with you over time. Interactions with (games played with) a cheater will have different choices and payoffs as time passes.

“Cheaters never prosper” is short and sweet and as these things go, a very good expression. If an academic game theorist had come up with it the expression would have been, “noncooperative choices will evoke noncooperative choices by other players in all future games and you will get a suboptimal outcome over your lifetime.”

Thank God Granny got there first.

Forever Prisoner's Dilemma Games and Social Structure

Mortality is a “last-period” problem. It’s the same problem that Jane’s or John’s had when they were suddenly in financial distress: they re-instated a last period, stopped making the cooperative choice, and started being self-interested. The short-run gains of self-interest are higher than the long-run costs if you take away forever.

That’s the case for people when they make life choices with a too-short horizon in mind. If you live somewhere that’s crime-ridden or facing other life-threatening realities, you often make choices that have short-run benefits and long-run costs, simply because the long run may never get here. You live in the constant reality of a last period. Put it in game theory terms: you assign a high probability to the short run occurring and a low probability to the long run occurring. Repeated theft of food from a restaurant might put the place out of business, and it’s risky in the long run. But, a hungry person is hungry in the short run. In humanity, short run almost always takes priority.

Ultimately, the problem is time. A human being’s time horizon, how much time they have to repeat a game, affects how they play. In repeated games, cooperative behavior begets cooperative behavior, and returns are better for everyone. So, how do we get people to think in terms of **good** and **forever**?

Family honor.

“Your good name.”

“Think of the Children...”

Many long-term perspectives help fix the problem that mortality creates in decision-making.

One of the most direct tactics to solve the last-period problem: promise people an afterlife.

Most religions have tenets that help society achieve the cooperative outcome every day. With some exceptions, religions typically have three things in common: an omniscient, supreme being; a set of do's and don'ts; and an afterlife. All three can be game-theoretic.

First: do's and don'ts specify cooperative and noncooperative behavior (aka, sins). This helps societies and communities set boundaries for certain types of choices and better specify daily games.

Second: an omniscient observer means you don't need a sign in the window or worry about direct communication; God is always watching.

Third, the afterlife: well... it completely eliminates the last-period problem. The afterlife removes the incentive to make a "bad" choice simply because your time on earth is ending. Bonus points for creating an afterlife that depends on you making good choices before you get there.

It's worth noting that the afterlife isn't always heaven and hell. Reincarnation is another form of perpetuity that allows the spirit or soul to live forever. In religions that believe in reincarnation, how you live one life (sometimes called your Karma) will determine your reincarnated form in the next. It's the presence of an afterlife, and its parameters, that rein in the human instinct to act in self-interest, preserving the cooperative solution indefinitely.

Have you noticed that most "do's and don'ts" in organized religions are rather rigid? These laws aren't presented as ambiguous, nor optional, and they don't allow for a lot of exceptions. Why? Allowing people to find exceptions would defeat the purpose. Take what Granny called a "little white lie." A white lie is a lie perceived as having few—if any—consequences. They're used when lying is perceived as better than telling the truth. Understanding that there is a case to be made for white lies, they are still lies and they can change the outcome of future games. So, rules were written to forbid lying outright, regardless of the circumstances.

The social structure of religion—sins, a supreme being, an afterlife—is meant to move people away from being entirely self-interested and toward a more cooperation. The hope is that society's outcomes are better in the long run, and society prospers as it grows generation after generation. Religions don't exist only to solve games, but they do impact the quality and longevity of the human experience.

Forgiveness and the Prodigal Son

Game theory and social and religious constructs share the study of a key idea: forgiveness. In repeated games, players find that reverting to the noncooperative outcome forever makes them all worse off, and they try to avoid that fate. Sometimes, that means allowing players to make bad choices—be noncooperative—once in a while. Jane's and John's may occasionally stray and cut prices, but they always see the error in outcomes and revert to the cooperative solution next time. To err is human, after all.

Granny said, "give a little grace." That's the idea behind forgiveness. In repeated games, though, frequency of forgiveness matters. The limits of forgiveness depend on the specifics of the game and the particulars of the indiscretion each player is forgiving.

In religion, why is forgiveness a major theme? After all, the rules are rigid and set—why bend them? First: it's not bending the rules; forgiveness happens *after* the game happens. It's not a pre-approval to lie, steal, cheat, or otherwise sin. Second: once a person believes he has broken so many rules he can't possibly reach heaven, society has a last-period problem. If someone doesn't think he can ever achieve a good outcome, he'll revert back to self-interest and do a lot of really bad things. Ultimately, religions are trying to get people to act cooperatively most of the time, with built-in safeguards to avoid the last-period problem. Forgiveness helps manage human error by placing players back on the cooperative path, even after they fall off.

As game theory concepts go, forgiveness is rather fluid. Every player and every repeated game (simultaneous or sequential) uses forgiveness differently. Granny may have said, "forgive and forget." President John F. Kennedy said, "forgive, but remember their names." The implication of the latter can be seen in the "price" of playing games with noncooperative players. If their behavior is truly a one-time indiscretion, your loss is small (in the Price Cut game, smaller one-period profits). If the noncooperative behavior is repeated, it will cost you and the other player much more, since you'll both make the noncooperative choice forever. Kennedy was on to something.

Parents are often tormented by concerns about the correct application of forgiveness. The Biblical approach, as familial as it sounds, is more about the religious family (a larger community) than a nuclear family. Future payoffs are aggregated for the entire group, so rules and punishments are cut and dry. In childrearing, however, the future payoffs are more ambiguous and related to the "wiring" of the child. As parents know, there isn't a concrete rule book for setting a child up for a future of good choices.

In practice, we often combine forgiveness and punishment to maximum effect. One acts to correct the behavior in the short run (punishment), and one acts to change the game in the long run (forgiveness). The combination helps the noncooperative player focus on both short- and long-run strategies, potentially changing their behavior forever.

The Good Lord Works in Mysterious Ways

If there's an occasional bad outcome, how do you keep cooperative players in the fold? We get used to accepting the occasional bad outcome and figuring it won't happen very often, especially when probabilities are concerned. Granny might have had a well-known bumper sticker with a variation of "Stuff Happens." Every once in a while, you make the cooperative choice and another player doesn't, and you get a really bad outcome: your car crashes, you lose an investment, you miss a flight. The rationale we often apply in a religious structure is to say the outcome was part of a grand plan. The supreme being operates in a way that we just don't understand. Some say, "stuff happens;" others say, "the Good Lord works in mysterious ways."

Either expression is better than saying "you can't trust anyone out there, so always look after yourself, the end."

Having a mentality of forgiveness or, at the very least, "forget-ness" when you get the occasional bad outcome will keep you and the societal game from achieving the best long-run outcome. Like it or not, attributing a bad outcome to a supreme being's mysterious plan is a great way to explain a game-theoretic world where it just doesn't turn out right sometimes.

The Devil Made Me Do It

Among the more compelling characters in most religions is the devil. We don't depend on the threat of eternal damnation to dissuade people from noncooperative choices. We associate these choices with powerful, ugly, evil images. We try, quite literally, to scare the hell out of you.

When you teach young children about the devil, you are trying to reinforce good behavior in a person who is probably too young to comprehend the afterlife. Why? To a child, the afterlife is too far away to encourage cooperative behavior. The devil? That's a little more immediate. When it comes to short-run motivation, the devil is to bad behavior what Santa Claus is to good behavior. Not a short-term punishment, nor a long-term promise, the devil

is a personification that provides a medium-term guide toward cooperative solutions.

Cleanliness is next to Godliness

A lot of what Granny was doing was modeling good behavior.

Having taught children about a divine being—the omnipotent, omniscient, entity who controls our fate in the afterlife—the concept applies to anything. Washing your hands and bathing don't seem especially important to children. They tend to focus more on getting dirty and picking up pretty much everything that they see so that they can ask you what it is ten times.

When the parents are around, hand-washing and other forms of hygiene are usually imposed. How do you get your kid to wash their hands *consistently* when they are in kindergarten? Voila, “cleanliness is next to Godliness.”

On Mortality

As we've seen, game theory gives us a new way to understand structures and relationships that exist in the world around us. At the same time, it helps to explain how the complicated interaction between biological and economic realities has developed over time. One such case is the “last-period” human problem: mortality.

Even with an afterlife, there are tangible things that happen as a person approaches death and ultimately dies. For close friends and relatives, few things are as challenging as watching a loved one going through this process. Those two games—the stage of the game being played by the person who is dying, and the ongoing game being played by the survivors—are very different. The person who is dying is approaching the end of their participation in all games (afterlife or not), but the others around him are not.

New, complex games arise for the survivors that again, invoke things like an intergenerational Golden Rule: do unto the dying as I want done to me when I'm on my way out. It's another heuristic that tells us to provide comfort, support, and other personal things to the dying, so that we can be

treated the same when it is our time. Empathy is the emotional, biological version of tit-for-tat.

How much of a person's time and money do relatives spend on a relative's end-of-life issues? How much of the dying person's money does the relative spend, if they know the remaining money will pass on to them, the living? These are decisions that are not straightforward nor easy. If you spend \$1,000,000 of Dad's retirement money on medical expenses that might extend his life for a few weeks or months, what will Mom have left to live on? This game even includes the problem of who should be making those choices.

We have an expression and a lot of literary references for achieving immortality the wrong way. Channeling classic works, we often hear references to making "a deal with the devil." The devil, in most versions of this story, offers us a short-run success (ascending a throne, getting rich, relieving pain) in exchange for our soul—forever in hell. These parables are old because the problem is old. Immediate returns in exchange for something so far in the future we can't even imagine it... seems like an attractive choice. Convincing people to forgo short-run glory because the cost will be eternal damnation is difficult if they don't have a concept of eternity.

Mortality, But Not People

Let's switch the example to something much more palatable: the death of a Fortune 500, multi-billion-dollar, international conglomerate. The company is floundering, people are jumping ship, investors are panicking. What should you, as an employee, do?

For a person, there's a direct way to close up shop when things get too bad: a DNR (do not resuscitate) order. It is a commitment a person makes well before a life-or-death decision needs to be made. If the person is too far to save, a DNR says let them go; the choice is removed from all parties involved.

Unfortunately, there is no DNR for major corporations. There is no commitment to let the company die with dignity. And because of that, the game is often harder to solve.

You know that if you and everyone who believes the company is dying decides to leave, the company will die faster. Before you play "stay or leave," ask yourself this: does staying matter? Put differently, does you staying or leaving have an effect on whether the company survives?

This brings up an important part of the setup that we haven't yet discussed. Game theory is used when you and others make choices that together determine the outcome. Just the existence of other decision-makers does not

mean you are in a Game-Theoretic situation. If there are a lot of others making choices, but no one of you makes an *appreciable* difference in the outcome, then you aren't in a game-theoretic situation. If your decisions—all of you—can't affect change, you're not playing a game.

Now, your choice is: how much of time, money, energy, and emotions (as an employee) should you spend on the dying company? Loyalty may exist in good times, or even challenging times, but finality changes how most people play a game.

Repetition and Reputation

Any game in which reputation can be used is, by definition, a repetitive game. Reputation is an established behavior that reinforces your resolve to make a certain choice. Put another way, it is a screen others use to decide how you might choose, and it is reinforced in each game. An easy way to incorporate reputation is to have a game that does not have dominant strategies and requires other approaches to reach a solution.

Deadlines, Deadlines, Deadlines

We've unpacked a lot of expressions in this book so let's unpack a single word—deadline. We use this word so frequently that we don't usually notice that it combines dead—our last period—with line, something we draw; something we impose.

The “last-period” problem isn't always a problem. It can also be a chance. Whatever you want to call it—a “line in the sand,” a “do-or-die moment”—a deadline incentivizes players to do something they don't want to do.

Imagine a repeated game in which one choice is best or correct, but you don't want to make it. It's not going to feel great, even if it's the best option. If you specify the payoff matrix in jollies, the choice involves a negative number. You could just do it, get it over with, and taking the “loss” of jollies. The alternative is procrastinating, avoiding that choice. You may get a smaller negative numbers of jollies in the first game, but you'll compound that negative every time you play. Over time, procrastination means your negative jollies grow over time. Better to rip the band-aid off now.

A deadline can push you to do the right thing. You are better off making the choice, submitting the document, closing the door, finishing the thing.

Human nature doesn't add well. If you put things off indefinitely, the high negative jollies tend to add up. A deadline stops the pattern, to your benefit.

Sometimes, we impose a deadline on ourselves. It's a "soft" deadline, a suggestion, a goal. For many of us, a soft deadline can become more of a theoretical aspiration pretty quickly. Last periods must be real if they are going to change our behavior. To affect the outcome of a game, a last period has to be real and imminent.

Conclusion

Repetition does a lot more than teach you how to play games throughout life. It can fundamentally change the outcome of games. Pushed to the limit, playing forever, payoffs are now the collection of all plays—the "sum total" of how you've lived your life. We've shown that some combinations simply can't happen every time in repeated play. Because of that, repeating some games can change the outcome for the better over time.

"Forever" isn't real for any mortal decision-maker. So, we find a way to make it real. We are taught about an afterlife, family honor, and other ways that the benefits of short run, self-interested behavior are dwarfed by the long-run costs. The closer you make the difference between short-run gains and long-run losses, the more likely people will be self-interested.

For all the expressions, social structures, and other ways that we try to get people to ignore death, there are plenty of reminders to enjoy life. "Eat drink and be merry, for tomorrow we die," "Gather ye rosebuds while ye may," and "There's no time like the present" are all reminders that life is short and finite. Balancing those two perspectives is left to the individual and the situation.

The occasional indulgence is sometimes considered life-affirming, an acceptable choice of short-run jollies. Constant indulgence, on the other hand, is one of the seven deadly sins—gluttony—which together with its six colleagues (lust, envy, wrath, pride, greed, and sloth) are all self-interested game-theoretic choices that we are taught to avoid.

Questions

- 8.1. Repeated games usually have a "last-period" problem. Arguably, it is basically the same as the problem you face when you play a one-period game with a total stranger. Explain.
- 8.2. Drinking (alcoholic beverages) and driving is problematic at best, deadly at worst. Unlike many drugs, though, alcohol is legal (in the US). Say

somebody gets drunk, drives, and causes an accident. What went wrong? Assume they knew about the worst-case scenario before they started drinking and ultimately driving before they caused the accident.

- 8.3. Why do we often observe lawless behavior in war-torn countries? Or, why do they always show a lot of looting in movies when an asteroid is about to hit earth?
- 8.4. An interesting tweak on the theme of this chapter is repeated games but with different players. Arguably, a company is chartered with the intention of being in business forever. Theoretically, its officers should make cooperative choices that allow it to thrive most of the time into perpetuity. What's wrong with this assessment?
- 8.5. Theologians have argued that hell, the place of eternal damnation, was not originally part of the biblical story (the foundation of many religions). So, there was heaven—eternal life after death in a good place, for the good actors. Otherwise, you just got “death,” game over, end scene. In what way does the *absence* of hell make the game of life more problematic?



9

If Everyone Jumped Off a Cliff: Brinkmanship

When we talked about threats in Chapter 7, we discussed the importance of credibility. The most important determinant of credibility was that the receiver could verify it would make the threatener better off, and then change their choice. The threatener also needs to follow through, which dramatically impacts reputation in repeated games. The perceived creditability of the threatener is a major factor in whether a threat is believable and effective.

Credibility problems can also be with the threats themselves: they can be “too big.” If a parent says to a child, “if you don’t clean your room, you can’t go to Disneyland,” the threat might be effective because it’s proportional, manageable, and likely. If you say “if you don’t clean your room, we’re moving,” it’s not so proportional, and really not likely. That threat won’t change your kid’s choice because it’s just too big. You’re not moving. They know it, you know it, and the room stays messy.

How do you analyze a game when the only available threat is “too big?”—something seemingly disproportional to the game being played. The situation seems unusual, but it happens all the time. You don’t have to threaten to do something catastrophic *immediately*, you might threaten to *start* a process that could lead to a catastrophic outcome. You could set players up for a repeated game of escalation.

Many decades ago, as game theory was finding its formal footing in academia, there was trouble brewing between the US and (then) USSR. Political strategizing and military mind games paved the way for a sub-discipline we call brinkmanship.

(Note: it’s not brinkSmanship. No “s.” Don’t you dare put an “s” in the middle.)

Brinkmanship is a series of strategic moves intended to get what you want by taking a situation to the brink of catastrophe, where both players face unconscionably bad outcomes, and somehow avoiding this disastrous endgame. It involves a last-period problem, threats, promises, and reputations—a delightful mix of factors circling a common question: are we (the players) going to survive this game or not?

Brinkmanship always has one payoff that's really bad. Like, catastrophically bad—and it affects all players. Think nuclear war or a complete economic shutdown. The existence of that potential payoff actually changes how players play. All players know the game could end badly for everyone. They can see it, and it's plausible, so that is an effective and credible threat. Brinkmanship is often associated with Cold War-era events, famously the Cuban Missile Crisis. That international incident is generally considered to be the closest the world has come to nuclear annihilation.

Game theory is fun!

The Cuban Missile Crisis: A Brief Synopsis

In the Fall of 1962, the US government discovered that the Soviet Union was building nuclear missile launch sites in Cuba and arming them with missiles capable of reaching many major US cities in about five minutes. Eek! The geographic proximity gave the Soviet Union a very big first-mover advantage.

Politicians, military generals, and armchair academics in the US argued that that advantage dramatically increased the likelihood that the USSR would launch a preemptive attack on the US, hoping to eliminate the ability of the US to retaliate. For this (and many other reasons), the US demanded the USSR remove the missiles and dismantle the sites.

The Soviet Union refused.

Plot twist: the US also had missiles on the other side of the world capable of inflicting the same damage on the Soviet Union. Double Eek! No wonder they wouldn't remove their missiles. Well, at least arms on both sides should help neutralize the threat, right?

Not exactly. That each country had relatively comparable levels of “first-mover advantage” actually made the catastrophic outcome of the game *worse*. Instead of a first-strike followed by world domination, the game tree included a first-strike, a second-strike, a third strike, and a series of choices that led to total global destruction.

Triple eek.

US leaders had to find a way to get the sites in Cuba dismantled (to them, the only acceptable way to neutralize the major threat) without provoking the USSR to start World War III. Some decision-makers, mostly in the military, advocated for an invasion of Cuba or bombing the sites. Political leaders, including President John F. Kennedy, worried that any such provocative move could start the very nuclear war that they were trying to stop. Armchair academics wrote paper after paper.

The US chose to impose a naval blockade off the coast of Cuba in international waters to block further Soviet military equipment from reaching the nuclear sites. While still risky, a blockade on neutral territory could limit the potential for damage and loss of life. It is what's characterized in the military as a passive strategy. One player puts up barriers to control the other's reaction in a subgame. The other player (in this case, the USSR) would have to make the active choice, putting all responsibility for nuclear war on *them*.

In decisions large and small, we often choose to abdicate to another player. In some games, like poker, we check the bet to the next player as a way to get information about their potential hand. If you're taking your new girlfriend's parents out for dinner, you let her pick the restaurant. Delegated authority in decision-making absolves you of any fault that comes out of the decision. If her parents hate the food, you can always say you didn't choose it.

This is a game employees and managers at firms across the world know well. You and your manager know you're not happy with your job. Both of you have a choice: to leave/stay (for you), and to fire/keep you (for the manager). The manager might keep you even if they want you out. Why? If you choose to leave, it's your decision. The manager gets what he wants without the guilt (or cost) of firing you.

Ultimately, the blockade worked. The USSR agreed to remove the missiles and dismantle sites in Cuba. In exchange, the US removed missile sites in Turkey.

Whew!

Diplomacy helped secure a mutually beneficial outcome—but smooth-talking politicians were not the reason for success. The passive blockade subgame *allowed* diplomatic negotiations to commence in the first place. When the game was simply “who launches first,” diplomacy wasn't even an option.

Brinkmanship is not an ideal situation; we wish we never had to discuss a “nuclear” option in our daily interactions. Yet, it highlights an important aspect of the decisions we make. Recall, we said we always have another option in any game setup: we can choose not to play. Brinkmanship shows us that we also have another option, in certain circumstances: we can change

the game. As Granny may have said, if she listened to conventional political wisdom: “if you don’t like what’s being said about you, change the conversation.”

The Cuban Missile Crisis also revealed a fancy logical conundrum. Having built up massive arsenals of nuclear weapons, would the leaders of either country ever really “push the button” and annihilate millions of people? If they weren’t willing to do it, why build the weapons in the first place? Why spend on an arsenal if it wouldn’t constitute a credible threat.

The Crisis has been studied in game theory for half a century, and most academics agree: the arsenal itself wasn’t the threat. The *building* of the arsenal was the game-theoretic move. It was the first action in a subgame that could have taken the world down a really bad path. Brinkmanship doesn’t start with the nuclear option (in this case, literally). It is a series of strategic baby steps that iterate to a nuclear last period. The arsenal moves each player toward the brink slowly. The last-period result is clear from the start, with the hope a better outcome may be found by players along the way.

Brinkmanship stems from a seemingly... well... weird part of human nature: apparently, we’re willing to entertain the possibility of annihilation just to see if we get something better along the way. We don’t always do this, though. Sometimes, we draw a line and agree to not cross it. When you were being particularly whiny about dinner, Granny would say “don’t start with me.” When we sign a contract for services, we’re limiting how much each party can escalate, thereby protecting everyone. Humans have created ways to manage subgames to ensure we don’t go over the cliff because of every little thing. Maybe we don’t have a death wish as a species after all.

Layers of Players

At the heart of brinkmanship situations are “too-big threats” that have consequences so bad that no responsible person would ever directly choose them. Starting a nuclear war clearly applies. There’s another unique aspect of brinkmanship games: each player can have a different perspective of the game.

Players can come from the highest level (world leaders, corporate executives, heads of labor unions, etc.) and the front lines (factory workers, military personnel in the field, perhaps even everyday citizens). In the Cuban Missile Crisis, decisions were made by political leaders with input from military generals. The situation didn’t just impact those groups, though. History is fraught with imagery of bare store shelves, families huddled together, and school children required to do nuclear war drills (hiding under desks, as many

of us remember). The threat impacted preferences, choices, and payoffs of citizens on both sides, en masse. The Crisis didn't happen in a vacuum. Each decision-maker—on either side—were left to make choices that affected an entire civilization.

When we delegate authority willingly to a counterpart—somebody with the same level of decision-making power to alter a game that we have—we usually seek to absolve ourselves of any fault for the consequences. When citizens delegate to a *higher* authority—such as elected or appointed officials, the assumption is that “delegate” will make decisions in our best interest. Most election campaigns feature a common message: “we will work for you.”

In practice, higher-level decision-makers will act differently, think differently, and react differently than the people they represent—the people on the front line. Sometimes, we describe leaders as being more rational. In a game-theoretic sense, this means their behavior should be more predictable or follow a recognizable set of parameters. Is this correct?

The difference between leaders (corporate, family, or otherwise) and followers (general population, employees, children, etc....) is the level of exposure to the consequences of a choice. Leadership may be responsible for making the decision, but it's the followers who have to deal with the fallout. Leaders act with longer-term outcomes in mind; followers (like soldiers) have to deal with short-term outcomes on the front lines. What do we call the long-term? The Big Picture: a concept that often has clearer, defined outcomes. Yes, a leader operating with the Big Picture in mind should be more straightforward, clear, and rational.

It's a Slippery Slope

In a brinkmanship game, each choice brings players closer to or further from “the brink.” In the case of the Cuban Missile Crisis, the decision was taken out of the hands of the military, whose short-term goal was to neutralize an immediate threat. The short-term gain of military action would have initiated the long-term, catastrophic outcome. We may have won the battle, but would have started World War III.

Granny's phrases like “slippery slope” come from brinkmanship. As a situation unfolds, there's a point beyond which things cannot be stopped. In a strategic series of decisions, subgames, the players decide whether to take steps toward or away from the nuclear option. Blockades and diplomacy were steps in the right direction, bringing us “back from the brink.”

Brinkmanship can be used to examine other types of decisions. The key is finding a game where following through on a threat makes everyone a lot worse off, but making the threat still has value.

Headlines in 2023 warned that a strike by Danish unions could close the entire country's economy. How can this possibly be good for anyone in Denmark? Because so much of Denmark is unionized (about 2/3 of its workers), it is certainly a threat that could be effective if carried out. But, if unions were to follow through, the entire country—from production to commerce to schools to grocery stores—would shut down. Is this threat credible?

Recall the very nature of negotiations. Two parties are creating some kind of joint value and deciding how to divide the final product. A too-big threat is tantamount to saying, “meet my demands, or the joint venture disappears.” Just like the Cuban Missile Crisis, the game starts before the “nuclear option”—the strike—is on the table. In fact, an effective negotiation tactic involves a *hint* of a strike. Danish union leaders never said they were going to strike: the newspapers did. And to the negotiation table we go.

When unions directly go on strike, like the Writers Guild of America (WGA) did recently in the US, there is usually another strategy at play. The strike lasted months, during which negotiations restarted multiple times and ended with the WGA getting the majority of their demands. A strike galvanizes union membership toward a common goal, against a common enemy. However, going over the brink has consequences in repeated games. In this case, the “winning” jollies were considerable, but they may get smaller and smaller in each game. Next year's contract negotiations will look considerably different. The world on the other side of World War III is irrevocably changed.

No Payoffs? What Gives?

We haven't included an actual payoff matrix or game tree in this chapter yet—and we're not going to. Brinkmanship is a subset of game theory (and if you're into math, we can point you to some great books on the topic), but it is not easy to show graphically. Brinkmanship typically involves multiple players with a *series* of choices. We don't need to show numbers to prove there is a wrong—a very wrong/bad/terrible—outcome, and players aren't trying to get there.

Situations that involve brinkmanship are legitimate uses of game theory, but there isn't a useful purpose in simplifying these complex situations enough to make them look like our earlier games. By now, the idea of game theory should be well rooted; the lightbulb should be on. We can see that there are players, choices, and outcomes and we can analyze Brinkmanship game without a formal structure.

Following through on the "too big" threat is something that is made possible in a multi-layered (different players), multi-period game if things go too far. Let's recap the Cuban Missile Crisis and try to see the game without any formal structure.

A tense situation between military superpowers could have escalated into the use of nuclear weapons. One decision, and players are, all of the sudden, too far down that slippery slope to stop the nuclear outcome. Both sides will let the public situation get closer to the bad outcome while more reasonable negotiations proceed.

Negotiations are where information (not all, but some) is exchanged. It's where players start to understand possible outcomes and the other side's preferences. What Granny called a "no win situation" is one where negotiations aren't beneficial, maybe not even possible.

World War III did not happen, but union strikes happen all the time. Brinkmanship is always tricky, occasionally catastrophic, and not driven by any one payoff matrix.

Tiananmen Square, Waco Texas, and Rationality

When brinkmanship does go wrong, we are tempted to say one or both sides were irrational. According to what we've described, the failure to resolve a situation like this leaves everyone worse off. We don't just lose the product of a joint venture; we lose everything.

Recall, though, that irrationality in game theory doesn't mean the other side is *wrong*; it means their jollies aren't specified the way yours are. If a brinkmanship game ends with the worst possible outcome, one side erroneously applied their jollies to the other side. It is a failure to consider other ways of thinking.

In the case of Tiananmen Square, the protestors believed that world opinion would dissuade their local government from using force to end a protest. The government, unfortunately for the protestors, needed to make it

clear that they would never give up control to a comparatively small fraction of an otherwise huge country. To do so would invite insurrection everywhere. Preserving human life maximized jollies for one side, but not for the other. Protesters standing in front of tanks—even with cameras watching—were killed.

The WGA strike in July 2023 effectively shut down all California-based production for movies, television, and online content. One side wanted financial strength, the other wanted longevity. In this case, the financial strength of many production companies, networks, and streaming services was weakened. The pie that these unions want a bigger piece of was made smaller. Although the conflict was resolved, and work resumed, the damage to longevity was done and the entire industry, on which the WGA depends, has shifted.

At the heart of most brinkmanship situations is a layer of decision-makers who have different incentives, desires, or approaches to the ongoing problem. In the Cuban missile crisis, the leaders of the US and the USSR wanted to avoid nuclear war while getting a strategic advantage. The risk that *both* sides saw when the Kennedy administration imposed a blockade on Cuba was that a ship captain *at the blockade* would feel like his life was in danger, fire a shot, and start a conflict. The presence of different players in a single game injects a type of uncertainty information may not be able to help. Kennedy removed this uncertainty by bringing the interests of both sides—the human interest in staying alive—closer together.

The Straw that Broke the Camel's Back

Brinkmanship situations are played out in small scale all of the time—we just rarely call them that. Taking small steps toward a dramatic—usually bad—conclusion is the idea behind the expression “the straw that broke the camel's back.” A single straw weighs nothing, but a critical mass of straws can add up and exceed a camel's capacity.

No one knows which straw will be the back breaker, but everyone knows there will be one. We try to play these games carefully, fully aware that at some point, you're over the brink, and everyone loses. You can push an employee to perform over and over, but eventually, burnout sneaks up and you're hiring again.

For some games, it's not always bad.

Parenting has little brinkmanship moments. Your 3-year-old is acting out (as they are prone to do), and you use a lot of patience, reasoning, and other

good parenting skills. Hopefully, it works. Once in a while, though, their behavior doesn't change. Mama is the broken camel. (Granny says, "oh, now you've done it!"). Toys are taken away, voices are louder, the toddler is crying, and Mom is... well, she's not happy.

Did the payoff actually accomplish anything?

On one hand, your toddler, who is just now developing the skills of strategic decision-making, knows that games can turn out very badly. Threats of "I'll take that toy away if..." might be more effective the next time. On the other hand, you may have emotionally scarred your child. The "nuclear option," as parents put it, is almost never a risk we want to take; the downside is just too severe.

We say "almost" because there are exceptions to every rule. Yelling damages a child, yes. It's probably not a great way to get her to clean up her room. If she's about to run into traffic, though, maybe yelling (and running and scooping her up) is appropriate. In that case, emotional damage isn't the worst outcome possible in the game.

The Tipping Point

Like the straw that breaks the camel's back, the tipping point is a point beyond which a game changes dramatically and often for the worse. The situation and the imagery are different from the ones discussed in brinkmanship, but the results are similar. Rather than a slippery slope where a situation slides irrevocably toward a bad outcome, a tipping point is more like a switch or a shift in weight that moves the situation from one side to another like a playground see-saw.

Tipping points are described in heavy-duty mathematics and physics. A process is stable for many possible data points, but if observations exceed a certain level, the rules change (rules like gravity, inflation, what medication will be effective, etc...). If we were to combine tipping points and brinkmanship, we'd be trying to figure out the point beyond which sliding down the slippery slope cannot be stopped. During the Cuban Missile crisis, both sides recognized that any large-scale military strike would be a tipping point, and they sought to avoid it.

Tipping point is one of several game-theoretic expressions that has become overused and often misused. Not everything that ultimately goes to hell—has a bad outcome—did so after passing a tipping point. One player making the cooperative choice and the other making the self-interested choice in a Prisoner's Dilemma style game gives the cooperator a bad outcome. That's just the result. There's no slow, iterative approach in play. It's just what happened.

Furthermore, the mere presence of a temporal element—sequential games or repetition of simultaneous games—doesn't mean a tipping point exists. In the Nancy Reagan game “Just Say No,” trying drugs allows for the *possibility* of becoming a drug addict, but there's no single point beyond which you are officially addicted. This point depends on a range of human physiological factors. It's why the strategy is recommended: we can't say for sure where the point-of-no-return is, so just don't start.

The key aspect of a tipping point is the dynamics of the decision-making process. Like all of our games and all of the results that we've described, this tipping point outcome depends on the structure of the game.

Avoiding this problem, staying on the correct side of the tipping point, and fixing this problem (getting back on the right side of the tipping point after you've passed it) requires more than just an understanding of tipping point dynamics. It requires an ability to monitor the situation as it's happening. There needs to be a way for decision-makers to see the point coming and understand that reversing the outcome will require more than just taking out the last straw. Once a camel's back is broken, the fix is much harder.

Is This the Hill You Want to Die On?

This Granny saying, like many others, needs a bit of unpacking. At the individual level, dying is the very bad ending; the nuclear war; the breakup; the firing; the worst possible outcome. It is the too-big sacrifice for what you're getting in exchange. The hill, in this case, is the problem or the issue; it's what you're playing the game for. Granny's expression is trying to dissuade you from playing by pointing out the irrelevance of the problem. She might pair this with another Granny classic: “you're making a mountain out of a molehill.”

When you take the first steps down a path that has a slippery slope, perhaps one you don't know about or didn't expect, you might have the opportunity to ask “why does this situation have to cost me my life?” We want our social structure, our laws, rules, and incentives, to be such that there aren't a lot of hills that need that level of sacrifice. Granny is asking you to consider whether the game you're about to play is worth playing, given the possible bad outcome. Not so subtly, she's suggesting you don't even have to play. It's a maximin strategy where the way to avoid the worst payoff is to not play the game.

Based on a True Story

A Fine Arts graduate student is heading home with a \$200,000 painting in her folio. The curator of the museum said she could take it home for the weekend to continue research she was doing on the piece. “Just be very careful with it and bring it back on Monday,” she was told.

The student was exiting a subway car when the folio got caught in the automatic sliding doors. Before she knew what to do, the train started moving. Still trying to extricate the folio, she was flung into a support column in the station which caused her grave bodily damage.

She sued the transit authority for millions of dollars. City lawyers simply said, “You should have let go of the folio.” Her lawyers said, “Why was your subway system designed in such a way that she had to make that difficult, split-second decision?”

Much like a brinkmanship game, people are sometimes put into a situation that quickly begins to spiral out of control. Correct choices are hard to make. Lawsuits like this one are decided by juries who focus on whether the plaintiff is reasonable or unreasonable (not right or wrong); they gauge the *rationality* of the decision at hand. It’s easy for jurors to put themselves in the student’s predicament and to say to themselves, “she shouldn’t have had to (almost) die on that small of a hill.” The student won the lawsuit.

Whistleblowing

You see something illegal—and possibly dangerous!—happening at your company. When you report it to your boss, he tells you to mind your own business. When you insist that this problem could hurt customers or employees, he makes up a reason to fire you. The system, the structure of many companies, allows that to happen.

But you were doing the right thing. Why did that have to be your hill? In cases like this, we don’t want people to think that doing the right thing is going to cost them their job. So, the US Federal Government (and many states) decided it shouldn’t have to be. In 1989, the US government enacted laws that protected whistleblowers from retaliatory punishment. Rules were written to protect good actors from facing what was, to them, a too-big threat. Society decided that doing the right thing shouldn’t be a punishable offense.

If I Have to Stop This Car...

There they were: five kids under 10 years old in a 1965 station wagon. It is a world without smartphones or iPads for a two-hour trip. Mom's up front, under-caFFEinated and weary. Dad's calculating the cost of gas and counting miles. They both just want to reach the destination with as few delays as possible.

Little Fred kicks the back of the driver's seat. "Fred, don't kick the seat."

Sister Jane starts whining, "Fred stole my yo-yo!"

"Fred, give back the yo-yo."

A little later:

"Mom, Julie won't stop chewing gum in my face!"

"My crayon broke!"

"Bobby's not sharing!"

The rearview mirror-monitored mediation continues to very little effect until the Mom says in a calm, clear tone, "If I have to stop this car...."

The consequences are unspecified, but does that matter? The older kids understand well; they've seen this game before. They shush.

Precocious 2-year-old Julianna, on the other hand, says, "well... what IF you stop the car, Mommy?" The others shush Julianna.

Brinkmanship works because the children believe something very bad could happen. Perhaps they have experienced a car being turned around, or toys being taken away, or the promise of ice cream disappearing.

Road trips highlight that the bad outcome doesn't even need to be known explicitly on a payoff matrix for the threat to be effective. Again, the bad outcome isn't the deterrent. Rather, it's starting down the *path* that changes behavior. Brinkmanship can be played effectively without ever harming a soul. The threat just has to be credible enough to convince the players not to go any further down the slope.

Hell and Brinkmanship

Let's talk about hell again!

Recall that religion and a belief in the afterlife help bring us to a cooperative solution. Hell—eternal damnation—is clearly a very big threat. Religions struggle with the question of what kind of behavior warrants eternal damnation. How bad is bad *enough* to end up in hell? If it's seriously horrible stuff, then hell won't be a great deterrent from doing lesser bad stuff. Hell might deter murder, but people still might lie.

That's a problem, theologically speaking.

You could try to convince people that sinning is a slippery slope. Lying leads to mass murder. They might not see this the same way, though. So, hundreds (or thousands, depending on the religion) of years ago, religions made the too-big threat smaller: purgatory. In some religions, it's the waystation for non-egregious sinners on the way to heaven. It's a subgame. Hell deters big bad behaviors, and purgatory deters the smaller ones. Society, overall, is better off.

Brinkmanship and Credibility

A too-big threat is one that is so disproportionate to the action, it just doesn't make sense. There are other reasons, though, that a threat may not be credible. When a harried parent threatens to burn the house down, it's both disproportional and unrealistic. If General George Washington had said to the British during the Revolutionary War, "surrender now or we'll fire a nuclear missile at London!" the British (really, everyone in 1775) would have been really confused.

Like any threat, a too-big threat needs to be credible to work. Kids know Mom's not going to burn down the house, and the British don't even know what nuclear technology is.

A more difficult issue is whether a threatened action has to have been done at some point to be effective. It has been suggested that the US did not have to drop the atomic bomb on Japan; it just had to show them that it existed and was horrible. Similarly, does a labor union have to have, at least once, gone out on strike? Will that help the employer believe that threat? Does the parent have to, at least once, stop the car or raise her voice?

In all cases, the sad fact is, no one knows. A child could have heard another parent yell at their kid and thus know it isn't much fun. The union could show the employer how much money they would lose if the workers went out on strike. The US could have dropped the bomb on an uninhabited island close enough for the Japanese to see it, rather than kill millions. Historians have focused on this last strategy quite a bit, wondering why the US dropped two nuclear bombs, 3 days apart, before Japan finally surrendered.

Conclusion

“Taking something to the brink” is a phrase that people use all the time. Brinkmanship in a game-theoretic setting is most often associated with complex situations featuring many people, or even groups of people (like governments and executive boards). There are layers of decisions and a potentially enormous, often catastrophic outcome. The point of brinkmanship is leaders getting their “people” geared up for some titanic move.

While the discipline originated in studies of nuclear war, Brinkmanship is applicable to many less dramatic situations. Taking a situation to the brink, what some people might remember as “playing chicken,” can have good outcomes as well.

Brinkmanship is an area of game theory that explains the presence of a too-big threat. As humans, we often wish humanity wouldn’t have to get to a certain point. We don’t want nuclear war “on the table” in any way, shape, or form. When a brinkmanship game fails, when the threatened action takes place and we experience the enormous loss, we get the sense that there has to be a better way. Your optimistic authors hope that some improvement in humanity—better approaches to negotiations, more perspective, more information, or a way to remember we are all in this together—could help.

Questions

9.1. Here’s a phrase you’ve probably heard: “cooler heads prevailed.” Short, simple, and to the point, the phrase is saying that all of the decision-makers were invested, but the outcome was determined by unemotional players. While game theorists certainly appreciate that mentality, it might not be entirely correct.

Return to our discussion of high-level leaders vs. frontline followers. Why might somebody on the front lines *not* have a cool head? When are high-level leaders in a better position to decide? When are frontline followers?

9.2. Here’s another relevant expression; “diffuse the situation.” Describe a stick of dynamite or a firecracker from the lighting of the fuse to the end. Does that sound like a brinkmanship situation? If so, why is the only good solution to the situation diffusing it?

9.3. You could flip the brinkmanship situation upside down. Suppose the player with the long-term horizon is *not* the high-level leader, but the

frontline player. What does this mean for the decision-making capabilities or the rationality of the high-level individual? (hint: think about the horizon of an elected official.)

- 9.4. Egomania, megalomania, and narcissism are ways of describing a love of oneself beyond all reason. Self-glorification is not irrational, but it doesn't make for a very effective leader. Why?
- 9.5. In studying brinkmanship, you will hear phrases like "chickening out," "blinking," and "flinching." In all cases, the player that backed down is portrayed as weaker. Consider the game known as Chicken in which the players drive cars toward each other, usually at great speed, and see who swerves first. Aside from "strong," what else would you call a player who never, under any circumstances, swerved? (Answer: eventually, dead!) Given that, what's the point of creating an image of weakness for players who occasionally "blink?"



10

All for One, and One for All: Collective-Action Games

What if the “other player” isn’t one player, but a whole group? game theory principles still apply to situations where you are making choices in a game you’re playing with an amorphous “everyone else.” Granny knew about this problem. You said to her, “you have to let me go to the party, everyone will be there!” She might have responded, “If everyone was jumping off a cliff, would you jump too?”

Early in life, you are warned against mindlessly following the herd, drinking the Kool-Aid, being the lemming that swan dives off the cliff with the others (bringing your species to an end). Until now, we’ve focused on more individual games, where self-interest and cooperation strategies yield the best outcome for you. Group-based games are a bit different, because the goal is to align interests *between and within* each group. This group-think area of game theory is called collective-action games.

Game theory deals with situations where one player makes choices, another player makes choices, and the two combine to determine an outcome. Usually, that means there aren’t very many decision-makers. It may be easy to understand the motivations of one or two co-decision-makers. It’s a little harder with three or four. When there are many, many simultaneous decision-makers, you really have no clue about each person’s preferences. In these games, you could make choice without regard to everyone else, since there isn’t a way to specify the game correctly anyway.

Or is there?

Collective-action games are ones where you explicitly consider the choices of everyone else when making your decision. These games often show up when the outcomes, the payoffs, are societal. It could be something as simple

as building and paying for a city park, a new highway, or a state-of-the-art battleship. As individuals, we feel like our choice doesn't matter. We are teeny, insignificant, too small to matter. If we align with others and form a large enough group, though, we might be able to change the outcome. Anyone who's ever worked on a group project really knows this is easier said than done.

What Collective-Action is NOT: Perfect

Recall that two or more players making choices together determines an outcome. The existence of another player is a requirement for a game. It is not a requirement for strategic thinking. There are situations that include multiple decision-makers at one time, but each can ignore the others' preferences. It happens when there are so many participants that no one player can change the outcome: one student on a campus of thousands, one voter in a national election, one farm of hundreds growing crops nationwide.

Game theory is a subset of Economics too (Economics and Mathematics go hand in hand, probably because they're equally mind-numbing). When a market has so many producers that no single one can affect the price of a good, Economists call it perfect competition. In sunny Southern California, there are at least 100 independent strawberry farms. The decision to increase prices by any one of them is not strategic, it's not a game. Why? The move by one farm does nothing to change the payoff matrix of another. A price increase by one farm simply means it will lose customers (in the classic version of this economic model, it actually loses all of its customers). By definition, perfect competition means no one producer can play a game.

Until these strawberry farms find a way to coordinate their pricing decisions (recall, that's legally problematic), they will have to accept that they have no ability to change their individual farm's price. This is an interesting case where a cooperative decision in their pricing game would actually make them all better off, but alas, price collusion is illegal.

There's a similar concept in Finance (only slightly less mind-numbing than Economics) known as efficient markets, where a single investor cannot make excess profits in actively traded financial markets. Finance professors explain market efficiency in many ways, but it's actually pretty simple: in modern markets, there are so many investors, no one entity is likely to "stumble" onto some "magic" mis-priced stock. Everybody is looking for the pot of gold, so nobody will find it. Granny used to say "you'll never find a one-hundred

dollar bill on the sidewalk.” She knew someone else already found it and picked it up!

Economics, Mathematics, and Finance use tightly controlled models to predict human behavior. These disciplines are allowed to use words like “perfect” and “efficient,” because they control the parameters in mathematical models. Game theory, as an exploration of human behavior, does not. We have many examples of situations where an individual does, in fact, take the choices of the group into account when they make their own decisions. Firms do make profits, people make money in the market, and there is such a thing as a premium brand.

Representative Government

In some cases, collective-action games are ones where people have found ways to remove the irrelevance of single players. Put another way, they have figured out how to row in the same direction to maximize effect. A representative republic is one such system. In the US, we’ve been doing this for so long that we don’t really think about what it means. In theory, one person represents the interest of thousands (or millions), and decision-making gets a whole lot simpler.

Creating this super-entity, the government, by choosing representatives to make decisions for the group gives a small number of people a lot of power. They are supposed to find a way (usually a very un-analytical way) to maximize the jollies of the people they represent. We give these people titles to remind them they work on behalf of thousands of people, not just themselves.

Of the people, by the people, and for the people

Game theory reminds us that government representatives are also human beings who make choices that are imperfect and sometimes overly individualistic. As such, there will always be a push-pull between the jollies of the individual serving and the jollies of the group that is being served.

The democratic ideals that Abraham Lincoln invoked in his brilliant speech, “The Gettysburg Address” are deeper and more subtle than we imagine. In using the word “people” three times in short order, he was certainly highlighting the notion that we elect one of our own when we decide who will govern us.

But nowhere in this ideal does he say that government is superhuman or infallible. Government is comprised of people who have bad hair days and hangovers and egos and get their feelings hurt. Games that involve governments are very much games that involve people and people's jollies. Lest you think that is not true, remind yourself about the governments of Napoleon, Hitler, Mao, and Stalin. Ego is a thing.

Put differently, elected officials, might be empowered to represent their constituents but they also know that their time in office is short relative to the life of the country. Being liked by the voters means pandering to desires of the current population. That's often done at the expense of future generations. After all, those future generations can't vote yet. Short- and Long-term outcomes matter greatly to the motivations of elected officials.

Follow the Leader Redux: Getting Lost in the Crowd

In collective-action games, Granny calls the idea of noncooperative decision-making "getting lost in the crowd." When you don't come to a complete stop at a stop sign or park illegally while waiting to pick someone up, you know you'll probably escape the consequences because there are so many people around. This behavior becomes more problematic if it escalates to a "mob mentality," when people loot stores, riot, and vandalize.

A group provides protection from negative jollies via anonymity. Collective-action games are unique, in that they can motivate the individual in two ways: "I am part of a group, I can make a difference;" or, "I am part of a group, what difference could I possibly make?" Collective-action games can be empowering or induce apathy.

Empowerment requires two things: power and coordination. Power comes from numbers; the more individuals who align with a particular goal, the better. Coordination means everyone is rowing in the same direction. Collective-action games rarely stop there, however. With enough rallying, group-think takes over. This mentality makes players believe their individual interests are inextricably tied to the actions of the group. They get more jollies when their group gets what it wants. We needn't look far to see this in action: people often feel better, get jollies, when the sports team they love wins. After the trophy is awarded, fans will say "we won!" Actually, you did nothing

besides cheer at your TV. The athletes did all the work, but you're part of their "group," you're after the same goal, and you find jollies in their success.

The same mechanism is at work in political elections—for better or worse. You can be politically apathetic (which, statistically, most people in the US are), completely abdicating decision-making jollies to elected officials. Or, you can feel empowered as part of a political party. Sometimes, as trends change and information becomes more available, individuals find themselves misaligned with the choices of the groups they consistently voted for. In recent presidential elections, for example, many voters have stated they are disenfranchised with all available candidates and want something "new."

Cartels

Let's step away from the political grandstanding (get it?) for a moment. There are plenty of business or economic examples of collective-action games too.

In Economics, we learn that a monopolist—a single dominant firm in an industry—has the ability to change market output and thus market price. Monopolists can alter supply by producing fewer units which leads to higher market prices. In most circumstances, monopolies see higher profits than firms in more competitive industries.

A cartel is a group of companies (or countries, in the case of OPEC) that collude with each other to set prices for the market. They communicate directly, so they don't have to suffer lower prices on the open market. For this reason, Economists sometimes call them a "group monopoly".

Forget for a moment that price collusion is illegal in most places. International law is a little fuzzy on this topic.

Suppose there are five equal-sized firms, each producing one thousand pencils a day. Current market demand says, together, the firms can sell five thousand pencils at \$5 each. At this point, in a competitive market, we say this means \$5 is the break-even price: no firm is making a profit. \$5 covers the cost of producing each pencil, and no firm makes excess money above the cost of production.

To make excess profits, the industry needs to reduce output to (say) 4500 pencils in total, across all 5 firms. This would increase the market price to \$12 per pencil, which represents higher revenues and lower costs (because they are decreasing production). This approach would certainly maximize industry profits. So, each firm agrees to produce $1/5$ of 4500—or 900 units—and evenly split the new industry profits.

Does the game end there? Do all firms make profits forever? Nope!

It turns out, while this level of production might represent the maximum profits for the industry and the firms split profits evenly, it does not represent the maximum profits for each firm. Consider Firm #3. It would say: “if I alone increase my output a just little—from 900 to 910—that wouldn’t decrease the market price because I’m such a teeny part of the whole industry, BUT, I can sell ten more units and get slightly higher profits.” Firm 3 increases production this “negligible” amount.

Then firms 4, 5, 1, and 2 follow. They all see the same thing. This is an individual versus group problem. When all five firms produce “a little” more, prices and profits fall for the whole group, and the entire cartel is worse off. Every individual firm has incentives to do things that are good for them and bad for the group.

Most collective-action situations have this problem. Individual incentives are not well-aligned with group outcomes. Notice: when these individual incentives have negative group outcomes, even the individual is worse off in the long run. The individual can make a choice that impacts the entire group by starting a chain reaction. Cartels operate in repeated games.

With a completely different setup, we have effectively recreated the Prisoner’s Dilemma result. When individuals think too much about themselves and choose in a self-interested way, the group—society—is worse off.

Labor Unions

Once upon a time, a remote mountain town’s largest employer was a coal mine operator. Because it was large and the only shop in town, it could take advantage of dominance in the local labor market to keep wages low. “If you don’t take this job at what we’re offering, there’s a dozen other people who will.” With no other choice, this approach kept many laborers poorer than they had to be (and working in terrible conditions). The coal mine operator was a monopoly, and it could set prices (for labor, meaning wages) wherever it wanted.

Then 1 day, someone came along and said: “hey, join a labor union and get better wages for all of you.” Basically, if all qualified mine workers band together, they reclaim power to set wages. They can tell the mine operator what to pay them, or else. They now have what’s called collective bargaining power.

Forming and acting like a coordinated group changes power balances in a game. Coordination games have different distributions of power and information which can change the outcomes. Power shifts are not calculated, though;

we simply recognize that a great number of individuals can affect greater change. Collective bargaining gives many unions the ability to demand more than just a fair wage; they can threaten the ability of a company to survive.

Fair Representation

We are not going to talk about the headline-grabbing election issues that have plagued the US and other countries. Instead, we want to note some interesting, game-theoretic aspects of election procedures. To do this, we have to go way back... to the origins of modern governments.

People owned land and farmed/grew everything they needed. Then, they started to interact. I'll trade you a chicken for a few ears of corn. Humans interacting can cause problems sometimes, and they would deal with those problems like human beings: killing each other, imprisoning people, acting animalistically. Still, societies grew. Skip forward a millennium or so, and we have fiat money: currency, dollar bills, and pieces of paper we use instead of exchanging a chicken for wrapping paper. Things cost money. We also deal with things a *little* better... call the police if somebody breaks into your store.

At this point in evolution, people start demanding things that are good for society: better roads, safe places to gather, ways to make sure food is safe. People elect representatives to make decisions for an entire group because, well, the group is pretty large at this point. They elect a single official (or team officials known as public servants) to do the research and come to a beneficial outcome. People do this so they can focus on their own lives and livelihoods, delegating authority to others to make informed decisions for the greater good.

People expect these things—rules, parks, standards, whatever—to be available to everyone. So, you ask the people: “how much are you willing to pay for these things?” Everyone says: “well, nothing.” Maybe taxes, but even those have an acceptable limit. How much do we want to pay? Nothing.

In a large society, we observe what's called a free-rider problem: individuals want the benefits of government choices without paying for it. Problem is, if no individual pays for it, there's no money and there aren't any government programs.

We solve this problem by electing representatives who know that they speak for many. We give them money, and they—having agreed to abide by the choices and decisions of their electorate—decide where/when to spend that money. Perfect! What could go wrong?

Well, choosing a “representative” representative isn’t that easy. If a place (country, city, municipality... family unit) bases elections on majority rule, then the winner really only represents the jollies of just over half of the electorate. This applies to any referendum that requires a simple majority.

Can we get closer to a more representative coverage of the people? Ranking systems are one alternative some countries prefer.

Suppose there are four candidates for mayor, and you are asked to rank them one through four. Points are given in reverse order: the candidate ranked first gets four points, the candidate ranked second gets three points and so on. The selection process is certainly more representative, but it might result in a leader no one group actually liked the best. Let’s see how that’s possible.

There are four candidates for mayor: Ed, Fred, Pat, and Wimbley. There are also four types of preference rankings in the voting public, in equal numbers. Preferences can come from many different things: skills and capability, personality, charisma, business acumen, height—anything the individual voter cares about.

(a) Preference Ranking #1	Ed, Fred, Pat, Wimbley
(b) Preference Ranking #2	Wimbley, Fred, Pat, Ed
(c) Preference Ranking #3	Pat, Fred, Wimbley, Ed
(d) Preference Ranking #4	Ed, Fred, Wimbley, Pat

You’re running the election. You assign each candidate a number—4 for somebody listed first, 3 when somebody’s listed second, etc., and you add up the numbers. The final results are:

$$\text{Ed} = 4 + 1 + 1 + 4 = 10$$

$$\text{Fred} = 3 + 3 + 3 + 3 = 12$$

$$\text{Pat} = 2 + 2 + 4 + 1 = 9$$

$$\text{Wimbley} = 1 + 4 + 2 + 2 = 9$$

Fred wins! The fact that no voting group actually liked Fred the most is irrelevant! Put differently, Fred doesn’t actually represent anyone the best. He’s everyone’s second choice.

What does this system give a society? Well, in ranked systems like this, candidates learn that they should not reveal things about themselves that would be problematic for any one voter. It encourages more middle-of-the-road, less-polarizing policies that are digestible by more groups in the population. In basic math, more “higher” numbers sum greater because extremely “low” offsets extremely “high.”

Ranking systems have not made their way into the mainstream, as elections go. While they seem to be more “fair,” they assume that the difference in magnitude between the candidates is straightforward—one point for each difference in the rankings. Voters might not want to give any points to some candidates, or they want to give all of their points to another. That’s been tried as well, and it has the same potential flaws. Suppose each voting group had a strong preference for a different candidate. They would each get four points, and no one would win!

There’s also a problem with primary elections. Suppose the system says Ed and Fred will face off in one primary and Pat and Wimbley in another, with the winners of each going head-to-head to determine the overall winner. To make this interesting, suppose these are the voter’s preferences for all possible head-to-head combinations (recalling that these preferences could be based on anything):

- (a) Ed is preferred over Fred
- (b) Ed is preferred over Pat
- (c) Wimbley is preferred over Ed
- (d) Fred is preferred over Pat
- (e) Fred is preferred over Wimbley
- (f) Wimbley is preferred over Pat

Ed will win his primary. So will Wimbley. Then, Wimbley will win the final election. Fred could have beaten either Pat or Wimbley, but he couldn’t (didn’t) win his primary. Elections with primaries are, in effect, two-step, sequential games. Voters who are being strategic need to think about winning the final election (the last period), not the primary. But, history has shown us that’s easier said than done.

Agenda Setting

If someone is in charge of the *order* of head-to-head choices, that person can set the agenda—the timing of choices—to get the final result that they prefer. If Fred faced Pat in the primary, we may have a different outcome.

Let’s do a homey example of this kind of gamesmanship.

Suppose the family—Mom, Dad, Timmy, and Betty—are going to watch a movie together. They’ve narrowed the choices down to four possibilities: Star Trip, Undersea Adventure, Cowboy Times, and Baseball Blues.

Mom is in charge of the final selection, and she knows two things: her personal preference (Undersea Adventure), and the head-to-head preferences for the family. The head-to-head preferences are:

- (a) Star Trip is preferred over Cowboy Time
- (b) Star Trip is preferred over Undersea Adventure
- (c) Undersea Adventure is preferred over Baseball Blues
- (d) Baseball Blues is preferred over Star Trip
- (e) Undersea Adventure is preferred over Cowboy Time
- (f) Cowboy Time is preferred over Baseball Blues

Mom is going to ask the family to make pairwise choices which will eliminate one movie and bring the other on to the final round. Mom is really competitive. She sees that the only thing that could beat Undersea Adventure is Star Trip. So, she sets the first pairwise choice between Baseball Blues and Star Trip. Once she's done that, Undersea Adventure can beat Cowboy Time and Baseball Blues. If she starts with any other pair, she won't get to see her movie (and Mom, like Granny, knows game theory).

Run pairwise eliminations on the combinations above to see how each movie could "win."

Threats Versus Persuasion

We've talked about a lot of things that we encounter in our day-to-day life like rules, laws, and regulations that are similar to threats. In essence, they say "do something the right way or else...."

Consider the ultimate collective-action game problem: getting people to vote. How do we approach *that*?

Some countries have mandatory voting, and it's usually based on a threat: vote, or you don't get your tax return (or another form of government benefit). Others take a gentler approach: peer pressure. They put advertisements on TV and post signs everywhere; volunteers and others go door to door. When they vote, they get a little sticker that says "I Voted." It is intended to be a badge of honor. Subtly, it says "I Voted...Did You?"

Why do some countries use persuasion as opposed to threats to get people to vote?

Arguably, you want people to vote full stop. You want them to *want* to vote. You want them to care. From the long-run version of the Prisoner's Dilemma to many of the issues in collective-action games, our societies

are structured to try to get people to care about something other than themselves—not because they have to, because it’s the right thing to do. These threats and forms of persuasion are intended to shift cultures from self-interested to cooperative outcomes.

The Median-Voter Model: Hot Dog Stands on the Beach

Classroom game theory does have a few absolute classics. The Prisoner’s Dilemma is certainly one. Another is Hot Dog Stands on the Beach. The setup (which is absolutely crucial to the final result) goes like this:

There are two identical, but independent, hot dog stands on a one-mile stretch of scenic, celebrated Newport Beach in sunny Southern California. Beachgoers are equally distributed along the beach. Since they’re humans, and the hot dogs quality, Yelp rating, condiments, and temperature of the surrounding sand are all the same: people will always walk to the nearest stand to get a hot dog. There aren’t any price differences, seating, umbrellas, or other reasons any person would walk to a stand farther away.

After the beach closes at the end of each day, the stands are allowed to move around. The beach is open every day for 3 months of summer. At the end of the first week, where with the stands be along the strip of beach?

Instinctually, un-strategically, you might say the stands will space themselves out: something like Stand 1 will locate at the $\frac{1}{4}$ mile point and Stand 2 will locate at the $\frac{3}{4}$ mile point. Those locations are best for beachgoers, because no one has to walk more than $\frac{1}{4}$ mile to get a hot dog. Figure 10.1 is a good starting point.

At the end of the first day of the week, the two stands will have split the market equally and they will be making the same profits. But are they *individually* maximizing their profits? Nope!

After the stands have both closed, the owner of Stand 1, Benny, starts partying. The owner of Stand 2, Jenny, starts strategizing. Jenny thinks, “given where Stand 1 is located, I can increase my market share and my profits by moving my stand up the beach, more or less right next to Stand 1.” So, during the night, she does moves to Fig. 10.2.

Jenny’s Stand 2 is now getting $\frac{3}{4}$ of the market and making $\frac{3}{4}$ of the profits. Even though he’s been partying, Benny can still see how to respond. While Jenny is out celebrating her strategic success, Benny moves his stand just past Jenny’s and captures most of the market on Day 3, in Fig. 10.3.

DAY 1

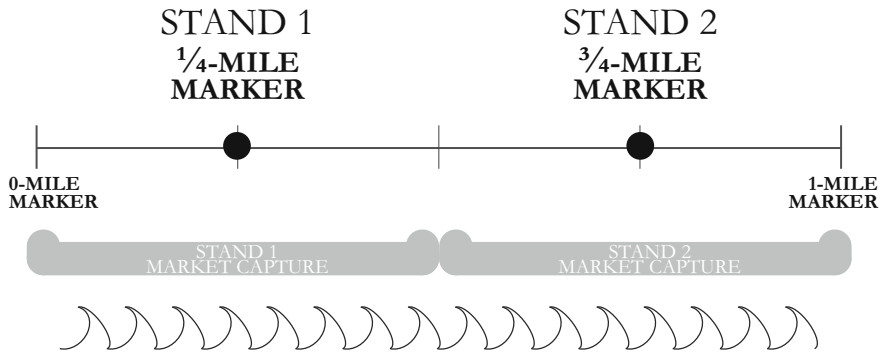


Fig. 10.1 Hot dog stands on a beach, Day 1

DAY 2

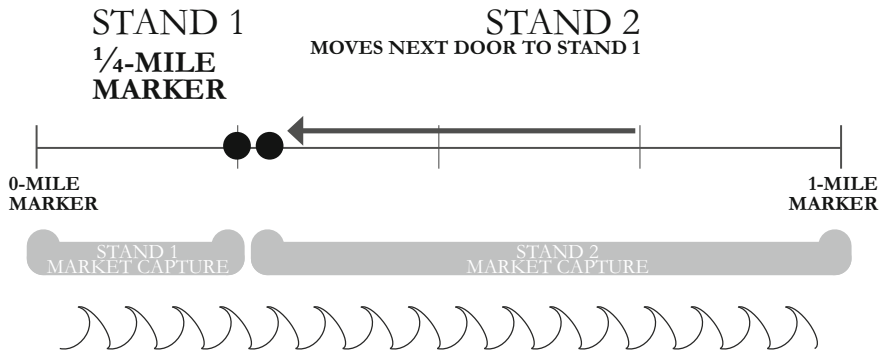


Fig. 10.2 Hot dog stands on a beach, Day 2

DAY 3

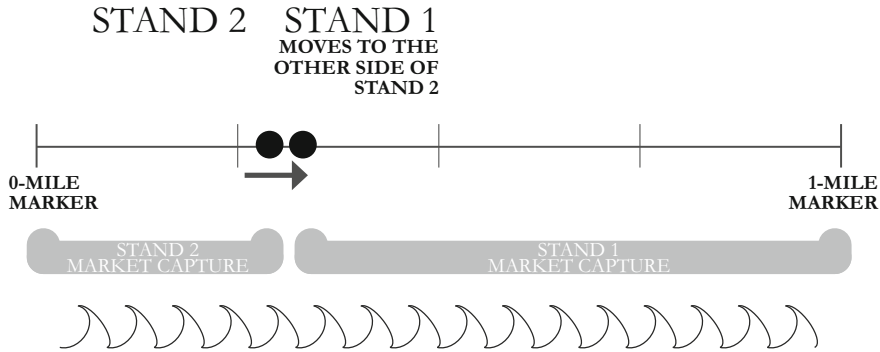


Fig. 10.3 Hot dog stands on a beach, Day 3

FINAL

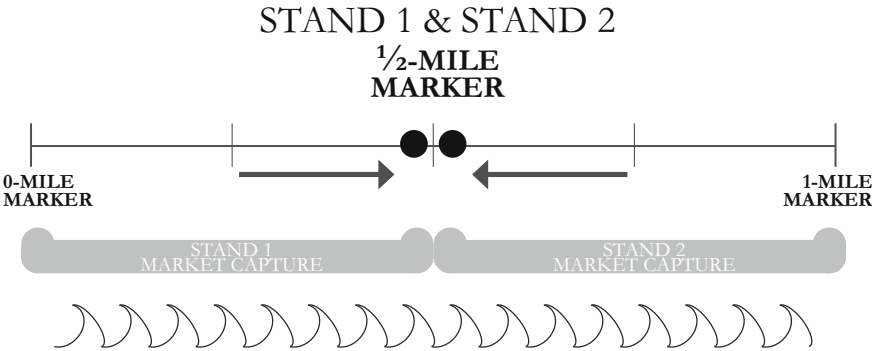


Fig. 10.4 Hot dog stands on a beach, final result

We could do this process for a couple more nights, but we can also jump ahead and ask: is there anywhere they could be located where they would *not* want to move again? (What do we call this? That’s right: a Nash Equilibrium! Whoo hoo!)

The Nash would be right in the middle of the beach. Benny and Jenny both locate at the 1/2-mile marker. Once again, they split the market and have equal profits—and they do *not* have an incentive to move (Fig. 10.4).

Hang on... When we started the game, Jenny and Benny equally split the market and profits. At the end, they also equally split the market and profits. What’s the difference? Well, beachgoers now, on average, have to walk farther to get a hot dog. When the stands were at the 1/4 mile location and the 3/4 mile location, the most anyone had to walk for a hot dog was 1/4 mile. In the final version, some customers have to walk 1/2 mile. That’s unfortunate and to many readers, it probably seems unfair. Game theory doesn’t pass judgment. Given the setup of the game, the final outcome might not be fair for those not playing. We set up the game initially in the interest of beachgoers, but... they’re not the decision-makers. They delegated hot dog-making authority, but the hot dog stands played their own game.

Maybe we can do better, though. Perhaps there’s a surf break or restroom beachgoers congregate around, so they’re not evenly distributed along the beach.... Actually, the hot dog stand result can be “fixed” with something: a regulation. The government can pass a law that requires the stands stay at the 1/4-mile and 3/4-mile markers; the stands are no worse off and customers are better off. If you think this type of regulation is far-fetched, ask yourself why restrooms are evenly spaced at most beaches. If you think the whole game is far-fetched, ask yourself why food courts exist in most malls. Auntie Anne’s and Cinnabon are usually right next to each other.

The hot dog stand problem is a game theory classic, and it explains a lot of things. Countries around the world elect their leaders in many ways. The US, like many, popularly elects its President. In most years, that comes down to a vote between a nominee of the Democratic Party and a nominee of the Republican Party. Historically, when people compare the political platforms of these candidates, they find a lot more similarities than differences. The candidates have, like the hot dog stands, tried to locate in the middle.

This is sometimes called the median-voter model.

The median voter is the voter who falls, statistically, in the middle of the distribution of political preferences. In modern politics, we use the terms liberal and conservative; the electorate on the far right are very conservative, and on the far left very liberal. In the middle is the majority of voters, and we sometimes call them centrist. Winning an election usually requires the candidate who can “win the middle;” that is, sway median voters toward a particular side.

Let’s revisit one other option for the hot dog game: polarization. What if, instead of gravitating toward the middle, the stands went the opposite way: one at the zero-mile marker and one at the 1-mile marker, like Fig. 10.5

If the game setup holds and people go to the stand closest to them, the result would be the same: each stand splits the market and profits equally... on Day 1. On Day 2, Jenny (Stand 2) figures out she can capture much more of the market if she moves just inside Stand 1, to the 0.01-mile marker. Over time, the game iterates to the middle the same way it did before.

Our Modern Times: A New Middle?

Political science pundits have argued that the most recent US elections have been unusual, inconsistent with the median-voter model, because they were more about personalities than political positions. Charisma

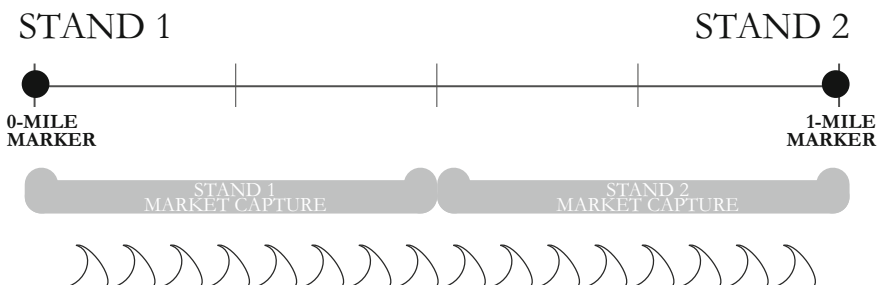


Fig. 10.5 Hot dog stands on a beach, polarized

become more important to voters, to be sure, but there's nothing about the median-voter model that says the middle has to be about the *political* middle. A so-called *populus* candidate, "a person of the people," quite literally appeals to everyone. This person is likely to capture the party and the median voter to secure a win.

When does this approach—the appeal to the median voter—not work? Quite simply: if you don't need them. If voter turnout is low in a two-party system, the winning party only needs to get more voters than the other. Put another way: the median-voter model assumes there are more voters toward the center of the political spectrum. If those with median political preferences don't vote, they are removed from the game, and they don't really matter to the candidate. A polarized candidate could win if her far-right/far-left voters are the only ones who vote.

Third-Party Candidates and Parliamentary Governments

In another example of "be careful what you ask for, you might get it," many frustrated voters in the US blame the mediocre outcome of elections on the lack of choice in a two-party system. In that vein, we frequently hear calls for a third party: specifically, an independent third party that has about the same size and volume of appeal as the other two.

Let's go back to the beach and see how that might turn out. We'll use the same setup (and no regulations about locations), but now there's a third hot dog stand, shown in Fig. 10.6.

Start with each stand spaced equally— $1/6$ mile, $1/2$ mile, and $5/6$ mile points—so each stand gets $1/3$ of the market.

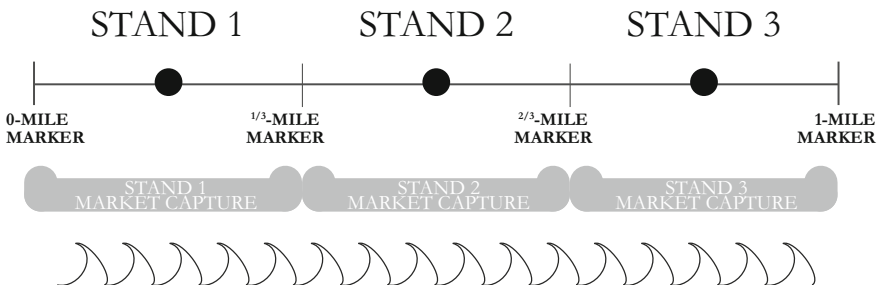


Fig. 10.6 Hot dog stands $\times$ 3

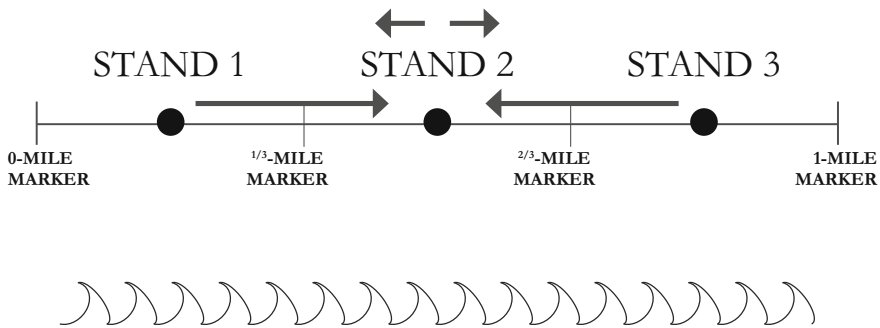


Fig. 10.7 No equilibrium to be found...

When they move around, trying to get additional customers, the two stands on the ends will move toward the center, getting about half of the market. This lasts for exactly one day, after which the stand in the middle moves to one end or the other and re-captures some of the market. Then, stands 1 and 3 follow. Stand 2 returns to the middle, and the process repeats without end. Check out Fig. 10.7 for the dynamics.

For students of game theory, the game shown in Fig. 10.7 has no equilibrium. The stands never stop moving around. In a purely game-theoretic setting, we would say this version of the model is unstable. In a mathematical, classroom game, there is no solution.

How does this translate to political parties? Imagine the spectrum above isn't a pleasant stretch of sand, but political leanings (left-right, liberal-conservative). The winner in a three-party race never has to appeal to anything at all. They would randomly move around between sets of political preferences; each party would switch platforms on political issues from election to election. Why wouldn't they adhere to a set of principles? Their goal is to be elected. In any system where elections happen more frequently than other outcomes (i.e., elections happen every 4 years, but economic policies take six or more years to affect change), you won't see the preferences of society represented well at all. You might see *voter* preferences represented in each election, but candidates will shift based on their whims. In a three-party (or more) system, this flip-flopping is amplified.

Generally speaking, the same results occur when the leader of a country is chosen from the party that wins a majority in the representative body: a parliamentary system. If there are two parties, the government in charge—the majority party—usually has time to fix things and solve problems because they have appealed to about half of the population. If there are many, similarly sized parties (some countries have ten or more such groups), the group in charge must form a coalition to gain any power and enact any change.

This is the situation facing many representative parliaments that stipulate the parliament has to represent every individual faction of society, usually by religion or caste. True, there's more accurate representation of the populous, but there are too many voices in the room for consensus building, policymaking. From a game theory perspective only, this could hinder societal progress.

Aha, we invoked time. Can that help find a solution?

Well, no. Playing the game repeatedly to infinity is assumed in elections, since no society anticipates or plans for its demise.

Here's an interesting thought... We showed in the hot dog game that a three-party system iterates too much for voters. It's unstable because candidate and party leanings change so much. We can't say how *much* these things change. If the shifts cause wide swings in each election, each year, or multiple times a year, this could destabilize the whole of government. If shifts in preferences happen over decades or more—if the iteration is much slower compared to human life—the effect would not be as destabilizing. Indeed, they could simply represent fluctuations in society. Believe it or not, in the US, there was a time when the Republican Party stood for environmental preservation and the Democratic Party wanted to uphold segregation. Times change.

Boards of Directors and Stewardship

Collective-action games and information problems collide on some occasions, putting decision makers in a difficult spot. Consider the predicament that the Board of Directors of an organization faces.

Boards are representatives of shareholders, and they exist to act in their interests. In the case of a corporation, they represent the owners—the stockholders. In not-for-profit organizations, they represent all future beneficiaries of the organization's mission. In either case, Boards must navigate the short-run problems and decisions as *stewards* of the company, maximizing the chance that the organization is passed on to future generations. A Board is playing a long-term, repeated game into infinity. They are elected or appointed officials charged with making decisions on behalf of followers.

All of this becomes tricky when you consider a Board's primary source of information about the organization is its current leaders and employees. Those players are critical to the organization in the short run, but they have a clear short-run bias when it comes to information: employees want to keep their jobs. They aren't likely to reveal bad information

that the Board might need for maintaining the long-term health of the organization, especially if that information threatens their job.

Human nature is embedded into complex sequential games, often with information asymmetry. Boards need to act on reliable info, which the employee has but probably doesn't want to reveal. Boards have to rely on screens, signals, and numbers to inform each decision. If that all sounds hard to figure out, welcome to the world of the Board of Directors.

Whose Team Are You On?

Collective-action problems occur when a player is a member of more than one group at any one time. At work, in your family, and in general society, your decision is combined with the decisions of others to determine the outcome for everyone.

Recall in Chapter 3 we discussed what makes people tick and we argued that the payoff matrix is best specified in jollies. Team leaders often spend as much time (or more!) learning, mitigating, or changing how to measure their team's jollies as they spend developing a "winning strategy."

... and this assumes the leader knows all of the teams that players are playing for.

When talking about a team—any team—it helps to draw on the tenets of collective-action games. Here's an intriguing expression: "whose team are you on?" Managers call "team meetings," and they'll say things like "we're all playing on the same team" or "we need to row in the same direction." All of these cliches are the subject of memes and the invocations of a coach or team leader when some players don't seem to be "giving it their all."

Now, think of all of the things that could cause a manager to mis-specify the payoff matrix of each individual player: they don't have faith in the strategy, but you think they do; they don't have faith in their teammates, but you think they like everybody; they don't have a team mentality, but you put them in charge of sub-teams; they prefer to work alone, but you keep assigning them group work; they stopped drinking caffeine and are really ornery, but you schedule them for 8:30 meetings; they don't see a path for forward advancement, but you think they're in it for the long-haul. "Getting people on board" and other ways of inculcating a team mentality is part of leadership, but success depends on understanding each player's jollies. As

a leader, you might be working out ways to gather the team together not knowing one key member is trying to become a free agent.

A person at work is usually on a team there and a different team at home—their family. A person at work is also, always, a member of their own team—a free agent. Most people have some degree of personal glorification in their payoff matrix. “Being a member of the winning team” is, hopefully, a major motivator, but knowing you are an important member of that team win or lose can be a stronger one.

In professional sports, a player’s ability to change teams—to sell their services to the highest bidder through free-agency—is often viewed as a dollars-and-cents thing (players being motivated by money). Many free agents will tell you it is usually about control. The desire to change teams might come from motivators like respect, independence, mobility, or a degree of personal glory. The decision may result in more money, but it is rooted in an individual’s *ability* to choose for themselves.

In the workplace, free-agent movers might be looking for an organization that better understands the many teams that the player is on in addition (cough*remote work*cough), rather than higher pay. An organization’s responsibility is to consider the skills it needs to grow, the ability of each player, and the player’s payoff matrix. In almost all cases where a team member can choose whether to play for you or someone else, success depends on the ability to understand all of the teams that the player is on.

Team Conflict

When a person’s job regularly requires them to risk her life, she often finds herself having to make decisions that conflict with other teams she is on. For the US general public, the national divorce rate remains at about 50%. The rate for First Responders (police, fire, fire-medical) is closer to 75% (firstrespondersinitiative.org). As much as we wish otherwise, a person who is “dedicated to family” is going to find it difficult to risk her life every day in a risky profession.

As a team leader, you don’t have to be part of every other team that your players are part of to understand that you need to include motivations that exist in your players’ payoff matrix. A team leader might be single and have no children while the players are all married with kids; that works fine as long as the complicated choices and dynamics that affect players with families are things the leader can understand. Collective-action games often overlap and even collide. This is when we

have to remember that all groups are composed of people, and human constraints usually win over other motivators in the long run.

Conclusion

Collective-action games are different to be sure. The individual, in many cases, finds out that his choices and decisions do not change the outcome. Sometimes that means you can “hide among the crowd,” be one member of the mob, the herd, or the throng that’s acting inappropriately, figuring you won’t get caught. Other times, it means you can feel powerless: your vote alone doesn’t count or won’t change the outcome, so why vote?

To mitigate self-interested outcomes, we’ve created societal norms, rules, and laws to dissuade the individual from many of these self-interested beliefs. Rules and laws were created when people saw the advantage of inventing governments to take the place of individual controls. Standards—generally agreed upon—moved societies away from a more lawless, random existence. In the modern world, we aggregate individuals who are themselves too small to have any power into a social superstructure that we imbue with power.

Regardless of how it operates—and regardless of its political philosophy—a government is an invention of people created to solve problems for people, and many of those problems come from game-theoretic situations. Government isn’t a thing unto itself. It is people who work for the government that creates the structure of government, and those people have human, often self-interested, approaches to problem-solving. They are also just people: people who have gas, a hangnail, or their sock sliding down in their shoe. As much as we wish the employees of the government were perfect or superhuman or completely benevolent, they are just people after all.

Questions

- 10.1. There’s a lot of gamesmanship in taxation. At any given time, there are things that the government wants to do or to buy, hopefully because those expenditures make society better off. Governments must pay for things and that usually means we all have to pay taxes. As it turns out, we don’t *all* have to pay taxes; the government can and usually does impose a different tax burden on different citizens for a variety of reasons.

Suppose the citizens of a city elect their representatives democratically and the city is comprised of lower-income people and upper-income people (two groups with no one in between). Representatives always try to lower or eliminate taxes for the people who elected them. What would go wrong if the population was comprised of 80% lower income and 20% upper-income voters? What would happen if the city was 80% upper income and 20% lower income? Given these results, should anyone vote for a candidate who promises to only tax the “other person?”

- 10.2. Party politics are policies that one party supports specifically to annoy (or hurt) the other party (as opposed to simply having different approaches to better society). Assuming we all want a better society, should we vote for candidates who support party policies?
- 10.3. Voting is one of those things that we take for granted in the US. We don't always think hard about what we're voting for as much as who. In current media, personalities rule.

The US also includes a bi-cameral legislature. Fifty states get two senators each, plus a number of house representatives based on state population. The House of Representatives number hovers around 435 members; the Senate has a simple 100 members. One part of the legislature is rooted in equity between states; the other is rooted in equity within each state. Laws need to be passed by both bodies of Congress. So: who should you care about more when you vote: your house representative or your senator? Why?

- 10.4. Consider the Hot Dog Stand on the Beach game as applied to elections and suppose people are distributed along the beach (left, center, and right). If there are candidates from the two major parties (let's call them Left and Right) who, during the national campaign, have gravitated to the political middle and a third-party candidate enters the race, who would win if the candidate was left leaning? If the candidate was right leaning? Given these results, which national party would support which third-party candidate?
- 10.5. Mob mentality is often a justification for bad collective actions, like rioting. It is difficult to be a voice of reason if your voice can't be heard. We've talked about carefully defining the term *irrational*. Focusing on looters: are they actually acting irrationally?



11

Granny's Gold

Granny was really smart. She, along with all the creators of parables, songs, and sayings, gave us a lot of good advice.

In these pages, we tried to formalize and contextualize this timeless wisdom and illustrate just how much game theory there is in the world around us. Game theory is a process, a way of looking at situations—arguably, most decision-making situations that you find yourself in—and systematically figuring out the structure of the interaction. What do you know about the other decision-makers? What makes them tick? Do you know the possible outcomes of the game? Are they best represented by something tangible like money or something intangible like jollies? What are the rules? Who chooses what and when? Are there ways to manipulate the game and its outcomes? Can we use commitments, promises, or threats?

We also looked at the ways human nature influences how we play games. How do you play differently when you're tired, hangry, or in a hurry? How do these very human feelings lead us to inferior outcomes?

Game theory is just a way to formalize what Granny was trying to teach us all those years ago.

Once we open this door, we find many more game-theoretic sayings and lessons in the world around. In this chapter, we'll visit just a few more.

The Good Parent Game

Taking care of your offspring is part of human nature. In younger years, you get jollies from knowing your child's needs are satisfied: they are fed, clothed, clean, and happy. When they grow and get ready to go to college, do you act differently? Is your threshold for maximum jollies the same?

Your kid, once a bundle of non-talkative joy, will soon need to know how to do laundry, cook, clean a bathroom, and manage a mini-household. Suddenly, the jollies of a parent grow from making sure they're fed and warm to knowing your child is ready to take on the world on their own.

"It's not going to clean itself," you might have said, hoping the idea of self-care would magically germinate in your offspring. Inside the kid's mind (but probably not out of their mouth) the response is, "so, you do it!"

You want your kid to miss you when they go away to college. You want them to experience some amount of homesickness; you want them to miss your home cooking. You also don't want your kid to be helpless. A very big part of going away to college is learning how to take care of yourself, and it's your responsibility to make sure they can.

Is this actually a game-theoretic situation? As the expression goes, the proof of the pudding is in the tasting. Can we get better results, understand the problem and the choices better if we approach it like a game? At heart, The Good Parent Game is just the Nancy Reagan game—with a twist:

As is often the case in this book, we've made some outcomes outlandish to make a point. In thinking about this game-theoretically, what good could possibly come from *not* preparing your kid to take care of his needs once he goes off to school? Well, nothing—except the parent's short-run jollies that they get from taking care of the kid will last to perpetuity.

When we follow the decision tree in Fig. 11.1, we see a simplified argument for introducing responsibilities at some point in childhood. Game theory helps us illustrate the choices we, as parents, make every day.

Turn Signals

One of the best examples of a game theory signal is an actual signal: a car's turn signal. On a good day, the device lets you know what the car ahead of you is about to do so that you can plan accordingly. That's the whole idea.

Think about how frustrated (and angry!) you get when someone leaves their turn signal on for an inordinate amount of time. You're driving down a city street or zooming down the freeway and the person in front of you has

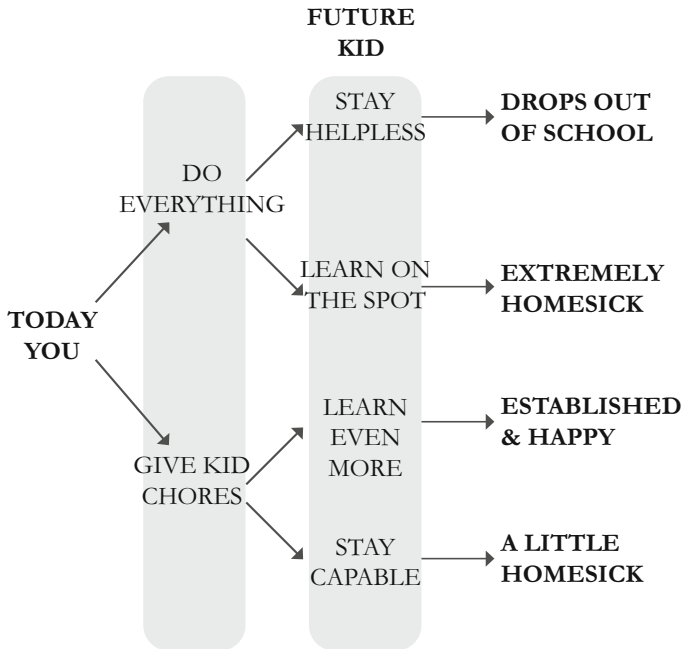


Fig. 11.1 The good parent game

had their left turn signal on for ten miles. You're probably showing your irritation by saying inappropriate things about the other driver's obliviousness. What's the real problem? The person's signal isn't telling you anything! A turn signal signals action, but since you have no idea of their intentions (when and where they're going to turn), it's not helpful information at all.

Do you believe the signal and slow down? Or, do you avoid passing? Do you ignore it, like we said before, and call it a diluted signal? If you have to guess what a signal means, it's unreliable.

Secrets and Lies

It's a fine line, as they say, between keeping a secret and telling a lie. Yet, being revealed as a liar and not being able to keep a secret both get you tagged as unreliable. How can that be? Secrets and lies are both ways to hold on to relevant, accurate information.

Like so many things that we've discussed, you learn the subtleties of secrets and lies through experience, but you can see the difference between the two using game theory. Consider a sequential game in which you are planning

someone's birthday celebration. You buy the presents and invite people to the party today, and you don't tell the birthday girl. You keep it a secret. You are actively trying to give her more jollies (through surprise) on her special day. You often have to lie; "no, I didn't get you anything special and we aren't having a surprise party," because the game is sequential. And your jollies in the future are connected to the jollies that you've been able to create for the birthday girl!

Similarly, you write a recommendation letter for a colleague who is "secretly" looking for a new job. You keep the secret (let's assume you do) because....? Again, there's a sequential game here in which you want to be thought of as *both* a reliable friend and trusted colleague. You might need to ask for a letter as yourself someday in the future, and you can now ask that friend because they know they can trust you. Secrets are just another form of asymmetric information in sequential games that you *don't* reveal because revealing the information would lead to a bad outcome for you in the future.

Lies are also forms of asymmetric information. They can change outcomes in both sequential and simultaneous games. The innocuous "white lie" and the acceptable bluff are intended to change the other players' choices in a way that might even make the other player better off in the long run. Telling a small child a white lie about where grandma had gone (forever) might be better for him than telling him the truth, simply because he's too young to comprehend death. Bluffing is lying to get something for yourself—a raise at work, the pot in a poker game—but both parties know that's part of the game and it doesn't always work.

Less innocuous lies are far *more* problematic. They can change the outcome of games, often in your favor and at the expense of others—but over time, they tag you as untruthful and unreliable. In repeated games, lying will likely make you worse off in the long run. Aesop knew the difference and tried to communicate it in the fable "The Boy Who Cried Wolf." What a child is being taught in this (slightly horrifying) story: if you outright lie for no good reason whatsoever, you might get eaten by a wolf.

When the child who hears this story grows up and says to mommy, "what's the difference between the little boy who cried wolf and you when you lied to me about Santa Claus," what will you tell them? You probably won't use these words, but what you'll tell them is something like "the situation, the game, is different."

The lightbulb moment referred to in the introduction is upon us: game theory is everywhere, even without the specific numbers and formal aspects of a mathematical game. If this book has worked, you may be running through a game theory analysis for simple things like the question above.

Threats and Secrets

In Chapter 7, we illustrated how a threat can change all future games involving the same players. Let's connect the dots a bit. Consider how things change when someone threatens to reveal a secret. This is a powerful strategic move. The person being threatened now knows two things about the other player: they aren't above using threats to get what they want, and they don't keep secrets very well.

Suppose an old college buddy is applying for a job at your company, and you get to make the final choice. He's got the right set of skills for the job, but there are five other equally skilled candidates. When you interview him, he reminisces about a fraternity party in which you used illegal drugs. He's reminding you that he knows something secret about you. Then he says if he doesn't get the job, he'll tell the company higher-ups about this secret; if he gets the job, he'll never tell anyone.

What would you do?

Here's the good news. Your old friend is proving to be a lot less of a friend than you had thought. Your Granny might have referred to someone as a "snake in the grass." This expression generally refers to someone who proves disloyal behind your back. In some ways, that's harder for you to deal with; believing someone is a friend when they are not will almost always lead you to a bad outcome.

In the case of the threat to reveal a secret, at least you know what a person is saying they will do and, more or less, you know they are not your friend. And here's why that's important in real-world game theory. The threat itself isn't particularly credible; after all, how would hurting your reputation at work be good for your friend? Suppose you didn't hire him, and he told your boss about your indiscretion and the worst happened, you got fired. Is your friend legitimately better off because of that? He still doesn't have a job!

The threat revealed additional information for you, though. Now, you know that a secret isn't safe with this person. From now until forever, you won't play any games with this person the same way.

One Man's Secret Is Another Man's Lie

The comparatively easy case of Santa Claus vs. "crying wolf" belies a much harder reality about the problems surrounding information and games. Let's go back to the colleague who asks you to write a secret job recommendation. Just for fun, let's add something to the structure.

Your colleague asked your mutual boss for a reasonable raise and, as part of the negotiation, said she would have to start looking for another job if the raise was turned down. Thinking this was a bluff, your mutual boss turned her down, she did start looking, and found this potential new job. That's why she's asking you for the secret recommendation letter.

Losing your colleague will be bad for the company and maybe even you, from a "team" perspective. If we look back on the game, we can see everyone had an opportunity to respond to accurate, full information—they just made bad choices. What do you do? If you tell your boss it wasn't a bluff and he's about to lose this important person, you might make a difference. Then again, your colleague will know you can't keep a secret...

Now, before you throw up your hands and say this is a no-win situation, think about all of the choices you could have made. You could have tried to tell your boss you don't think the colleague was bluffing before you knew she found another job. You could have told your colleague that you don't want to write the recommendation letter because it puts you in a difficult position. It's true, you're caught in the middle, but that also means you have more choices in front of you. Classroom game theory doesn't give you these chances to "make something right," but the real world does.

An Option Is Only Worth More Alive Than Dead, in Finance Books!

This isn't a granny expression; it's a finance classroom expression. It means that an option, like an option to buy a company's stock, is intrinsically valuable. A stock option (strictly speaking, a call option) lets you pay a small, fee known as the premium in exchange for the right to buy a stock at a fixed price for a fixed amount of time. If you have a hunch that a stock price might rise soon, you can bet on that hunch by buying the option. The advantage of the option is if your hunch turns out to be wrong, the most you can lose is the premium. If you buy the stock based on your hunch and you're wrong, the price of the stock might fall, and you could lose considerably more.

If your hunch is correct and the price of the stock rises, the premium on its options also rises and you can actually make a profit on your hunch by simply selling your option to another investor. The "more alive than dead" expression means you can usually sell your option and make more money than you would if you used the option to buy the stock right away (there are details that we're omitting here for the sake of brevity.)

That's all interesting stuff if we're talking about investments but what does it have to do with game theory? Let's relate this expression to mixed signals. Suppose the stock you could make money on produces something you don't approve of (cigarettes or land mines or something like that.) If your hunch is correct, you'll make money on the option but your jollies might not rise; you feel bad getting rich in certain ways. In that case, hunch or not, you would be better off without the option.

Now, recall the student that gets accepted by Hoity-toity University. He doesn't have to go there; he simply has the option to do so. Let's assume if he's good enough to get accepted at Hoity-toity, he also got in at many other excellent schools. The option to attend Hoity-toity can't hurt him, right?

Here's where you should think about the future, uncertainty, and lots of things that are hard to know. In game theory, we've known that removing some choices today in a sequential game can make a person better off in the long run.

The granny expression that does fit here is "be careful what you ask for, you might get it." Arguably, the student's problems—at least difficult decisions—started when he applied to Hoity-toity. Asking for that option puts the student in the difficult position of having to make the correct choice today in a game that won't be obvious it has results for many years.

An option is just a mixed signal. Recall the student that gets accepted by Hoity-toity University. He doesn't have to go there; he simply has the *option* to do so. If he's good enough to get accepted at Hoity-toity, he probably also got accepted to many other excellent schools. Surely, just the option to attend Hoity-toity can't hurt him, right? Put another way: is having too many choices a problem?

In game theory, the answer is yes. As we saw in some of the more complex games with more than one choice, there's a better chance of making the right choice (in both the long- and short-run) when we can at least remove some choices today. In a sequential game where so much is unknown about a person's future after college, having the *option* to attend Hoity-toity University may not really make the student's life easier.

The Granny expression that fits here is "be careful what you ask for, you might get it." Arguably, the student's problems—in the form of difficult decisions—started when he applied to Hoity-toity. Asking for that option puts the student in the difficult position of having to make a choice (to attend or not to attend) in a game that won't obvious have results for many years.

Note: the current state of higher education in the US necessitates that students apply to any and all programs they remotely qualify for. This is a costly process, not only in time and grooming, but financially. In fact,

college admissions advisors have created a lucrative (read: exploitative) cottage industry. We are not saying it's better off to only apply to one or two colleges, especially in this competitive landscape. We just suggest that if long-term success in some unforeseen future is dependent on a decision today, it helps to make that decision as easy as possible.

Laws, Rules, and Regulations

This book showed how we, as a society, have built structure in the form laws, rules, and regulations. These limitations are often very little more than ways of preventing people from making self-interested choices. When you see the result of the unfettered hot dog stand on the beach game, you see that no one is actually better off at the end of repeated games; some people are the same (the stores) and others are worse off (the hot dog-consuming public). Each game ends with players having exhausted their self-interested choices and non-players, the rest of society, is worse off. So, government implements laws, rules, and regulations stop the game from even starting. They remove the option to play—to alter the competitive landscape—to keep consumers better off.

Why do we need laws to stop people from littering? Everyone knows that if you allow people to throw their trash on the ground, your community (/city/state/country/planet) will soon look like a trash dump. But, that bad endgame is the result of *everyone* littering; not just one person. Not just little ol' me. If one person throws one candy wrapper out of the window of their car as they drive down the highway, likely no one will ever see it. Ten people do that, then a hundred, then its bags from the burger joint, then...who knows. So, you pass a law to stop the *first* candy wrapper from being tossed out of the window. That some of the laws are disproportionate (ex, \$5000 fine for a single candy wrapper) is relevant too. As a deterrent, a city is in effect saying "don't start," to avoid the slippery slope mass-littering would cause.

This also takes us back to the difference between rules and norms. A well-accepted societal choice is often called a social norm. They are informal, often learned behaviors taught to you by others. A rule or law, by contrast, is a formalized, codified norm. It's a norm that society finds so important, we write it down and enforce it.

How do people in different cultures (or countries) greet each other? In some places, you shake hands; in others, you kiss one cheek; still others, you kiss both cheeks; and sometimes you bow. When you visit a new country for the first time, you are often surprised by these differences, perhaps even

embarrassed when you kiss a handshaker or bow to a kisser. These social mistakes don't end with anyone going to jail or paying a fine; sometimes the mere embarrassment at "violating" a social norm is enough to "correct" your behavior.

If you hurt someone while driving a car incorrectly, it could end up very differently. There are laws that govern driving in every country. They are not all the same and they are all codified and enforceable. So, when someone from the US goes to Great Britain, they have to drive on the other side of the street (drivers are on the right side in the US and on the left side in Great Britain); if they get it wrong and cause an accident, they get a ticket. Incidentally, calling it the "wrong side of the street" is funny when you realize that right and wrong are just formalized social norms.

Rather famously, some countries have strictly enforced laws about things that other countries don't even have strict norms for. Consider Singapore's law against spitting in public, which is (as of this writing) enforced with a \$1000 fine. There are other countries that have laws against adultery, punishable by prison time or worse. Noncooperative choices, what we sometimes call anti-social behavior, is governed around the world by a variety of norms, rules, and laws. The notion that some of these seem to us like "driving on the wrong side of the street" means that we specify jollies in different ways. Different cultures have different ways to manage behavior that a society finds unacceptable.

There is a clear delineation between how different societies manage noncooperative behavior, in the examples above, and repressive control mechanisms. Punishments for being a woman or non-binary or of a different sexual preference are not meant to control noncooperative, anti-social behavior. On the contrary, they are meant to remove entire groups of players to increase control of dominant groups.

Last Clear Chance

There's a famous, somewhat problematic cliché: "rules are meant to be broken." That could be the topic of an entirely different book. Let's consider the related possibility that rules or laws might be abused by a player.

Suppose a driver makes a right turn on red while another car is approaching a few hundred feet away and the approaching car actually speeds up to hit the car that turned. Why? Because the driver of the second car saw the possibility of a lucrative lawsuit. The turning car is legally at fault; the right-turn-on-red law says that the turning car must yield to oncoming traffic. The approaching driver could have prevented the accident by slowing

down, and they didn't. It doesn't matter. The approaching car was trying to take advantage of the law in order to sue the other driver, and the law seems to be on their side.

The legal justice system has had to deal with this possibility so many times, a doctrine emerged called the "Last Clear Chance" rule. Roughly speaking (and this is a legal gray area), the ability to win such a lawsuit requires that the approaching driver show they couldn't have reasonably prevented the collision.

"Gaming the system" is what we would call this, and it's a risk anytime a strategic move is codified into a rule or law. Laws run the risk of being interpreted too literally. The driver who could have prevented the accident is applying an erroneous version of "kill or be killed." The courts can and have applied their own doctrine to avoid abusing the letter of the law: "an exception to the rule." One driver made a mistake, the other driver caused an accident on purpose. Two wrongs don't make a right.

Ignorance Is Bliss

There are an amazing number of expressions that talk (or warn!) about information. The world has certainly morphed into a better-informed place, with technological innovations related to AI and the world wide web.

In a lot of ways, opportunities and information conflict with important game-theoretic concepts like commitments. You have a mid-level employee with a few years of experience. Your company is going through a "rough patch," and you need "all hands on-deck." In past decades, you, as an employee, ramped up your performance for a sense of loyalty to the organization, and to avoid the "pain" of a new job search. Today, on the other hand, a list of available jobs in your industry, specific to your experience, is right there on your phone. Now, if you get wind that your company is downsizing, you can weather the storm and score a new job in your pajamas over a holiday weekend.

Concerns about disturbing the established social order with easier access to information go back centuries. The expression "how are you going keep them down on the farm" was a World War I song title that worried about sending farmhands overseas. (Even then, the concern was winning the war!) Movies, radio, and television all changed the availability of information and there were concerns in each case about how the new information that they provided would change people's choices. Repressive mechanisms often rely on restricting information—such as access to the unregulated internet—to

control oppressed populations. Information and change go hand in hand, whether by equipping players to make better choices or adding new players to the game.

The Curious Case of Mixed Strategies

We've actively avoided decision-making based on probabilities throughout this book. That approach is the stuff of statistics. There is an area of game theory that has to actively combine strategic decision-making and statistics and, complicated as that sounds, you probably already do it every day. Rock, paper, scissors; odd or even; pick a hand: you do these games with your hands and fingers, and they are both random and game-theoretic.

Let's try the odd or even game. Two people will make a fist, count one, two, three (usually with some gesticulations), then put out either one finger or two (could also be no fingers, but let's limit it here to two choices.) Simultaneously, one of the players will say odd or even. You then add up the fingers. If you both choose the same thing, it'll add up to an even number. If you chose different things, it'll be odd.

Now comes the strategy...sort of. If you always say odd, and you always put up one finger, you will (eventually) always lose. The other player will figure out your consistent strategy and they also always put up one finger and, we all know, one plus one is two—even!

So, you have to mix up your call (odd or even) and your finger choices. If you randomize your choices, statistics says you should win half of the time. Consider how hard that is! You need to be truly random at your choices.

Random means sometimes you put up one finger four times in a row. Sometimes you alternate. Similarly, sometimes you say odd a couple of times in a row then alternate. Any pattern that you fall into could be figured out and used by the other player to win more often. Being random isn't actually a human strong suit.

When someone says, "you're becoming too predictable" and they aren't trying to be kind, they mean your pattern has become easy to figure out. Like all of our other expressions, we need to find a context where unpredictability in a game is useful.

Let's go back to sports. In baseball, the hitter has an advantage if they know what type of pitch is coming. If you have a good fast pitch, a slow pitch (called a change-up), and a curvy pitch, you'll have to learn how to "mix them up" in a pattern that's not just fast, then slow, then curvy and repeat. Being unpredictable is a competitive advantage. In tennis, you can't

always return the ball to the right, then to the left, and repeat (or deep, then short, etc.). In basketball, you can't always dribble to the right one time down the court then to the left the next time.

In these examples, a successful strategy is a set of probabilities that lead to scoring the most points or giving up the fewest hits. Optimizing a set of probabilities—creating order in your randomness—is part of what's called a mixed strategy. In game theory class, we often create games in which we can calculate the probabilities precisely to show how these games work. In the real world, you are almost always determining the probabilities of choices by trial and error.

The correct probabilities would incorporate a difference in, say, your select choice against the other person's choice. A pitcher's best pitch might be his fastball. If he throws it every time, the hitters will eventually hit it. So, he mixes in the occasional slow or curveball. The optimal probabilities might be 80% fastballs, 10% slowballs, and 10% curveballs. The key to these mixed strategy situations is not having a pattern even as you use your probabilities. The pitcher might throw two slowballs in a row every once in a while just to, as they say in baseball, keep the batter guessing.

Lest you think that mixed strategies are only for sports and simple games, there are life circumstances that allow you to use these skills. Spontaneity in relationships is occasionally good. So is getting out of your comfort zone or being the designated driver. As long as it doesn't sacrifice reliability or predictable safety, "changing it up" is a tried and true strategy in many human relationships.

Holding Grudges

It makes sense that you are more likely to make a cooperative choice and expect the cooperative choice will be made in return, when the other player is a family member, neighbor, or long-time friend. As long as the games are Prisoner's Dilemma style games, both of you are better off in the long run (many repetitions) when you both make the cooperative choice. It also makes sense that you're much less likely to make the cooperative choice when the other player is a total stranger and you're only playing the game one time.

What about people that you cross paths with on occasion, off and on during your lifetime; perhaps professional colleagues or industry peers. You might still play the game many times, but neither of you know how many times or when the last play might be. In cases like this, it's not uncommon for one player to make the noncooperative choice, getting the biggest one-period

payoff and saddling the other player with the lowest one-period payoff. After all, who knows when, if at all, you'll interact again?

What would the player who got the low return do after this happens? That's easy: they would hold a grudge. A grudge is a comeuppance that you hold onto until the next time you interact with the other player. There might never be another chance for revenge, or you might "get over it." For some people and some indiscretions, a good grudge can last a lifetime. The cautionary tale of game theory is this: when you think about making the self-interested play in an intermittently played, Prisoner's Dilemma style game, remember—your punishment might come many years later.

Player Beware: Grudges Over Time

Holding a grudge can also lead to bad future decisions. Most people have had the experience of a game not going their way. You want a job or a promotion (your choice was to apply for that new position), and the other players made a different choice, and someone else got that job. Human nature makes people feel bad about that outcome; you feel unappreciated or inadequate. Those bad feelings don't always go away over time. There's a great expression about this: the result "sticks in your craw." (Roughly speaking, this means something gets stuck in your throat... we have adapted it to mean the slight sticks in your brain for a long time.)

The problem with holding onto this type of grudge is it can lead you to make incorrect choices in future versions of the game. Let's say life goes on after the decision; both you and your employer have to live with the decision. Now, suppose things don't go well at the firm. For whatever reason, they made the wrong choice, and you get another bite at the apple. You're offered that job again. You feel vindicated; your inner voice wants your outer voice to say, "I told you so." Your long-festered grudge tells you to take the job now that you have a chance. Should you?

Probably not. At the very least, be aware of the reality that this is a different version of the game with a different payoff matrix. Getting jollies from being right—feeling vindicated as we said—comes from getting the offer. *Taking* the job means you have to deal with the state of the firm *now*. This is not the company you originally applied to. Your jollies going forward (based on your decision to accept the job) will be determined by your ability to succeed in this new environment—not in the earlier version of the game that didn't go your way. We've used this

expression before, but it applies here more than ever, “be careful what you ask for, you might get it!”

Follow the Leader/Blind Leading the Blind

Being a leader, it has been argued, means very little more than having followers. That simple definition is useful in differentiating between people with titles and power and *actual* leaders. There are many books on leadership and many expressions related to leadership, but they aren’t usually connected by way of game theory. Let’s see if we can fill that void.

Most people have heard of the expression and children’s game “follow the leader.” In that game, the leader is chosen arbitrarily, and the point is to do things that the followers can’t copy which eliminates them.

Grown-up leadership is exactly the opposite. The leader is more like a first mover that takes the group down a particular path. That should sound familiar; that’s a sequential game where the leader goes first. A good leader, therefore, needs to be well-informed and decisive. Looking at leadership as a sequential game lets us see many things. Leaders aren’t expected to be perfect. Some choices that were correct given available information don’t work out. Leaders should be expected to be human. They aren’t, however, supposed to be random. The idea of a sequential game is to look ahead and make choices now that should lead to the best possible future. If today’s choices are guesses, then leaders aren’t really leading.

Consider, then, another expression: “the blind leading the blind.” The imagery here says it all. When you’re the first mover in a sequential game that will determine the outcome for the entire group—the organization, the company, or even the country—you really need to make informed choices. Bad outcomes will result if you don’t, and this could compromise your role as a leader. The expression “the blind leading the blind” isn’t one that has long-run success for the leader associated with it.

Think of leadership as the first mover in a sequential game that involves information. Success can be increased if a leader has more relevant skills, a longer education (informal or formal), and better processing of information. Advisors also heavily influence leaders and their abilities. Once more, we can invoke the expression “you’ll be known by the company you keep.”

Gaslighting, Brainwashing, and “Psyching Out”

Bluffing and lying are well-known, ancient ways in which information is manipulated to change the outcome of a game. Famous stories like the Trojan Horse and the talking Satanic snake in Genesis in the Bible are oft told tales that remind us to be wary of information—even from seemingly trustworthy sources.

More modern manipulations are present in the algorithms that determine what and when we see in news and social media. A handful of players in modern society have an ability to control wide swaths of information available to us. The outcome of games with information manipulations challenges our ability to define rationality.

In the extreme, players who have been “brainwashed” or “gaslit” have, literally, a warped sense of reality and might make choices that others would not consider rational.

What’s good, and bad, about these situations is that they do make sense with a jollies-based payoff matrix, in the mind of the manipulated player. Like a person using what we call mind-altering substances (many illegal drugs for example), they believe they are getting more jollies from their choices. We wouldn’t call their choices irrational, just very much misinformed.

On Being Fooled

Granny loved talking about fooling and fools:

Fool me once a curse on you fool me twice a curse on me.
A Fool and his money are soon parted.
Everybody plays the fool sometimes.
We won’t be fooled again.
The Greater Fool Theory.

If a game-theoretic situation does turn out the way you hoped or expected, it is sometimes referred to as being fooled. It happens on occasion. You cannot possibly know every person and their payoffs in every situation. That fact, by itself, does not mean you did anything wrong; you’re human.

What we should have learned from this book is being fooled on occasion doesn’t make you a fool. Being fooled in the same situation, by the same person, in the same way, *repeatedly* does. Einstein famously said, “Doing

the same thing over and over and expecting different results is the definition of insanity.” Humbly, game theory can teach us ways to enhance our information, learn in repeated games, and get better outcomes over time.

Being fooled might be an application of what we called asymmetric information. The other player had information that was important to your outcome, and they chose to withhold it or lie about it. Being fooled might also mean you weren’t using game theory in a situation that called for it; you believed that your choice alone determined the outcome when in fact, the choices of other players also mattered.

You can even extend the idea of fooling players in a sequential game. If every player makes their choice—buys something—because they were fooled, let’s say a little, and then turned around and sold the thing in question to a new player by fooling them a little more, the end point is usually fooling the last buyer a lot. That’s the nature of what’s called the Greater Fool Theory. The last person who mistakenly choses something that’s actually not the correct choice will likely be the biggest fool in the group.

In any case, you hopefully learned from Granny, from scriptures, from Aesop, and from Einstein the importance of thinking and acting differently when you’re in game-theoretic situations and considering how important accurate information is in these circumstances.

A No-Win Situation

In real-world situations, you are sometimes put in the awkward position of being unsure about what to say; what to signal in a game-theoretic situation. In an application of the expression “discretion is the better part of valor,” it is sometimes better to avoid the situation rather than have to make the choice.

Consider this scenario: a colleague at your company asks you to go to lunch and says they want to talk to you about making a professional connection for them with a close friend of yours, the CEO of a hot new tech company. You don’t really think this colleague is a good fit (or maybe you think even less generous things), so you don’t want to make the connection for him. You also have to interact with him at work on occasion. What do you do? You can lie to him or lie to your CEO friend about him or some other unconformable thing. Or, you can politely turn down the lunch offer.

In life, there are games you don’t want to play. We often refer to these as being in a “no-win” situation, but it’s only no win if you play.

Information, Expertise, and Decision-Making: "Physician Heal Thyself"

To make choices that lead to the best outcomes in sequential games, we need accurate, useful information. This puts decision-makers in a difficult spot. Who or what do you believe? Who do you trust? Credibility comes in many forms: titles, experience, educational attainment, and other factors. A decision-maker that has become a leader should have skills and competencies that allow them to better filter information, navigate uncertainty, and make correct choices.

Granny would have prioritized things like common sense and intuition. As a decision-maker, perhaps a leader, what do you do when the expert advice that you were given is at odds with your intuition or what you consider common sense?

Consider the expression, "if it seems too good to be true, it probably is." This is a cautionary phrase telling you that there aren't many easy answers to hard questions (or in the case of game theory, hard choices). Most of what looks "too good" has been tried or maybe just won't work. If you're offered twice the market rate to take a job, dig deeper. You might find out that the position has been held by six people in 3 years, and you should run for the hills.

A really good expert can almost always explain things in a way that aligns with your common sense or your intuition—and the corollary is also true. The somewhat cynical biblical proverb "physician heal thyself" can be interpreted as saying "hey doctor (or expert), why should I believe you if you can't even cure the disease that YOU have!" Obviously that's not always fair—there are incurable diseases—but when heeding advice, a person who has successfully followed the advice that they are giving you (in similar circumstances) could be considered a reliable, credible resource of information.

Game Theory Goes to the Doctor

Players in sequential games have an interesting temporal problem that we often refer to as short-sightedness: a tendency to focus too much on the outcomes from the first choice rather than the outcome of the complete game. That might come from human nature. We see it in a lot of places. In fact, one of the most tried and true rules in Finance says that outcomes in the future are not worth as much as outcomes today (everything else the

same). We prefer outcomes we can see today rather than esoteric, phantom outcomes that might happen tomorrow.

The medical profession has given us another way to think about this problem. Granny was talking about a difficult decision once and you heard her say, “that’s a hard pill to swallow” (or possibly “that’s a bitter pill”). As a concerned grandchild, you asked if she was sick and she said “no, dear.” What Granny meant is, there are choices in the short run that are unpleasant, but they lead to the best possible future outcome. Making the hard choice today goes against our nature, but it may still be the right choice. All the more reason to learn game theory.

Except, swallowing a bitter pill is not automatically a game-theoretic situation. A game needs two players. To be a one-player *sequential* game, the player has to make a current and future choice. That can be a game. Take two players: a patient and a medical doctor. The doctor tells the patient to take a pill every day to be healthy in the future. The patient, who isn’t sick, thinks it’s a waste of time and money. Short-run negative jollies are clouding much higher long-run positive jollies. We should take vitamins, but few of us do.

Ask any toddler who’s pitching a fit about taking the bad-tasting medicine that they need to cure their illness (or better still, getting a shot). Making the correct first choice is difficult if that choice is unpleasant. Finding ways to make treatments less onerous is important for drug producers, doctors, and other healthcare providers. As part of the game of healing, they need to be aware that the negative jollies people perceive from that “hard pill” are inhibiting, sometimes even preventing, a desirable outcome for the patient.

Consider the soft skill known as bedside manner. Having a good bedside manner means you’re good at communicating and connecting with your patients. You can empathize with them. You understand their negative jollies. From the doctor’s perspective, only being right or technically trained doesn’t always get the patient to make the right choices. The arrogant surgeon might tell the patient, “you have to cut to cure.” She might be right, but she’s also making the patient think much more about the cut than the cure. A good bedside manner helps doctors understand jollies from the patient’s perspective, helping get the patient past their fears and on the path to cure.

There’s another wise expression: “you have to take your medicine.” It refers to accepting punishment for a mistake. Medicine as punishment seems odd, but in earlier times, very little attention was paid to how medicine tasted (spoiler alert: it was bitter and gross). Getting a desired outcome in a healthcare situation could be accomplished by making the medicine seem less like a punishment; an easier pill to swallow.

Dieting

Most adults have had the experience of trying to lose weight in a world where the BEST foods we enjoy have a lot of calories. Food producers, our internal jolly generators, and game theory conspire against us during most diets. Knowing many of us have what we call “a weakness” for salty, sweet, fried, and crunchy foods, producers fill store shelves and café menus with them. With these readily available sources of short-run jollies at hand, we revert to human nature. The long-run path to health is uncertain, and the ice cream is right there! A lot of us make a short-run decision that takes us away from our long-run desired goals.

Being “penny-wise and pound-foolish” is an expression that sounds like its related to this behavior. (Pound, in this case, refers to British currency). It means you're making decisions that save you a penny at the expense of a more lucrative pound. Making a short-run choice to cheat on your diet may feel good today, but the sacrifice is not losing weight (and having to stay on the diet even longer) tomorrow.

Fairness, Future Outcomes, and the “From Here” Problem

We observed that the rules of most games in life are invented by some of the players somewhere along the way—and not always with fairness or equity in mind. Human nature often leads rule-makers to bias outcomes in their favor. When a society changes laws and rules that are biased, sexist, racist, etc...., these are often done in the name of increasing fairness for repressed groups.

There's no doubt these shifts are a marked improvement over the previous regulation. Progress is iterative, after all. The problem is where we go “from here.” Changing available choices to make outcomes more fair and less biased can be shown as a sequential game with some information issues. Once a rule is changed, however, we have two problems: how to implement the change going *forward*, and what to do about those *currently* operating under the old law.

If you own an apartment building, you aren't allowed to choose your tenants based on their gender, the color of their skin, and other factors that are unfair screens. Laws prohibiting discrimination allow many more people to get housing and secure better future outcomes. Here's the problem: all

rules in sequential games start with the *next* play. They make outcomes fairer going *forward*. They don't do anything about the past. In that sense, these laws and rules start "from here." You have to use more balanced criteria to select future tenants, but you don't have to evict the ones you already have.

One could argue that fixing these sorts of problems going forward is often the only thing you can do. Even in the most progressive society, you can't (shouldn't) force a landlord to evict people of one group to make room for another. Making sure things are fair going forward might be as easy as not allowing the use of irrelevant, biased screens. Fixing the imbalances that the unfair rules *already* created is much harder. Using the structure of a sequential game, you would have to go back to the first time the unfair process was applied, retrospectively change the rules, and figure out what outcomes should be. That's impossible in most situations.

The "from here" problem exists in a lot of places. There are many sequential games in which the starting point is arguably rather subjective or even arbitrary. Consider how many college sports teams are ranked (which impacts things like recruiting, NCAA funding, and their position in postseason tournaments). A group of "experts,"—head coaches in the sport or writers that cover the sport—rank teams in the preseason based on what they would say are informed guesses at future performance. After teams start to play, rankings change and become more realistic. The preseason, made-up rankings are the "from here" starting point and will affect rankings for the entire season. Being able to say that you beat a Top 10 team matters even though the team was not necessarily worthy of the Top 10 based on anything real.

Another, somewhat scarier version of this problem exists in asset markets (stock markets, housing markets, etc....). Stock prices react to news about corporate earnings and the overall economy. Unexpected, good earnings will make the price of the stock go up in most cases. That means the news will make the stock price go up "from here." What if "here" is wrong—perhaps already too high? Finance people often talk about asset prices that have moved too far too fast; a stock is said to be overpriced relative to its fundamentals. Asset markets—both stock markets and housing markets—have experienced what are called bubbles; valuations that are unsustainably high. Bubbles could be analyzed as sequential games with a "from here" problem. Just like the biased decision process earlier for rental housing, it's very hard to go back and figure out what things should be worth when an asset bubble bursts.

The Cold Medicine Problem

Your bestie calls you clearly suffering with a cold. She says, “I took some cold medicine, but it didn’t work.” Having learned about sequential games from our book (and apparently feeling a bit obnoxious) you say, “you don’t know that.” She says, “well I know how I feel, and I don’t feel any better!” So, you choose to bring her some soup and save your friendship.

You are right, though. Your friend doesn’t know how she might have felt if she hadn’t taken the medicine. She could feel a lot worse. The Cold Medicine Problem shows us it’s nearly impossible to correctly say what “would have been” had we made another choice. Once you’ve made a choice in a sequential game, the outcomes from the other choices are unknowable. Time—along with all the unknowns that can change a game—has moved on. Life is a constant stream of sequential games, but on your deathbed, you can’t waste your remaining moments saying, “if only I had made a different choice.” It’s a phantom reality you can’t really know.

The Airport Game

Having the tools of game theory can allow us to navigate problems that aren’t exactly games, although they share a similar structure. These are not interactions with others, but they do depend on some strategic thought. Here’s a good example:

A high-priced consultant is at a party, and you hear him say, “if you never miss a flight, you’re spending too much time at the airport.” Upon hearing that, you wondered if you were doing something wrong because you had never missed a flight in your entire life. You might also wonder if this guy has had a bit too much to drink.

You’re not doing anything wrong; this is a dollars-versus-jollies issue. You are choosing between your time and the value of your time. It’s not a sequential game because the future “choice” is actually just two possible future outcomes, but we can use the framework of a sequential game to explain the difference between your travel philosophy and the consultant’s.

Suppose the consultant—Robert—makes \$500/hour, and he makes all of his decisions based on money. He also can’t do any work while at the airport (for some reason, he never has access to wifi. The airport is complete downtime). He looks at the problem in something like Fig. 11.2.

Robert’s decision depends on money. If the time and cost of rebooking a flight on occasion is less than \$1000, Robert goes to the airport one hour

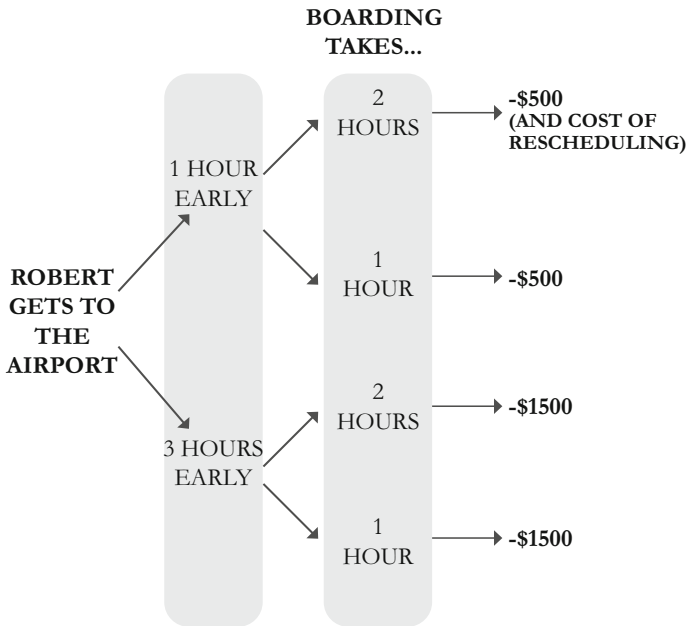


Fig. 11.2 The airport game

early and occasionally misses a flight. He's perfectly fine playing the odds of how long boarding takes. How often does one actually miss a flight?

You make much less per hour (sorry). Unlike Robert, you make decisions based on jollies which include income lost and other things, like your personal anxiety about missing a flight. We can convert dollar costs to jollies and still use the same hourly rate as Robert, but we can include negative jollies when you *worry* about missing the flight and a lot of negative jollies when you actually *do* miss a flight (Fig. 11.3).

This is still probabilistic, but you can understand why you and Robert make different choices. Nobody is right or wrong. It's completely rational for a person to pay more money to avoid a lot of anxiety.

This game also lets us end with an expression that isn't always right: "all's well that ends well." In this game, and in much of our book, we're highlighting the importance of how people *feel* about the outcome of a game: the jollies that they receive in the end. How you decide on a choice can and does affect your jollies. We've included things like honor, empathy, fairness, and now anxiety in the final payoff matrices. You can also include things like coercion, pride, and apathy. We consider these feelings when we specify a game for ourselves, intrinsically. We don't always know we're doing it, because we

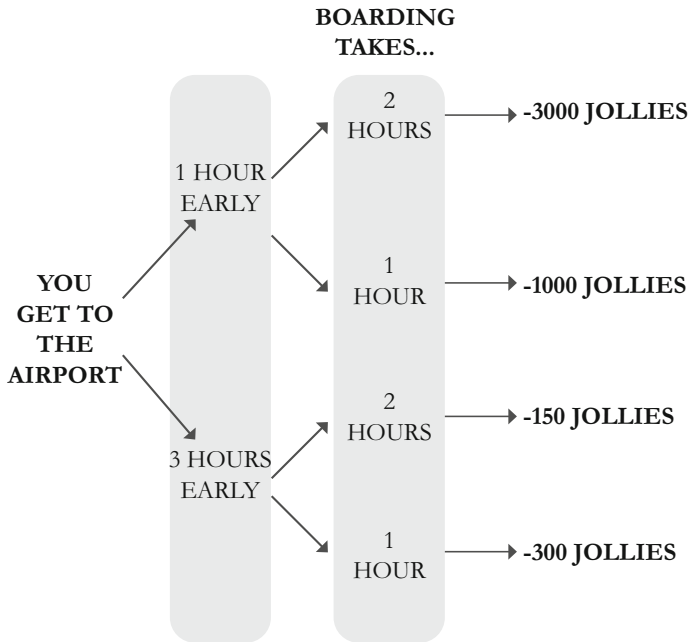


Fig. 11.3 The airport game, for you

don't always understand our motivations, but how you feel about making a decision always affects your payoffs.

Conclusion

Game theory is tricky and complicated even if you don't make it mathematical or overly analytical. You have to make choices in situations where other decision-makers are also making choices and the outcome depends on both of these decisions. These situations happen every day and by the time you become a functioning adult, you learn to navigate them without reading a game theory book or taking a game theory class.

Before looking to mathematics and economics to complete your game theory education, remember that you were taught to make decisions by many people who came before you. People who dealt with similar problems and situations and may have learned the hard way—by getting a bad outcome—that their approach to the situation was flawed. You have already learned from these previous mistakes—it's why we study history in all its forms.

Understanding the structure of a game and the determinants of payoffs (what makes people tick) requires insight and reflection. Learning game

theory and using it well is a life-long pursuit. If you want to help those who come after you, pass along what you've learned and make the lessons easy to remember—and never underestimate the power of a good expression!

Questions

- 11.1. Consider the following expression: “all’s fair in love and war.” Focusing on the choice of words in this well-known caution, what word makes you think that this has something to do with game theory? Which aspect of game structure is it addressing? Finally, if the expression is literally true, and knowing what you know about war, what is it saying about the stakes (the payoff matrix) of “the game of love?”
- 11.2. We discussed expressions throughout this book that are dated or have otherwise lost their context or reference point. “A stitch in time saves nine” and “You reap what you sow” are examples. Let’s try to connect some modern acronyms to what we know about game theory:
 - **YOLO.** How is this related to sequential games?
 - **FOMO.** How is this related to understanding your player?
 - **IMHO.** How does this defer responsibility to another player?
- 11.3. Our book touches on some very personal and private areas of life. Granny wasn’t as much of a prude as some people think; to paraphrase a bit, “she saw some stuff.” Suppose you told her that you and another person were “friends with benefits.” What would she worry about, aside from the biological consequences? What changed from her time to yours that negates some of her concerns?
- 11.4. Game theory tries to get people to think more clearly about the process of mutually dependent decision-making; your choice and mine together determine our outcome. Many other things are involved in the process of decision-making, including skills and statistics. Some people also rely on sketchy approaches to determining outcomes—like superstitions. We often do silly things because they worked once: I wore socks inside out during the World Series and we won, so I guess I’m doing that every time. As long as you don’t take these things too seriously, applying superstitions to a random process doesn’t do any harm. Does that logic extend to game-theoretic situations? What’s wrong with thinking, “as long as I’m wearing my lucky hat, I’ll get the best possible outcome” in a game?

- 11.5. Willingness to play a game has many components: willingness to accept outcomes that might not be desirable, willingness to “share” the determination of that outcome, and willingness to be “screened” by other players. This last issue is often the reason some people are unwilling to run for political office.

Few people are willing to have their entire past brought out and put on display. In the old days, it is argued, private lives were off limits. If you think of this as an information acquisition issue, is that really what's going on? If it's easier to find skeletons in closets, how can the electorate adjust to these modern realities? Here's a hint and one final expression, “let he who has no sin cast the first stone.”

Answers to Discussion Questions

- 1.1. Homework and practice are ways of building skills. The end result of these skills is better life outcomes; having the ability to excel at something that “pays off.” As we’ll see, life is a sequential game where choices and decisions today allow or preclude future outcomes. Parents know, because they’ve already gotten there, that today’s skill-building becomes tomorrow’s paycheck.
- 1.2. You didn’t listen to Granny about a fool and your money. The palm reader, on the other hand, is assuming that no one will do unto her as she has done unto you!
- 1.3. The biggest single problem—the one word that really makes this expression a very bad application of decision-making strategy—is “sight.” Considering the long-term nature of a relationship, the small and fleeting amount of information that one can glean from the “first sight” is insufficient to base a long-term commitment on.
- 1.4. “Lucky” Most of the skills you will need in that game are learned in class on probability and statistics.
- 1.5. Game theory is about decision-making in situations where there are multiple, inter-connected, decision-makers. Competitive sports often include game-theoretic decisions, but sports are usually based on physical skills.
- 2.1. Based only on things we’ve discussed so far (reputation, repeated games, and other possibilities could have something to do with this but we’ll talk about those things in later chapters) empathy outweighs money for you in this case.

- 2.2. “If” Game theory is the process of incorporating other players’ choices into your decision-making process. If you know with certainty what the other player is going to choose, this is much more straightforward.
- 2.3. In the real world, you must figure out what’s in people’s payoff matrix. What gives them jollies. They also have to figure out your payoff matrix! That the things with Janet work out reasonably well after twelve years is, you’ll pardon the neighborhood pun, a two-way street.
- 2.4. Coordination game results require a mechanism—a shared approach—for reaching a solution. Kids need to learn how to start games; how to make rules that are inclusive. These things require experience with starting and playing games. Adults help small children do this early in life and those successes give kids experience in structuring games.
- 2.5. Many people start—with help from an understanding restaurant—with choosing categories. Meat or fish or pasta or a big salad? Maybe appetizers for the meal? Once you’ve made your category choices, you can strategize with others at your table in hopes of getting a “taste” of things that weren’t in your category of choice.
- 3.1. The couple’s alcohol-fueled commitment skipped far too many steps. The couple did not have enough information about each other to for the marriage commitment to be successfully maintained. Put differently, one or both people almost certainly discovered that their marriage partner was not well-suited as a permanent source of everything that a happy marriage requires.
- 3.2. Obviously this question has many possible answers. Let’s assume that they want to watch something together (so the answer is not grandpa watches fishing on TV while grandson watches sci fi on his phone.) The “been there, done that” approach that we discussed would say grandpa should agree to watch something other than fishing. Perhaps the science fiction show, perhaps a third option. But grandpa knows what it’s like to be fourteen; grandson has no idea what it’s like being sixty-five.
- 3.3. We’re going to go with learned. There are so many things that change as a person gains experience that it’s hard to imagine a functioning decision-maker is relying on the small number of jolly generators that are hard-wired.
- 3.4. Understanding the source of the other decision makers’ jollies in no way requires you to share that source of jollies. The key is understanding the source of jollies and also respecting the other person’s tastes. We often hear the advice “love someone for who they are.”

Simplistic though it is, that's a much better recipe for a happy long-term relationship than judging everyone in your life by your tastes and standards.

- 3.5. People act differently in public. Your boyfriend is less likely to burst into tears, become verbally abusive, or many other unpleasant responses intended, perhaps, to change your decision.
- 4.1. Take a picture with a smartphone and text the photo to would-be rescuers. But this technology comes with a twist, early on in this story, after one or two fake sittings, the kid would be told to send a picture and without that confirmation no one would come. Heaven help the kid if his cellphone's battery ran out!
- 4.2. We have laws that penalize people (fines, imprisonment, etc.) for selling things under false pretense.
- 4.3. "Blocking and tackling" in football are important to the outcome of a game but they are among the least fun things that team members have to do. That reality transfers to the business world. In general, employees don't enjoy doing the jobs that the boss is referring to when she says, "we need to focus on blocking and tackling."
- 4.4. It isn't fun for anyone when you lie about things like your educational background or your legitimate skillset. No one wants to have brain surgery from someone who lied about going to medical school.
- 4.5. Singles bars. Feel free to replace "singles" with any hard-to-signal attribute and you'll get the idea. In most cases, these places do not require that customers are part of the group, but they provide a place that people can willingly and knowingly go to meet similar people.
- 5.1. Personal choice often conflicts with cooperative behavior. One can make a personal choice to be cooperative of course, so where's the problem? The problem is in the word mandate. Being told, or ordered, to be cooperative is different from choosing to be cooperative. And when the issue is a communicable disease, the importance of cooperation—either mandated or chosen—is clear. People other than yourself can die because of your noncooperative choice. Mask mandates are arguably less effective than vaccines, but they are visible, and they have the ability to reinforce a sense of seriousness about controlling the problem. In that way, mask mandates are but symbolically and at least to some extent literally a cooperative way of controlling communicable disease without mandating that people get a vaccine.
- 5.2. He could hold the door open for everyone. That seems like a reasonable solution but how would a woman, who was offended by this old-fashioned behavior know you do this?

- 5.3. There need to be resources available to catch and (ultimately) punish those who are being noncooperative. If a thief knows that there aren't enough resources in the local legal justice system to find and punish them, cameras don't do much good.
- 5.4. Paying in advance for the right to not wait in line is giving up dollar jollies to avoid time jollies. Paying a fine for speeding or stealing or some other noncooperative behavior is a systematic way of taking back some of your ill-gotten noncooperative jollies.
- 5.5. Any game in which competitive outcomes are important to maximizing overall results. That includes zero-sum sports games but also economic games that depend on the best player winning.
- 6.1. Making a simultaneous game sequential creates a situation that's stronger than a commitment. The second player in a sequential game knows with certainty what the first player's choice is. A commitment in a simultaneous game, as we've seen, could be a bluff. Agreeing to make the game sequential and then "going first" is an ironclad commitment.
- 6.2. Rules that limit voting to one gender or one class of people or one ethnicity are all good examples. And having created the situation where only your type of people are in charge, you have the ability to control and even abuse all other types of people in your society. Alternatively, if your society allows everyone to vote, rules and laws are less likely to restrict access to things like education and other social benefits to only the chosen few.
- 6.3. When some of the choices are unambiguously bad. In tic-tac-toe, no choice at the beginning of the game is guaranteed to be bad. On the other hand, giving a small child access to anything and everything that might want to try does include things that are clearly bad for them.
- 6.4. At the beginning of the second half of a game like this information has been revealed. Each coach has been able to observe the strengths, weaknesses, and strategies of the other team. If the point of receiving the ball at the beginning of each half is the score points, the team that receives the ball in the second half can approach that task with much more information.
- 6.5. You could write a payoff matrix with your current and future self as players and you could have one outcome that's very, very bad which leads to a Maximin approach—avoid the choice that could lead to that outcome. The part of the problem that would seem to be impossible is making simultaneous choices with your present and future self.

- 7.1. Cynical though it sounds, having lots of people at your wedding means there are lots of people watching you, seeing you at the local bar or restaurant with someone other than your spouse, and potentially reporting their observations to your spouse.
- 7.2. The problem is in using the word “promise.” Situations can always change, and promises do sometimes have to be broken but if you’re raising your child to use the word promise correctly, you might say you plan to take her to Disneyland instead.
- 7.3. Yikes! Well, a credible death threat potentially takes away all of your future payoffs so the choice between your money or your life is a no-brainer. Credibility is key. The implicit promise is you won’t be killed if you give up your money. If you have reason to doubt that implicit promise, your choice might be different.
- 7.4. If you give in to the bully, he will bully you forever. The advice-giver is seeing this as a repeated game where your decision today will establish your reputation.
- 7.5. In the setup of a simultaneous game we say that both players are choosing at the same time which makes promises and threats—what we called response rules—impossible. It has been argued that you could somehow pre-program your choices in a simultaneous game to align with a promise or threat. The promiser/threatener would program a computer or something like that to automatically choose something if the other player makes a choice. While this sounds possible, it’s hard to find any examples in the real world.
- 8.1. A one-period (simultaneous) game with a stranger has no way of incorporating reputation or long-run benefits of cooperation that counteract short-run benefits of self-interest. One-time-only games with strangers usually need to be played with a lot of caution and an assumption of noncooperation.
- 8.2. Assuming they knew about the worst-case scenario—the accident, etc.—what likely happened is that they drank enough to be mentally impaired and thus didn’t correctly incorporate the future downside.
- 8.3. There is a perception that there won’t be any future in which you can be punished for bad behavior.
- 8.4. The officers don’t live forever. Policymakers in situations like this often focus too much on the short run—they want to be liked or re-elected—at the expense of long-run success.

- 8.5. No hell means no lost jollies forever, just more jollies in heaven or no more jollies. In that case, a bad actor could try to accumulate enough short-run jollies during their lifetime to make forever jollies in heaven unimportant.
- 9.1 The frontline people are also the ones who are likely to be hurt or killed first. While not actively wanting to start World War III, a person on the front line will still do what's necessary to protect themselves. The high-level leader can limit things (or not). They can agree to let skirmishes happen on the front lines while not allowing nuclear missiles to be fired. That said, frontline decision-makers might know (sooner) that things have already escalated too far. The use of things like chemical weapons for example.
- 9.2. When the fuse is lit, it burns slowly but steadily toward detonation. Pull out or cut the fuse—defuse the explosive—and there will be no detonation. This is brinkmanship with a twist. The escalation of the situation is based only on time. Once the lit fuse reaches the explosive, there is no turning back. The bad outcome isn't exactly at the end of a slippery slope. The fuse burns steadily, and it can be cut right up until the explosion.
- 9.3. This is the situation with voting for representative policy makers in some circumstances. The voters are the frontline players, and they often want a happy-ever-after society in which to live for a long time. Elected officials are only in office for a few years so they might care more about the short run (being liked or being re-elected.) One might argue that this is a flaw with representative government.
- 9.4. Leaders should care about the people being led more than they care about themselves. The egomaniacal leader, in a brinkmanship situation, is more likely to start a war because he doesn't care about how many people he kills.
- 9.5. Eventually, you would call that player dead. If there is a point, it isn't in the game of chicken. Weakness and strength as concepts make a lot more sense when we think about situations where threats are being used to determine the outcome of repeated games where one player is trying to avoid a bad outcome every time they play.
- 10.1. If the city was 80% lower income, elected officials would impose most or all the tax burden on the rich twenty percent. Here's where we must remember that people usually have the choice not to play the game at all. The rich people could just leave; move to another city (we purposely chose a city rather than a country to make this easier to do.)

If that happens, there would be no tax revenue and no government spending.

If the city was 80% rich people, they would vote to lower their own taxes. They might vote to increase taxes on the poor 20% as well but those people don't have very much income to tax. In this case, and given the setup of the question, the city again wouldn't have much money to spend on government things. Voting for people who always promise to tax the other guy doesn't result in a well-funded government sector.

10.2. No. Next question.

... Okay, a little more on this. Party politics is a game, but it isn't one that helps any voter in the long run. Party politics is like the brawl that breaks out in the stands among soccer fans. The brawl will have winners and losers, but it has absolutely nothing to do with the reason you came to the stadium—the soccer game. The point of representative democracy is to form a government that serves most of the people most of the time. That doesn't have anything to do with party affiliation.

10.3. Ignoring the rules of the Senate and the House of Representatives, the ability to pass legislation in either body depends on getting slightly more than half of the elected officials on your side. With that in mind, 51 (a majority of the Senate) is easier to reach than 218 (a majority of House.) So, everything else the same, senators.

10.4. In this case, the third-party candidate is taking voters away from the candidate that they are more closely aligned with. So, a left-leaning third-party candidate helps the right-leaning candidate win and a right-leaning third-party candidate helps the left-leaning candidate win. Given that, third-party candidates usually get a lot of support from the candidate that they are least aligned with.

10.5. This is a bit of a trick question. Looting is never a good thing. But reasonable in this case might focus on the likelihood of being caught and punished for taking things without paying. When you are part of a mob looting a store, you probably feel like it's unlikely anything will ever happen to you, so you are getting free jollies.

11.1. Fair. Fairness often refers to the rules of games. Love, and to a lesser extent war, are strategic not random or physical so there is an element of game theory here. The implication is the love, like war, is a zero-sum game—a game with a clear winner and a clear loser.

11.2. YOLO is "you only live once." The usual application of this acronym is to promote choices that increase current jollies while ignoring or minimizing future jollies. The easiest pushback against this attitude

would be “but if you make very bad choices today, you don’t live very long!”

FOMO is “fear of missing out.” Players often play games or make choices because of peer pressure. A person literally loses jollies because of their perception of what others think of them. Not being part of the group (missing out) can make a person feel lonely or “uncool.” The pushback against this attitude is a reminder that being part of some groups is far worse than being left alone. There’s nothing cool about being part of a group that hurts or abuses other people.

IMHO is “in my humble opinion.” Players may use this term to delegate authority to another player. By transferring responsibility, the player also defers the consequences of the action. We do it all the time in conversation to absolve ourselves of choices made by friends and family.

- 11.3. Granny knows that “benefits” are hard to keep under control. “Benefits” often lead to attachments. She might say “how can you both be sure that your benefits can stay friendly?”
- 11.4. Thinking that wearing your lucky has changed a probabilistic outcome is innocuous because, well, it didn’t. At worst, you might play games of chance too often if you follow this logic, but your superstition didn’t actually affect the outcome. In a game-theoretic situation, your belief that a lucky charm of some kind always leads to good outcomes might make you less likely to figure out the game correctly and thus less likely to make the correct, game theory based choice.
- 11.5. The best approach to this change is for voters to accept and contextualize the flaws in those who are running for political office rather than pretend the person has no flaws. Wishing that—like the “old days”—no one wanted to (or could) find the skeletons in your closet allow people with very bad “skeletons” to become political leaders.

Glossary

Agency Theory, Principal-Agent Problem The area of study that deals with problems that arise when a decision-maker (the principal) delegates decisions to a subordinate (agent) where these decisions or choices will determine outcomes for both parties. The “problem” is how can the principal get the agent to act in the principal’s best interest and not their own best interest.

Assurance Certainty

Assurance Game A game with many possible solutions (equilibria) in which players have found a coordination mechanism that allows them to reach one of the solutions.

Asymmetric Information A situation in which one player has more information than other players.

At the Limit A holdover from mathematics, this refers to playing a game an infinite number of times.

Balanced Equal

Binary Choice See: choices

Bluff, Bluffing Announcing a choice that you don’t actually intend to make.

Brainwashing Psychologically manipulating someone into believing things that are not true or things that are not in alignment with a person’s existing value structure.

Break-even Price The price that exactly equals the cost of production.

Brinkmanship Situations where the only available threat is generally considered to be too big to be credible. Players often start down a path that could lead to the threat happening. Different players and different decisions move the process down what is often called a “slippery slope.”

Cartel A group of firms that coordinate their production and prices in order to maximize group profits. A cartel is, in effect, a coordinated “group monopoly.”

Cognitive Bias A thought process that is skewed, often inappropriately, toward a set of beliefs or a particular outcome.

Choice, Binary A choice set with only two options for each player.

Choice, Non-Binary A choice set with more than two options.

Choice Process The approach that a person uses when choosing between alternatives.

Choices, Choice Set The available alternatives that a player or players must choose between.

Cold War The period between the end of World War II and the early 1990s when the US and USSR were building large arsenals of nuclear weapons, purportedly for deterrence and defense, and using the threat of nuclear annihilation to destabilize the world.

Collective-action Games Games in which individual choices and outcomes depend on the choices and actions of a larger group.

Collective Bargaining (Power) Often discussed in relation to labor unions, people who agree to bargain as a group rather than individually. In many cases, the group has better bargaining power than any one individual.

Collusion players actively coordinating their decisions or choices. Famously, businesses might do this to keep prices from falling when they actively compete. This is called price collusion.

Commitment A choice that one player says they are going to make before a game is played. Commitments are usually made in simultaneous games, games in which all choices are made at the same time. Committing to a choice is saying, in advance, what you are going to choose.

Complements Things that are used together or at the same time. Often, things that augment each other.

Consistent Making the same choices or decisions under the exact same circumstances.

Context The situation under which a decision is being made.

Control Ability to determine the outcome.

Cooperation, Cooperative Games, Cooperative Solution An active alignment of the decision-making process by all players. We generally think of cooperation as a decision on the part of all players to make choices that are best for the group rather than any one player.

Coordination Game A game with many possible solutions (equilibria) in which players must find a way to reach one of the solutions. If players can find the "coordination mechanism," this becomes an assurance game.

Counterpart A different decision-maker who is part of the opposite player group.

Credible, Credibility A measure of believability.

Cuban Missile Crisis The thirteen days in 1962 when the US and the USSR dealt with the Soviet attempt to put nuclear missiles in Cuba.

Culture Clash A situation in which players have different backgrounds or belief systems and thus do not approach things in the same way.

Decision Tree See: extensive form

Delegated Authority Passing decision-making responsibility to other decision-makers.

Different as players, motivated by different extrinsic or intrinsic factors.

Dominant Strategy A choice that's best for a player regardless of what the other player chooses. In a game-theoretic setting, a dominant strategy is the choice a player should always make, given the possible outcomes.

Dominated, dominated choice A choice or some other alternative that never makes the decision-maker better off.

Duress A state of stress or extreme discomfort, sometimes caused by external pressures.

Efficient Market In economics and finance, a market in which what's often called excess profits—profits above what is required to take on an asset's risk—are impossible in the long run because markets are composed of many participants, all searching for excess profits. Forces of supply and demand move excess profits to zero over time.

Empathy Concern for the well-being of others.

Equilibrium A stable stopping point for a process.

Escalation Increasing magnitude or intensity.

Extensive Form A way of presenting a decision process as diverging branches, similar to the branches on a tree. Also, Tree Diagram, Decision Tree.

Factors Determinants.

Familiar, Familiarity A relationship where the person's preferences are more known to you, perhaps like a family member. Opposite: unfamiliar.

Fairness A general sense that people or players are equally well-off.

First-mover Advantage in sequential games, a setup that gives the player who chooses first the ability to determine his best outcome.

Focal Point The coordination mechanism that allows a coordination game to reach one of the solutions. A focal point is a shared belief, shared values, or a shared approach to decision-making that makes a coordination game an assurance game.

Forgive, Forgiveness The act of allowing a person who has made an inappropriate choice to play future games without any additional punishment or other penalty related to that choice.

Free-Rider Problem In economics, the situation in which one member of a large group can use or consume something without paying for it because the group is assumed to have paid for it.

Frequency the timing of events or occurrences.

Game Structure In a game-theoretic situation, this is the players, the choices, the payoffs, and the rules for making choices (simultaneous vs. sequential games, repeated games, etc.)

Game Theory Also known as strategic decision-making, this is the concepts and approaches for analyzing decisions and choices in situations where many players' decisions ultimately determine each other's outcome.

Gaslighting Similar to brainwashing, gaslighting is providing a person with false information which ultimately causes them to question their evaluation of the game.

Group-think A colloquialism referring to a crowd or herd mentality.

Guarantee To assure something.

Happiness A generally pleasurable human condition.

Heuristics A fast, often “hands on” way of setting preferences and making choices.
A learned mental shortcut.

Honor An individual’s personal, core beliefs.

Imbalanced Unequal

Immutable Unchangeable

Information Asymmetry See *asymmetric information*.

Information What is known about a player, a choice, a situation, or an outcome.

Irrational For our purposes, not following a consistent line or reasoning or not using a consistent set of preferences.

Irreversible Cannot be changed

Irrevocable See *immutable*.

Iterate A step-by-step process for reaching a conclusion or determining an outcome.

Jollies Happiness units. Used only metaphorically in the economics area known as utility theory.

Last-Period Problem In many games, the reality that players always act self-interestedly in the last period of a repeated game.

Lie Purposely and actively providing incorrect information.

Lightbulb Moment A revelation. The point in time when a person realizes or understands something.

Limit See *At the limit*

Maximin Maximize the minimum possible outcome. Making a choice to actively avoid a very bad outcome.

Median-Voter Model In political science, the idea that candidates will try to appeal to the average voter. If candidates have a political bias (often called right or left) and each candidate gets all of the voters in that bias group, the candidate who gets more of the middle—the median voters—will win the election.

Mixed Signal providing information that can have more than one meaning or interpretation.

Mixed Strategy In games that cannot be played or solved deterministically, this approach makes choices or decisions probabilistic.

Mnemonic A catchy way to remember something.

Model In economics and other disciplines, an abstract, often mathematical representation of a “real-world” activity or problem. A model is often a set of equations, graphs, or diagrams (like a payoff matrix in Game Theory) that captures the essence of the problem.

Monopolist A single firm that dominates its market and has the ability to control total market quantity and thus market price. A monopolist is able to make and maintain excess profits because there is no entry in their market.

Mutually Beneficial Outcome A game-theoretic process that has a final outcome that all players are happy with.

Nash Equilibrium A choice that is best for a player given what other players have chosen. A Nash Equilibrium is attained when no player can unilaterally make themselves better off given their current choice and all other players choices.

Negotiation The situation in which players try to divide the outcome from a joint venture.

Non-binary Choice See: *Choice*.

Observable Easily seen.

Original Boundaries The initial parameters or limitations of a situation or game.

Outcomes See *Payoffs*.

Passive Strategy The decision to not actively make choices.

Payoff, Payoff Matrix what players will get when all players have made their choices.

Perfect Competition An economics model in which there are many, small, homogeneous firms and easy exit and entry. No firm, in this model, is able to make excess profits.

Players In game-theoretic situations, the decision-makers.

Polarization A situation in which players gravitate to extreme political positions.

Populus Candidate A political candidate with broad appeal to the vast majority of the electorate.

Power The ability to exert influence over another person.

Preferences A ordering of outcomes based on a person's desires.

Prejudice A biased choice often based on spurious information or skewed beliefs.

Price Collusion See *Collusion*.

Principal-Agent Problem See *Agency Theory*.

Prisoner's Dilemma A well-known simultaneous game in which players must decide if they should confess to a crime which they committed with a partner not knowing if their partner will confess.

Promise in sequential games, a choice that the player making the second choice says they are going to make if the player making the first choice doesn't make a particular choice.

Prospering In general, doing very well or getting good outcomes repeatedly.

Proxy a player who is making decisions for other players and usually with the other players preferences in mind.

Pyrrhic Victory Winning a game but at a very high price.

Rational For our purposes, following a consistent line of reasoning or using a consistent set of preferences.

Reinforcement Strengthening a position.

Repeated Game Any game that's played more than one time with the same setup (same players, choices, payoffs, etc.)

Repressive Control Mechanism Any way in which societies prevent a subgroup from playing the game. We associate these mechanisms with rules that don't let some people vote.

Reputation In a general sense, what other players think about you.

Revenge Exacting retribution or imposing punishment for another player's inappropriate choice.

Risk See *Uncertainty*.

Rollback, rollback solution in sequential games, the ability of the player choosing first to find a path (a set of choices by both players) for achieving a particular

outcome. A rollback solution requires the first player to consider what the second player will choose under all circumstances.

Rules The procedures, usually pre-specified, for playing a game.

Rules of Thumb A colloquial expression referring to choices that work well in most situations.

Scale of Preferences The possibility that the ordering of preferred outcomes does not represent the full extent of preference. It is the difference between saying “better” and saying “a lot better.”

Screen In situations where there is a less-informed and a more-informed player, this is a mechanism for the less-informed player to learn some of the unknown information.

Second-best Solution In economics, a solution to a problem in which a superior solution exists.

Second-mover Advantage in sequential games, a setup that gives the player who chooses second the ability to determine his best outcome.

Secrets information known as to some players that they are not supposed to reveal.

Self-interest Incorporating only the interests and desires of yourself.

Sequential Game A game in which there is a pre-specified order in which players make their choices. Well-known games like chess, checkers, and Tic-Tac-Toe are sequential games.

Setup in Game Theory, the players, choices, rules, and payoffs for a game.

Short-sightedness A tendency to focus too much on outcomes that will occur sooner rather than later.

Signal In situations where there is a less-informed and a more-informed player, this is a mechanism for the more-informed player to reveal some of the unknown information.

Signaling Theory The body of work that analyzes the transfer of information from a more-informed player to a less-informed player.

Signal-to-Noise Ratio A mathematical colloquialism that refers to the accuracy of information. A high signal-to-noise ratio means that information is generally accurate.

Simultaneous Game A game in which all players make their choices at the same time.

Sincere Believable and trustworthy.

Skill An inherent or learned competitive advantage.

Social Norm In any society, the way people are expected to act. Social norms typically come from a societies culture, history, and shared beliefs.

Specify Indicate or dictate. When we specify a game, we understand the other player’s jollies and the potential outcomes. Specifying a game correctly leads to better outcomes. If a game does not land in your favor, or at an outcome you didn’t anticipate, you mis-specified the game.

Stalemate At the end of a game, the situation in which neither player has a move that allows them to win.

Strategic Advantage A situation such as superior information that allows one player to make choices that will move the outcome of a game in their favor.

Strategic Decision-Making See *game theory*.

Strategic Moves Saying what you're going to choose before a game is played. Commitments, promises, and threats are strategic moves. These are done in order to change the choice of other players and ultimately change the outcome of a game.

Structure See *Game Structure*.

Subgame A smaller game that is embedded in a larger game.

Suboptimal an outcome that does not represent the best possible outcome.

Thief A person who takes things that don't belong to them.

Threat in sequential games, a choice that the player making the second choice says they are going to make if the player making the first choice makes a particular choice.

Timing in Game Theory, refers to when players make their choices relative to each other.

Tipping Point A point beyond which a situation will irrevocably reach a conclusion. Numerically, this is often a point beyond which a process will irrevocably reach zero.

Transfer Portal In college sports, a virtual "door" in which athletes change the college they are playing for.

Trust Believing in another player's choices when direct observation is not possible.

Truthfulness Related to the act of revealing correct or accurate information.

Uncertainty Any situation in which outcomes are risky or cannot be accurately determined in advance.

Unilateral Decision Any decision that can be made without regard to other decision-makers.

Unilaterally On your own or by yourself.

Unstable See *Equilibrium*.

Any situation in which the current result isn't going to stay the same for very long.

Utils See *Jollies*.

Utility, Utility Theory In economics, the idea that the consumption of anything makes the consumer happy or satisfies a need.

Whistleblowing Reporting to higher authorities an illegal otherwise inappropriate behavior on the part of some decision-maker.

Zero-Sum Game A game in which there is an unambiguous winner and an unambiguous loser. Sports games are zero-sum games.

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